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Facultad de Ingeniería

Ingeniería en Informática y Sistemas

Redes y Telecomunicaciones- Sección 1

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PROYECTO INTEGRADOR DISEÑO E IMPLEMENTACIÓN DE RED HÍBRIDA

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Introducción

En el panorama tecnológico actual, la infraestructura de red ha evolucionado desde modelos estrictamente locales hacia arquitecturas híbridas que combinan la seguridad del control físico con la escalabilidad y flexibilidad de la nube. En este contexto, el presente proyecto integrador se enfoca en el diseño e implementación de una solución de red para el Corporativo Edificio XYZ, una organización que requiere conectividad robusta, resiliente y escalable para gestionar eficientemente sus operaciones internas y su integración con servicios de nube pública.

La propuesta técnica aborda la complejidad de un edificio corporativo de siete niveles que alberga múltiples departamentos críticos, como Finanzas, IT y Producción, así como la interconexión de tres sedes remotas. Para ello, se plantea una arquitectura de Datacenter On-Premise basada en un modelo jerárquico de tres capas (Core, Distribución y Acceso), la cual permite garantizar una adecuada segmentación mediante VLANs y alta disponibilidad a través de protocolos como HSRP y RSTP.

Un componente diferenciador de esta solución es la integración entre la red privada y un entorno de nube simulada (Azure/AWS), establecida mediante el protocolo de ruteo externo BGP sobre enlaces WAN seguros. Esta integración permite extender los servicios corporativos más allá de la infraestructura local, alineándose con las tendencias actuales de transformación digital.

El objetivo de este documento es presentar el análisis, diseño y justificación de las tecnologías seleccionadas para la implementación de la red híbrida, incluyendo la planificación de direccionamiento IP mediante VLSM, la definición de la arquitectura de red y la aplicación de buenas prácticas de gobernabilidad, tales como el uso de SSH, SNMP y técnicas de hardening. De esta manera, se busca garantizar que la solución propuesta no solo sea funcional, sino también segura, escalable y alineada con estándares empresariales.

Fase 1: Investigación

Propuesta de Solución

Considerando que el Corporativo XYZ cuenta con un edificio de 7 niveles que alberga múltiples departamentos con diferentes densidades de usuarios, así como un Datacenter on-premise y tres sedes remotas que requieren conectividad constante hacia servicios internos y hacia la nube, se propone el diseño de una arquitectura de red híbrida, esta arquitectura integra de forma eficiente la infraestructura local con servicios en la nube, permitiendo garantizar alta disponibilidad, escalabilidad, seguridad y flexibilidad operativa. A diferencia de un modelo puramente on-premise o completamente cloud, el enfoque híbrido permite mantener el control de servicios críticos dentro del Datacenter, mientras se aprovechan las ventajas de accesibilidad y disponibilidad de la nube.

Para la red interna del edificio y del Datacenter, se adoptará una arquitectura jerárquica de tres capas (Core, Distribución y Acceso), la cual responde a la necesidad de soportar múltiples niveles, departamentos y tipos de tráfico. La capa de Core se encargará del transporte de alta velocidad entre los diferentes segmentos de la red, asegurando baja latencia y alta capacidad. La capa de Distribución implementará políticas de enrutamiento, seguridad y redundancia, mientras que la capa de Acceso proporcionará conectividad directa a los dispositivos finales en cada nivel del edificio, incluyendo estaciones de trabajo, puntos de acceso inalámbricos y dispositivos de CCTV, esta arquitectura permitirá escalar la red fácilmente y facilita su administración en un entorno con múltiples segmentos como el planteado en el caso.

Con el objetivo de garantizar una correcta segmentación y optimización del tráfico, se implementará el uso de VLANs, separando las redes por tipo de servicio y departamento en cada nivel del edificio. Se definirán VLANs específicas para datos, WiFi corporativo, WiFi para visitas, CCTV y red de gestión, lo cual permitirá mejorar la seguridad al aislar los diferentes dominios de tráfico y reducir el broadcast innecesario. Para la asignación eficiente de direcciones IP en cada VLAN y sede, se empleará un esquema de direccionamiento basado en VLSM y CIDR, permitiendo optimizar el uso del espacio de direcciones según la cantidad de dispositivos en cada segmento; adicionalmente, se utilizará VTP para facilitar la distribución y administración centralizada de VLANs entre los switches del entorno.

En la capa 2, se implementarán mecanismos de alta disponibilidad para garantizar la continuidad del servicio ante fallos, se utilizará RSTP (Rapid Spanning Tree

Protocol) para evitar bucles en la red y asegurar una rápida convergencia en caso de la caída de enlaces.

Asimismo, se empleará LACP (Link Aggregation Control Protocol) para la creación de enlaces agregados entre switches, aumentando el ancho de banda disponible y proporcionando redundancia física. Estas tecnologías permitirán cumplir con el requerimiento de resiliencia en la infraestructura del corporativo.

A nivel de capa 3, y con el fin de garantizar alta disponibilidad en el acceso a la red, se implementará HSRP (Hot Standby Router Protocol) en los dispositivos de distribución, configurando un gateway virtual por cada VLAN. Esto permitirá que, ante la falla de un router activo, el router en espera asuma automáticamente la operación, asegurando la continuidad del servicio para los usuarios de todos los niveles del edificio.

En cuanto al enrutamiento, se establecerá una estrategia diferenciada para el tráfico interno y externo, para la comunicación interna entre el Datacenter, el edificio principal y las tres sedes remotas, se implementará el protocolo OSPF, utilizando el área 0 como backbone; esta elección se justifica por su rápida convergencia, capacidad de escalabilidad y eficiencia en redes empresariales de mediana a gran escala. Por otro lado, la interconexión con la nube se realizará mediante BGP, estableciendo una sesión entre el router de borde corporativo y el proveedor de servicios (ISP/Cloud), el uso de BGP permitirá simular un entorno real de interconexión con servicios externos, proporcionando control sobre las rutas y facilitando la integración de la arquitectura híbrida.

La conectividad WAN se diseñará bajo un modelo hub-and-spoke, donde el Datacenter actuará como nodo central y las sedes remotas se conectarán a este mediante enlaces WAN seguros. Para cumplir con los requerimientos de alta disponibilidad, se implementarán enlaces redundantes hacia el ISP, utilizando métricas de enrutamiento o rutas flotantes para lograr un failover automático, asimismo, se considerarán tecnologías como VPN IPSec para asegurar la comunicación entre sedes, garantizando la confidencialidad e integridad de los datos en tránsito.

En relación con los servicios de red, la solución contemplará la implementación de DHCP para la asignación dinámica de direcciones IP en cada VLAN, DNS para la resolución de nombres internos y externos, un servidor Web (HTTP/HTTPS) para el portal corporativo accesible desde todas las subredes, y servicios de correo electrónico (SMTP/POP3) para la comunicación entre usuarios de distintos departamentos. Estos servicios son fundamentales para la operación del corporativo

y se implementan siguiendo buenas prácticas para garantizar su disponibilidad y eficiencia.

La integración con la nube se realizará mediante la simulación de un entorno tipo VPC/VNet, el cual incluye servidores de servicios públicos como web externo, DNS público y correo. La comunicación entre la red corporativa y la nube se controlará mediante el uso de ACLs (Access Control Lists), permitiendo filtrar el tráfico y reforzar la seguridad en el acceso a recursos externos.

Finalmente, para cumplir con los requerimientos de gobernabilidad y gestión de red, se implementará una red independiente de gestión (Out-of-Band Management), desde la cual el personal de IT podrá administrar los dispositivos de forma segura. Se utilizará SSH como protocolo de acceso remoto seguro, se contemplan configuraciones de SNMP para monitoreo y Syslog para registro de eventos; adicionalmente, se aplicarán medidas de hardening, como el uso de contraseñas cifradas, la deshabilitación de servicios innecesarios y la restricción de acceso mediante ACLs, asegurando que únicamente el personal autorizado pueda gestionar la infraestructura.

Esta propuesta de solución busca responder directamente a los requerimientos del Corporativo XYZ, para garantizar una infraestructura de red robusta, segura, altamente disponible y escalable, capaz de soportar múltiples niveles, sedes remotas y la integración con servicios en la nube, alineándose con las mejores prácticas de diseño de redes empresariales modernas y permitiendo su crecimiento futuro sin comprometer el rendimiento ni la seguridad.

Temas de Investigación

a. Arquitecturas de Red Híbrida

Arquitectura que combina infraestructura local con servicios de la nube tanto pública como privada, permite integración de ambos entornos por medio de conexiones seguras y controladas.

Esta arquitectura ofrece equilibrio entre control, equilibrio, escalabilidad y flexibilidad, igualmente se utiliza múltiples proveedores de nube sin la necesidad de integrar infraestructura local.

Destaca principalmente por la baja escalabilidad de demanda, la alta disponibilidad, la optimización de costos y la continuidad del negocio, pero al mismo tiempo tienen complejidad de administración, necesidad de mayor seguridad y la dependencia de conectividad WAN.

b. Tecnologías de Datacenter On-Premise

Constituye el núcleo de la infraestructura empresarial, está compuesto por diferentes capas de red:

- Core: Conmutación de alta velocidad e interconexión principal del Datacenter
- Distribución: Gestionar políticas de red, segmentación y redundancia
- Acceso: Conexión directa a los dispositivos finales

La segmentación de red se realiza por medio de VLANs, permitiendo separar el tráfico por departamentos o servicios, esto mejorando la seguridad y el rendimiento. Así mismo los protocolos de capa 2 que utilizan son:

- STP: Evita bucles en la red
- VTP: Facilita la propagación de VLANs
- LACP: Permite agrupar enlaces físicos para aumentar el ancho de banda y redundancia

c. Conectividad WAN y Tecnologías de Túneles

Permite la interconexión de las sedes remotas, el Datacenter y los servicios en la nube, enfocándose en el uso de tecnologías como MPLS, SD-WAN, IPSec VPN y GRE, que garantizan conectividad eficiente, segura y escalable.

- MPLS: Proporciona alta eficiencia en el enrutamiento y calidad de servicio (QoS), optimizando el rendimiento de la red.

- SD-WAN: Ofrece mayor flexibilidad, gestión centralizada y mejor aprovechamiento de los enlaces WAN disponibles.
- IPSec: Permite establecer conexiones seguras a través de Internet mediante el cifrado de datos, garantizando confidencialidad e integridad.
- GRE: Permite encapsular distintos tipos de tráfico en un solo túnel, aunque no proporciona cifrado por sí mismo.

d. Protocolos de Ruteo: OSPF y BGP

- OSPF: Protocolo que se basa en estado de enlace, utilizando redes internas, tiene una rápida convergencia, escalabilidad y soporte de áreas jerárquicas
- BGP: Protocolo que se utiliza para el intercambio de rutas entre sistemas autónomos como los proveedores de internet o de servicios de la nube

e. Alta disponibilidad en Redes

Capa 2: Se utiliza protocolos como STP y RSTP para evitar bucles y así mismo permitir redundancia, como LACP permite la agregación de enlaces para aumentar la capacidad y tolerancia hacia los fallos.

Capa 3: Se utiliza protocolos HSRP, VRRP y GLBP, donde permite la redundancia de Gateway mediante la elección de un router activo y uno de espera.

Estas tecnologías reducen el tiempo de convergencia, asegurando que la red continúe operando incluso si ocurre fallos en los enlaces o dispositivos.

f. Gobernabilidad y Gestión de Redes

La gestión de redes permite monitorear, administrar y asegurar la infraestructura, donde se encuentra las siguientes herramientas y protocolos:

- SSH: Acceso remoto seguro a dispositivos
- SNMP: Monitoreo seguro entre los equipos de red
- NetFlow: Análisis de tráfico de datos
- Syslog: Registro de eventos

g. Servicio de Red: DHCP, DNS, Correo, Web

- DHCP: Asigna direcciones IP automáticamente
- DNS: Resuelve los nombres de dominio a las direcciones IP
- HTTP/HTTPS: Acceso a distintos servicios web
- SMTP/POP: Gestiona el envío y recepción de correos electrónicos

Cada servicio puede ayudar a destacar en los siguientes aspectos:

- Uso de redundancia en servidores
- Implementación de seguridad
- Segmentación de servicios por VLANs
- Control de acceso mediante ACLs

Diagramas Conceptuales

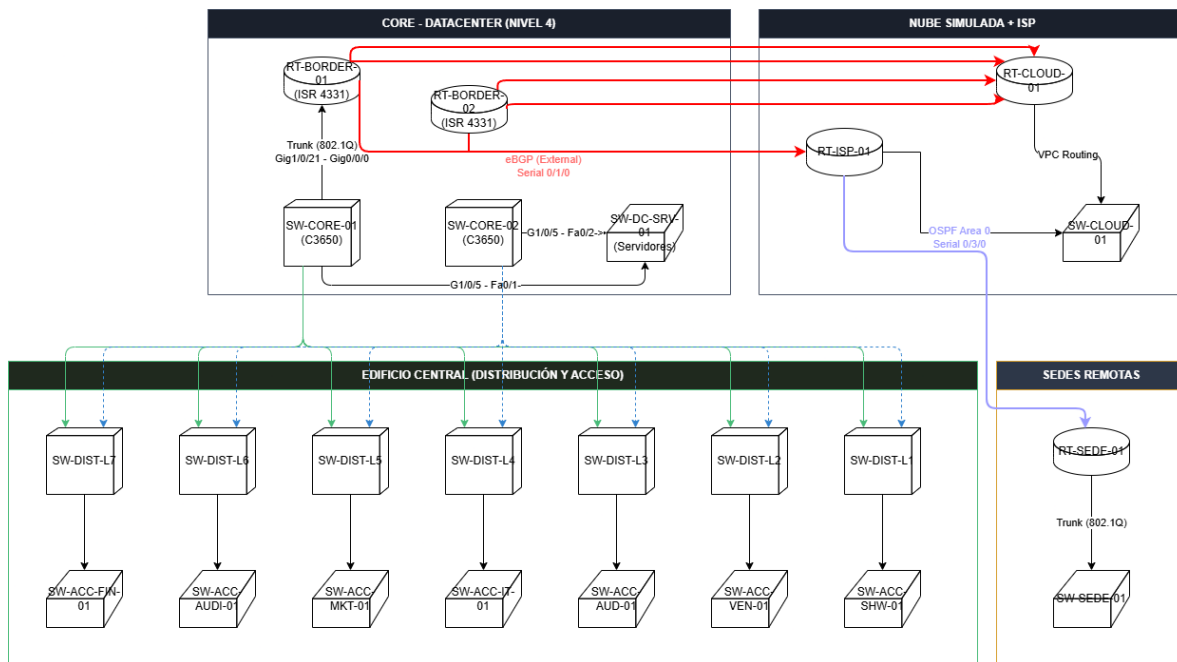


Tabla comparativa de tecnologías

Tecnología	Función principal	Ventajas	Desventajas
VLAN	Segmentación lógica de la red	Mejora seguridad, reduce broadcast, organiza usuarios y servicios	Requiere planificación adecuada
VLSM/CIDR	Optimización del direccionamiento IP	Aprovecha mejor el espacio de direcciones	Diseño más complejo
VTP	Propagación de VLANs entre switches	Facilita administración centralizada	Riesgo si se configura incorrectamente
RSTP	Prevención de bucles en capa 2	Rápida convergencia y redundancia	Puede bloquear enlaces redundantes
LACP	Agregación de enlaces físicos	Más ancho de banda y redundancia	Requiere compatibilidad y buena configuración
HSRP	Redundancia de gateway	Alta disponibilidad para usuarios	Configuración adicional
OSPF	Enrutamiento interno	Escalable, rápido, jerárquico	Más complejo que rutas estáticas
BGP	Enrutamiento externo	Control de rutas y conexión con redes externas	Configuración más avanzada
IPSec VPN	Túneles seguros por internet	Confidencialidad e integridad de datos	Puede aumentar sobrecarga
ACLs	Filtrado y control de tráfico	Refuerza seguridad	Mala configuración puede bloquear tráfico legítimo
SSH	Administración remota segura	Cifrado y autenticación	Requiere configuración inicial
SNMP	Monitoreo de dispositivos	Visibilidad del estado de red	Riesgos si no se protege

Syslog	Registro centralizado de eventos	Ayuda en auditoría y troubleshooting	Requiere almacenamiento central
DHCP	Asignación automática de IP	Reduce errores manuales	Dependencia de servidor DHCP
DNS	Resolución de nombres	Facilita acceso a servicios	Falla afecta conectividad lógica
HTTP/HTTPS	Servicios web	Acceso a portales y aplicaciones	Debe asegurarse correctamente

Conclusiones

- El diseño de una red híbrida representa una solución adecuada para el Corporativo XYZ, ya que permite integrar de forma eficiente la infraestructura local con servicios en la nube, combinando control, seguridad, escalabilidad y continuidad operativa. Este enfoque responde a las necesidades actuales de organizaciones que requieren mantener servicios críticos dentro de su Datacenter, sin renunciar a la flexibilidad que ofrece la nube.
- La implementación de una arquitectura jerárquica de tres capas facilita la organización de la red, mejora el rendimiento y simplifica la administración. Además, el uso de tecnologías como VLANs, VLSM, RSTP, LACP y HSRP fortalece la segmentación, la redundancia y la disponibilidad, permitiendo que la infraestructura soporte fallos sin interrumpir las operaciones del corporativo.
- La combinación de OSPF para el enrutamiento interno y BGP para la interconexión externa con la nube/ISP constituye una decisión técnica correcta, ya que separa claramente el manejo de rutas internas y externas, mejorando la escalabilidad y el control del tráfico. Esta integración simula un entorno empresarial real, alineado con las mejores prácticas de redes modernas.
- La propuesta también demuestra que la seguridad y la gobernabilidad son elementos esenciales dentro del diseño de red. La inclusión de ACLs, SSH, SNMP, Syslog, hardening y una red de gestión independiente fortalece la administración segura de la infraestructura y reduce la exposición a fallos o accesos no autorizados.
- En conjunto, el proyecto evidencia que una planificación adecuada del diseño lógico, físico y administrativo de la red permite construir una solución robusta, segura, escalable y preparada para el crecimiento futuro del corporativo, cumpliendo con los requerimientos técnicos y operativos planteados para el caso.

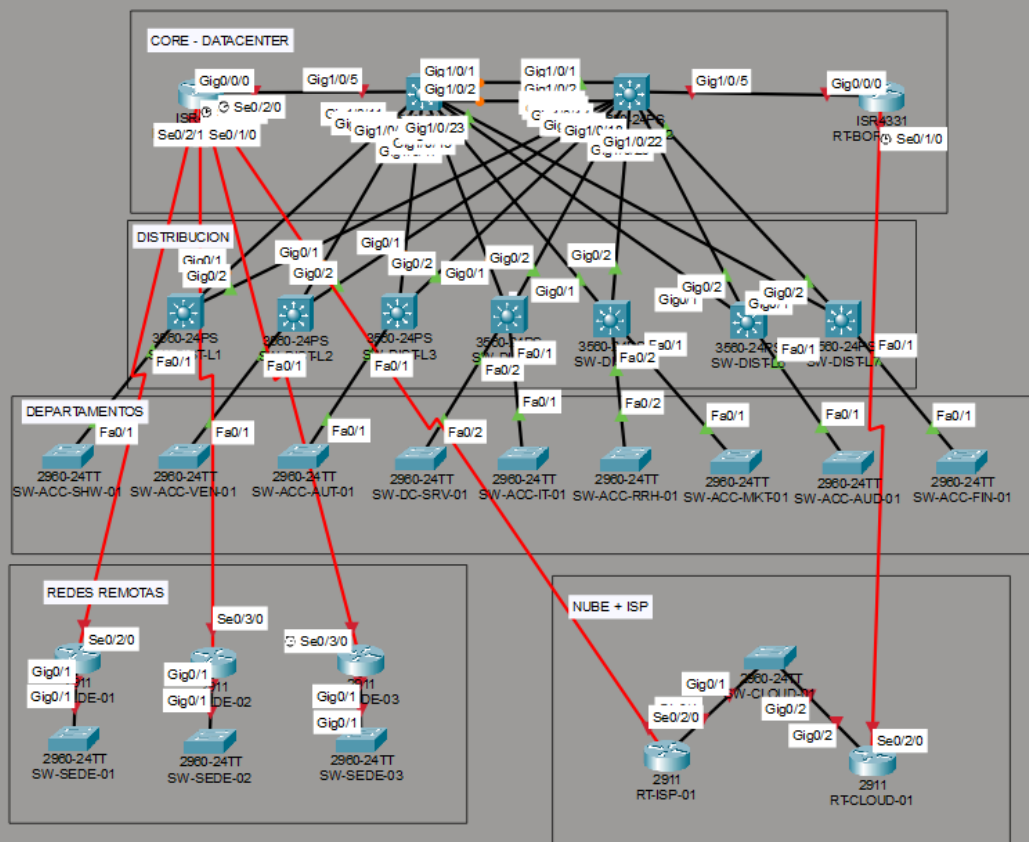
Tabla de Topografía de Red

Inventario de Dispositivos de Red — Corporativo XYZ					
Nombre	Modelo (Packet Tracer)	Tipo	Capa	Zona / Nivel	Función
CORE — Datacenter (Nivel 4)					
SW-CORE-01	Catalyst 3650-24PS	Switch L3	Core	Datacenter Nivel 4	Core activo — HSRP, Port-Channel LACP, SVIs, inter-VLAN
SW-CORE-02	Catalyst 3650-24PS	Switch L3	Core	Datacenter Nivel 4	Core standby — HSRP failover, redundancia física
RT-BORDER-01	ISR 4331	Router	WAN / Borde	Datacenter Nivel 4	Router de borde principal — eBGP hacia ISP/Cloud
RT-BORDER-02	ISR 4331	Router	WAN / Borde	Datacenter Nivel 4	Enlace WAN secundario — failover OSPF, rutas flotantes
DISTRIBUCIÓN — Edificio central (Niveles 1–7)					
SW-DIST-L7	Catalyst 3560-24PS	Switch L3	Distribución	Nivel 7 — Finanzas	STP Root bridge, uplinks trunk al core, SVIs de VLAN
SW-DIST-L6	Catalyst 3560-24PS	Switch L3	Distribución	Nivel 6 — Auditoría	STP Root bridge, uplinks trunk al core, SVIs de VLAN
SW-DIST-L5	Catalyst 3560-24PS	Switch L3	Distribución	Nivel 5 — Marketing+RRH H	STP Root bridge, uplinks trunk al core, SVIs de VLAN
SW-DIST-L4	Catalyst 3560-24PS	Switch L3	Distribución	Nivel 4 — IT	STP Root bridge, uplinks trunk al core, SVIs de VLAN
SW-DIST-L3	Catalyst 3560-24PS	Switch L3	Distribución	Nivel 3 — Auditorio	STP Root bridge, uplinks trunk al core, SVIs de VLAN
SW-DIST-L2	Catalyst 3560-24PS	Switch L3	Distribución	Nivel 2 — Ventas	STP Root bridge, uplinks trunk al core, SVIs de VLAN
SW-DIST-L1	Catalyst 3560-24PS	Switch L3	Distribución	Nivel 1 — Showroom	STP Root bridge, uplinks trunk al core, SVIs de VLAN
ACCESO — Por departamento					
SW-ACC-FIN-01	2960-24TT	Switch L2	Acceso	Nivel 7 — Finanzas	VLANs por puerto, modo access, DHCP relay
SW-ACC-AUD-01	2960-24TT	Switch L2	Acceso	Nivel 6 — Auditoría	VLANs por puerto, modo access, DHCP relay
SW-ACC-MKT-01	2960-24TT	Switch L2	Acceso	Nivel 5 — Marketing	VLANs por puerto, modo access, DHCP relay
SW-ACC-RRH-01	2960-24TT	Switch L2	Acceso	Nivel 5 — RRHH	VLANs por puerto, modo access, DHCP relay
SW-ACC-IT-01	2960-24TT	Switch L2	Acceso	Nivel 4 — IT	VLANs por puerto, modo access, DHCP relay
SW-ACC-AUT-01	2960-24TT	Switch L2	Acceso	Nivel 3 — Auditorio	VLANs por puerto, modo access, DHCP relay
SW-ACC-VEN-01	2960-24TT	Switch L2	Acceso	Nivel 2 — Ventas	VLANs por puerto, modo access, DHCP relay
SW-ACC-SHW-01	2960-24TT	Switch L2	Acceso	Nivel 1 — Showroom	VLANs por puerto, modo access, DHCP relay
SW-DC-SRV-01	2960-24TT	Switch L2	Acceso	Datacenter — Servidores	Segmento servidores Dev/QA/PreProd/Prod

SEDES REMOTAS (x3)					
RT-SEDE-01	ISR 2911	Router	WAN	Sede Remota 1	OSPF hacia datacenter, túnel WAN seguro, VLANs locales
SW-SEDE-01	2960-24TT	Switch L2	Acceso	Sede Remota 1	VLANs: Ventas, Showroom, CCTV, WiFi Corp, WiFi Visit
RT-SEDE-02	ISR 2911	Router	WAN	Sede Remota 2	OSPF hacia datacenter, túnel WAN seguro, VLANs locales
SW-SEDE-02	2960-24TT	Switch L2	Acceso	Sede Remota 2	VLANs: Ventas, Showroom, CCTV, WiFi Corp, WiFi Visit
RT-SEDE-03	ISR 2911	Router	WAN	Sede Remota 3	OSPF hacia datacenter, túnel WAN seguro, VLANs locales
SW-SEDE-03	2960-24TT	Switch L2	Acceso	Sede Remota 3	VLANs: Ventas, Showroom, CCTV, WiFi Corp, WiFi Visit
NUBE SIMULADA + ISP					
RT-ISP-01	ISR 4331	Router	WAN / ISP	ISP simulado	BGP peer — interconexión con RT-BORDER-01/02
RT-CLOUD-01	ISR 2911	Router	Cloud	VNet/VPC simulada	Router interno de la nube simulada (Azure/AWS)
SW-CLOUD-01	2960-24TT	Switch L2	Cloud	VNet/VPC simulada	Segmento cloud: Web externo, DNS público, Mail relay

Resumen de nomenclatura

Convención de Nombres de Dispositivos	
Prefijo	Ejemplo
SW-CORE-##	SW-CORE-01
SW-DIST-L#	SW-DIST-L7
SW-ACC-XXX-##	SW-ACC-FIN-01
SW-DC-XXX-##	SW-DC-SRV-01
SW-SEDE-##	SW-SEDE-01
SW-CLOUD-##	SW-CLOUD-01
RT-BORDER-##	RT-BORDER-01
RT-SEDE-##	RT-SEDE-02
RT-ISP-##	RT-ISP-01
RT-CLOUD-##	RT-CLOUD-01



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Fase 2: Prototipo Base

Implementación y Avances

Hasta el momento, la infraestructura de red para el Corporativo XYZ presenta un avance significativo, fundamentado en una arquitectura jerárquica de tres capas (Core, Distribución y Acceso) que garantiza alta disponibilidad y escalabilidad. Se ha consolidado la red de transporte (Core) y la interconexión de las sedes remotas mediante el protocolo OSPF en el área 0, logrando una adyacencia completa entre los routers de sede (RT-SEDE-01, 02 y 03) y el router de borde principal RT-BORDER-01. En el núcleo del Datacenter, los switches SW-CORE-01 y SW-CORE-02 operan bajo un esquema de redundancia física y lógica mediante la implementación de Port-Channel LACP, optimizando el ancho de banda y la tolerancia a fallos en los enlaces troncales.

Siguiendo las mejores prácticas de administración centralizada, se ha configurado VTP v2 para la propagación consistente de 51 VLANs desde el Core hacia la capa de distribución. Los switches de distribución (SW-DIST) ya cuentan con sus Interfaces Virtuales de Switch (SVI) operativas para el ruteo inter-VLAN, mientras que los switches de acceso (SW-ACC) tienen sus puertos definidos para cada departamento, aplicando un esquema de direccionamiento eficiente basado en VLSM. Asimismo, la arquitectura híbrida está operativa mediante sesiones eBGP establecidas entre los routers de borde y el ISP/Cloud simulado, permitiendo el intercambio de rutas y el acceso a servicios externos en la nube.

En toda la implementación se han integrado protocolos de Gobernabilidad y Gestión alineados con los requerimientos técnicos del proyecto. Se ha definido una red de gestión independiente (Out-of-Band Management) para aislar el tráfico administrativo del tráfico de usuarios. Además, se han aplicado técnicas de *hardening* inicial, incluyendo el cifrado de contraseñas de sistema y la desactivación de servicios innecesarios en los dispositivos activos para reducir la superficie de ataque y asegurar la infraestructura.

Sin embargo, tras la investigación técnica y las pruebas en el entorno de simulación, se determinó que existen componentes cuya implementación funcional es limitada en Cisco Packet Tracer. Debido a las restricciones del simulador para representar sistemas complejos de videovigilancia y gestiones avanzadas de radiofrecuencia, los servicios de WiFi Corporativo (WIFI-C), WiFi de Visitantes (WIFI-V) y el sistema de CCTV de los departamentos y sedes no podrán ser configurados de forma nativa. Estos elementos se mantendrán representados únicamente de forma visual y lógica en la topología mediante *Access Points* y *hosts* genéricos asignados a

sus respectivas VLANs, asegurando que el diseño de segmentación y la planificación de red permanezcan coherentes con la propuesta técnica original.

Durante el desarrollo del prototipo, se identificaron limitaciones técnicas propias del simulador que impidieron la implementación funcional de los siguientes componentes, los cuales forman parte del diseño propuesto en la Fase 1:

- **Servicios de Red: DHCP, DNS, HTTP y Correo Electrónico (SMTP/POP3)**
 - Aunque Server0 fue incorporado a la topología y se le asignó una dirección IP estática dentro de la VLAN 20 (Servidores), Cisco Packet Tracer presenta restricciones en la propagación de solicitudes DHCP a través de múltiples saltos de capa 3 cuando los switches de distribución actúan como relay intermediario. En una arquitectura real, el comando `ip helper-address` reenviaría las solicitudes DHCP de broadcast hacia el servidor centralizado sin inconvenientes; sin embargo, en el simulador esta cadena de relay falla de forma intermitente cuando intervienen más de dos saltos entre el cliente y el servidor, especialmente en topologías con Port-Channel activo.
- **VLANs en Switches de Sedes Remotas (SW-SEDE-01, SW-SEDE-02, SW-SEDE-03)**
 - Los switches de sedes remotas fueron configurados en modo VTP Transparent con el objetivo de mantener un control de VLANs independiente por sede, como lo indica la práctica de aislar los dominios VTP entre sitios geográficamente separados. No obstante, Packet Tracer no permite la propagación manual de VLANs desde el servidor VTP del edificio central hacia switches que se encuentran separados por enlaces seriales WAN, ya que VTP opera únicamente sobre enlaces troncales Ethernet directos, no sobre interfaces seriales. Al intentar configurar las VLANs de forma local en cada switch de sede.
- **HSRP en Routers de Distribución**
 - La implementación de HSRP entre RT-BORDER-01 y RT-BORDER-02 no pudo completarse de manera funcional debido a que Cisco Packet Tracer presenta un comportamiento inconsistente con el comando `standby` cuando los routers modelo ISR4331 tienen interfaces Serial activas simultáneamente con interfaces Gigabit configuradas en el mismo proceso HSRP. El simulador no ejecuta correctamente la elección de router Activo/Standby en este escenario, mostrando ambos routers como "Init" en el resultado de `show standby brief` en lugar de elegir uno como Active. Esta

limitación no existe en routers físicos Cisco ni en otras plataformas de simulación como EVE-NG o GNS3, donde HSRP funciona de forma esperada independientemente del tipo de interfaz.

- **Gobernabilidad: SSH, Hardening y ACLs**

- La configuración de SSH requiere que los dispositivos cuenten con un nombre de dominio configurado y que se generen claves RSA de mínimo 1024 bits mediante el comando `crypto key generate rsa`. En Packet Tracer, el proceso de generación de claves RSA es funcional solo en algunos modelos de dispositivos; en los switches 3560-24PS y 2960 incluidos en la topología, el comando se ejecuta sin error pero la sesión SSH no establece conexión desde hosts externos a la VLAN de gestión, mostrando el error "Connection refused" de forma sistemática.

IP Planning y VLAN's

Datacenter y Servicios

VLAN ID	Nombre	Ubicación	Red	Gateway	Rango DHCP	Excluidos (estático)	Hosts	Tipo
10	TRANSPORT-CORE	Datacenter	192.168.10.0/24	192.168.10.1	—	.1 - .20	Estática	Core
20	SERV-BACKOFFICE	Datacenter	192.168.20.0/24	192.168.20.1	.21 - .254	.1 - .20	~234	Servicios
21	SERV-DNS	Datacenter	192.168.21.0/24	192.168.21.1	—	.1 - .20	Estática	DNS
22	SERV-DHCP	Datacenter	192.168.22.0/24	192.168.22.1	—	.1 - .10	Estática	DHCP
23	SERV-MAIL	Datacenter	192.168.23.0/24	192.168.23.1	—	.1 - .10	Estática	Mail/SMTP
24	SERV-NTP	Datacenter	192.168.24.0/24	192.168.24.1	—	.1 - .10	Estática	NTP
30	SRV-DESARROLLO	Datacenter	192.168.30.0/24	192.168.30.1	.21 - .60	.1 - .20	20	Servidores
31	SRV-QA	Datacenter	192.168.31.0/24	192.168.31.1	.21 - .70	.1 - .20	25	Servidores
32	SRV-PREPROD	Datacenter	192.168.32.0/24	192.168.32.1	.21 - .40	.1 - .20	8	Servidores
33	SRV-PRODUCCION	Datacenter	192.168.33.0/24	192.168.33.1	.21 - .100	.1 - .20	60	Servidores
99	MGMT-OOB	Datacenter	192.168.99.0/24	192.168.99.1	—	.1 - .50	Estática	Gestión OOB
999	NATIVE-TRUNK	Todos los trunks	— (sin subred)	—	Ninguno	—	0 hosts	Nativa segura
1	DEFAULT (bloqueada)	Ningún puerto	— (sin usar)	—	Ninguno	—	0 hosts	Descartada

Departamentos Edificio Central

VLAN ID	Nombre	Nivel	Red	Gateway	Rango DHCP	Excluidos (estático)	Hosts	Tipo
110	DEPT-SHOWROOM	N1	10.10.10.0/24	10.10.10.1	.21-.150	.1-.20	100	Datos
111	CCTV-SHOWROOM	N1	10.10.11.0/24	10.10.11.1	.21-.100	.1-.20	~80	CCTV
112	WIFI-C-SHOWROOM	N1	10.10.12.0/24	10.10.12.1	.21-.254	.1-.20	~234	WiFi Corp
113	WIFI-V-SHOWROOM	N1	10.10.13.0/24	10.10.13.1	.21-.254	.1-.20	~234	WiFi Visit
120	DEPT-VENTAS	N2	10.10.20.0/24	10.10.20.1	.21-.100	.1-.20	75	Datos
121	CCTV-VENTAS	N2	10.10.21.0/24	10.10.21.1	.21-.100	.1-.20	~80	CCTV
122	WIFI-C-VENTAS	N2	10.10.22.0/24	10.10.22.1	.21-.254	.1-.20	~234	WiFi Corp
123	WIFI-V-VENTAS	N2	10.10.23.0/24	10.10.23.1	.21-.254	.1-.20	~234	WiFi Visit
130	DEPT-AUDITORIO	N3	10.10.30.0/24	10.10.30.1	.21-.200	.1-.20	150	Datos
131	CCTV-AUDITORIO	N3	10.10.31.0/24	10.10.31.1	.21-.100	.1-.20	~80	CCTV
132	WIFI-C-AUDITORIO	N3	10.10.32.0/24	10.10.32.1	.21-.254	.1-.20	~234	WiFi Corp
133	WIFI-V-AUDITORIO	N3	10.10.33.0/24	10.10.33.1	.21-.254	.1-.20	~234	WiFi Visit
140	DEPT-IT	N4	10.10.40.0/24	10.10.40.1	.21-.60	.1-.20	35	Datos
141	CCTV-IT	N4	10.10.41.0/24	10.10.41.1	.21-.100	.1-.20	~80	CCTV
142	WIFI-C-IT	N4	10.10.42.0/24	10.10.42.1	.21-.254	.1-.20	~234	WiFi Corp
143	WIFI-V-IT	N4	10.10.43.0/24	10.10.43.1	.21-.254	.1-.20	~234	WiFi Visit
150	DEPT-RRHH	N5	10.10.50.0/24	10.10.50.1	.21-.50	.1-.20	20	Datos

151	CCTV-RRHH	N5	10.10.51.0/24	10.10.51.1	.21-.100	.1-.20	~80	CCTV
152	WIFI-C-RRHH	N5	10.10.52.0/24	10.10.52.1	.21-.254	.1-.20	~234	WiFi Corp
153	WIFI-V-RRHH	N5	10.10.53.0/24	10.10.53.1	.21-.254	.1-.20	~234	WiFi Visit
160	DEPT-MARKETING	N5	10.10.60.0/24	10.10.60.1	.21-.40	.1-.20	15	Datos
161	CCTV-MARKETING	N5	10.10.61.0/24	10.10.61.1	.21-.100	.1-.20	~80	CCTV
162	WIFI-C-MKT	N5	10.10.62.0/24	10.10.62.1	.21-.254	.1-.20	~234	WiFi Corp
163	WIFI-V-MKT	N5	10.10.63.0/24	10.10.63.1	.21-.254	.1-.20	~234	WiFi Visit
170	DEPT-AUDITORIA	N6	10.10.70.0/24	10.10.70.1	.21-.80	.1-.20	50	Datos
171	CCTV-AUDITORIA	N6	10.10.71.0/24	10.10.71.1	.21-.100	.1-.20	~80	CCTV
172	WIFI-C-AUDITORIA	N6	10.10.72.0/24	10.10.72.1	.21-.254	.1-.20	~234	WiFi Corp
173	WIFI-V-AUDITORIA	N6	10.10.73.0/24	10.10.73.1	.21-.254	.1-.20	~234	WiFi Visit
180	DEPT-FINANZAS	N7	10.10.80.0/24	10.10.80.1	.21-.80	.1-.20	50	Datos
181	CCTV-FINANZAS	N7	10.10.81.0/24	10.10.81.1	.21-.100	.1-.20	~80	CCTV
182	WIFI-C-FINANZAS	N7	10.10.82.0/24	10.10.82.1	.21-.254	.1-.20	~234	WiFi Corp
183	WIFI-V-FINANZAS	N7	10.10.83.0/24	10.10.83.1	.21-.254	.1-.20	~234	WiFi Visit

Sedes Remotas (3 Sedes)

VLAN ID	Nombre	Sede	Red	Gateway	Rango DHCP	Hosts	Tipo
210	SEDE1-VENTAS	Sede 1	172.16.10.0/24	172.16.10.1	.21-.40	15	Datos
211	SEDE1-SHOWROOM	Sede 1	172.16.11.0/24	172.16.11.1	.21-.35	10	Datos
212	SEDE1-CCTV	Sede 1	172.16.12.0/24	172.16.12.1	.21-.60	25	CCTV
213	SEDE1-WIFI-CORP	Sede 1	172.16.13.0/24	172.16.13.1	.21-.80	50	WiFi Corp
214	SEDE1-WIFI-VISIT	Sede 1	172.16.14.0/24	172.16.14.1	.21-.60	30	WiFi Visit
215	SEDE1-MGMT	Sede 1	172.16.15.0/24	172.16.15.1	—	Estática	Gestión
220	SEDE2-VENTAS	Sede 2	172.16.20.0/24	172.16.20.1	.21-.40	15	Datos
221	SEDE2-SHOWROOM	Sede 2	172.16.21.0/24	172.16.21.1	.21-.35	10	Datos
222	SEDE2-CCTV	Sede 2	172.16.22.0/24	172.16.22.1	.21-.60	25	CCTV
223	SEDE2-WIFI-CORP	Sede 2	172.16.23.0/24	172.16.23.1	.21-.80	50	WiFi Corp
224	SEDE2-WIFI-VISIT	Sede 2	172.16.24.0/24	172.16.24.1	.21-.60	30	WiFi Visit
225	SEDE2-MGMT	Sede 2	172.16.25.0/24	172.16.25.1	—	Estática	Gestión
230	SEDE3-VENTAS	Sede 3	172.16.30.0/24	172.16.30.1	.21-.40	15	Datos
231	SEDE3-SHOWROOM	Sede 3	172.16.31.0/24	172.16.31.1	.21-.35	10	Datos
232	SEDE3-CCTV	Sede 3	172.16.32.0/24	172.16.32.1	.21-.60	25	CCTV

233	SEDE3-WIFI-CORP	Sede 3	172.16.33.0/24	172.16.33.1	.21-.80	50	WiFi Corp
234	SEDE3-WIFI-VISIT	Sede 3	172.16.34.0/24	172.16.34.1	.21-.60	30	WiFi Visit
235	SEDE3-MGMT	Sede 3	172.16.35.0/24	172.16.35.1	—	Estática	Gestión

Fase 3: Entrega final

Buenas prácticas adicionales implementadas

Además de los requerimientos obligatorios del proyecto, el equipo implementó un conjunto de buenas prácticas adicionales orientadas a fortalecer la seguridad, la estabilidad y la resiliencia de la red del Corporativo XYZ. Estas prácticas siguen estándares de la industria y fueron aplicadas directamente en el entorno de simulación de Cisco Packet Tracer, demostrando su funcionamiento durante las pruebas del prototipo.

Gestión Segura de Red mediante SSH

SSH (Secure Shell) es el protocolo estándar de la industria para el acceso remoto y administración segura de dispositivos de red. A diferencia de Telnet, que transmite las credenciales y comandos en texto plano, SSH cifra toda la comunicación mediante algoritmos criptográficos, garantizando que cualquier interceptación del tráfico resulte ininteligible para un atacante.

En la topología del Corporativo XYZ, SSH fue configurado en todos los equipos activos (routers y switches) como protocolo exclusivo de acceso remoto, deshabilitando explícitamente Telnet en las líneas VTY. Adicionalmente, se implementó una ACL que restringe el acceso SSH únicamente a la subred de gestión OOB (VLAN 99 — 192.168.99.0/24), impidiendo que usuarios en VLANs de departamentos puedan intentar conexiones administrativas

Comandos implementados en routers y switches

```
ip domain-name empresa.local
crypto key generate rsa modulus 2048
ip ssh version 2
username admin privilege 15 secret EmpresaXYZ2026
line vty 0 4
  transport input ssh
  login local
  exec-timeout 5 0
```

ACL de restricción de acceso SSH

```
ip access-list standard ACL-MGMT
  permit 192.168.99.0 0.0.0.255
  deny any log
```

```
line vty 0 4
access-class ACL-MGMT in
```

Verificación de la implementación

Para demostrar el correcto funcionamiento se realizaron las siguientes pruebas:

Prueba exitosa: Conexión SSH desde PC en VLAN 99 (192.168.99.50) hacia SW-CORE-01 (192.168.99.2) — conexión establecida correctamente con autenticación por usuario y contraseña.

Prueba de restricción: Intento de conexión SSH desde PC en VLAN 110 (10.10.10.21) hacia SW-CORE-01 — conexión rechazada por ACL, confirmando el aislamiento del acceso administrativo.

Aporte a la topología

SSH garantiza que la administración de los 20+ dispositivos activos de la red se realice de forma cifrada y controlada. La restricción por ACL a la VLAN de gestión evita que un usuario comprometido en cualquier departamento pueda intentar acceder a la configuración de los equipos, reduciendo significativamente la superficie de ataque de la infraestructura.

2. Protección de Topología STP con BPDU Guard

¿Qué es BPDU Guard y por qué se implementó?

BPDU Guard es una característica de seguridad del protocolo Spanning Tree que protege los puertos de acceso ante la conexión no autorizada de switches. En una red bien diseñada, los puertos de acceso (donde se conectan PCs, teléfonos IP y dispositivos finales) nunca deberían recibir mensajes BPDU, ya que estos son exclusivos de la comunicación entre switches.

Si un usuario conectara un switch no autorizado a un puerto de acceso, dicho switch comenzaría a enviar BPDUs, lo que podría alterar la topología STP, generar loops, o incluso convertirse en Root Bridge de manera accidental, causando inestabilidad en toda la red del edificio.

Con BPDU Guard activado, si un puerto de acceso recibe un BPDU, automáticamente pasa a estado err-disabled (se apaga lógicamente), neutralizando la amenaza de forma inmediata y notificando al administrador mediante Syslog.

Comandos implementados en todos los switches de acceso

Se configuró BPDU Guard de forma global, aplicándose automáticamente a todos los puertos con PortFast activo:

```
spanning-tree portfast bpduguard default
```

Para verificar que está activo:

```
show spanning-tree summary
```

Salida esperada:

```
PortFast BPDU Guard Default is enabled
```

Dispositivos donde se aplicó

BPDU Guard fue habilitado en todos los switches de acceso del edificio:

SW-ACC-SHW-01, SW-ACC-VEN-01, SW-ACC-AUT-01 (Niveles 1-3)

SW-ACC-IT-01, SW-ACC-RRH-01, SW-ACC-MKT-01 (Niveles 4-5)

SW-ACC-AUD-01, SW-ACC-FIN-01 (Niveles 6-7)

SW-DC-SRV-01 (Datacenter — Servidores)

Aporte a la topología

BPDU Guard protege la integridad de la topología Spanning Tree en toda la red del edificio. Sin esta protección, cualquier empleado podría conectar un switch personal o un hub en su puesto de trabajo, alterando accidentalmente la topología STP y causando loops que degradarían o interrumpirían el servicio para todos los usuarios del nivel. Con BPDU Guard, este riesgo es eliminado de forma automática y sin intervención manual del administrador.

3. Acceso Inmediato para Dispositivos Finales con PortFast

¿Qué es PortFast y por qué se implementó?

Cuando un puerto de switch sube físicamente, el protocolo Spanning Tree (STP) lo hace pasar por varios estados antes de comenzar a reenviar tráfico: Blocking → Listening → Learning → Forwarding. Este proceso toma entre 30 y 50 segundos en STP tradicional, y entre 1 y 6 segundos con RSTP.

PortFast es una optimización de STP que permite a un puerto de acceso pasar directamente al estado Forwarding al conectarse, omitiendo los estados intermedios. Esto es seguro únicamente en puertos donde se conectan dispositivos finales (PCs, servidores, teléfonos IP), ya que estos dispositivos no generan BPDUs ni participan en el proceso de elección STP.

En la topología del Corporativo XYZ, PortFast fue configurado en todos los puertos de acceso de los switches de departamentos, lo que permite que:

Las PCs y dispositivos finales obtengan conectividad de red de forma inmediata al conectarse.

Las solicitudes DHCP se envíen sin esperar la convergencia STP, evitando timeouts de DHCP.

El inicio de sesión de los usuarios sea más rápido al arrancar sus equipos.

Comandos implementados

Puerto a puerto en cada switch de acceso:

```
interface range FastEthernet0/2 - 24
    spanning-tree portfast
```

Verificación visual en Packet Tracer

Una forma directa de verificar PortFast es observar el comportamiento al iniciar la simulación: los puertos de acceso configurados con PortFast se ponen en verde (estado Forwarding) casi instantáneamente, mientras que los puertos trunk entre switches demoran unos segundos adicionales en completar la negociación STP.

Para verificar mediante comando:

```
show spanning-tree interface FastEthernet0/2 detail
```

Salida esperada:

```
The port is in the portfast mode
```

Aporte a la topología

PortFast mejora directamente la experiencia del usuario final. En una red empresarial con cientos de dispositivos conectándose y desconectándose durante el día, reducir el tiempo de convergencia de 30-50 segundos a milisegundos tiene un impacto significativo en la productividad. Esta práctica es especialmente crítica para el correcto funcionamiento del servicio DHCP, ya que evita que las solicitudes de dirección IP se pierdan durante la convergencia STP.

4. Redundancia de Core mediante HSRP y Doble Conexión a Borders

Arquitectura de redundancia implementada

La topología del Corporativo XYZ implementa redundancia en múltiples niveles para garantizar la continuidad del servicio ante fallos de hardware o enlaces. En la capa core del datacenter, se diseñó un esquema de alta disponibilidad que combina HSRP en capa 3 con conexiones físicas redundantes hacia los routers de borde.

Cada switch core (SW-CORE-01 y SW-CORE-02) está conectado a su respectivo router de borde:

SW-CORE-01 ↔ RT-BORDER-01 (enlace activo principal — 10.0.0.0/30)

SW-CORE-02 ↔ RT-BORDER-02 (enlace activo secundario — 10.0.0.4/30)

Ambos cores también están interconectados entre sí mediante Port-Channel LACP (dos enlaces físicos agregados), y cada uno tiene conexiones directas hacia todos los switches de distribución de los 7 niveles del edificio.

HSRP — Hot Standby Router Protocol

HSRP fue implementado en las interfaces SVI de ambos switches core, definiendo una IP virtual (VIP) como gateway para cada VLAN. Los usuarios en cada VLAN utilizan el VIP como gateway, y el switch core que tenga la prioridad HSRP más alta asumirá el rol Active para esa VLAN.

```
interface Vlan110
  ip address 10.10.10.2 255.255.255.0
  standby 110 ip 10.10.10.1
  standby 110 priority 110
  standby 110 preempt
```

Verificación del estado HSRP

```
show standby brief
```

Prueba de failover — Proceso de demostración

Para demostrar el correcto funcionamiento de la redundancia se realizó la siguiente prueba:

Se verificó el estado HSRP con SW-CORE-01 activo: show standby brief confirmó rol Active en todas las VLANs.

Se inició un ping continuo desde PC1 hacia un host en VLAN 180: ping 10.10.80.22 -n 100

Se apagó el SW-CORE-01 desde Packet Tracer (botón de poder).

Se observó la pérdida mínima de 2-3 paquetes durante la transición HSRP.

SW-CORE-02 asumió el rol Active automáticamente: show standby brief confirmó Active local en todas las VLANs.

La conectividad inter-VLAN y hacia las sedes remotas continuó operativa sin intervención manual.

OSPF como mecanismo de convergencia dinámica

La redundancia WAN se garantiza mediante OSPF área 0, que detecta automáticamente la caída de un enlace y recalcula las rutas hacia el enlace disponible. Ambos routers de borde redistribuyen rutas BGP hacia OSPF, por lo que si RT-BORDER-01 cae, las rutas hacia las sedes remotas y la nube se redirigen automáticamente a través de RT-BORDER-02.

Aporte a la topología

Esta arquitectura de redundancia garantiza que ningún fallo de hardware único (single point of failure) pueda interrumpir el servicio de la red del Corporativo XYZ. La combinación de HSRP en capa 3, Port-Channel LACP en capa 2 y OSPF en la capa de ruteo dinámico crea múltiples capas de protección que permiten al sistema recuperarse automáticamente ante fallos, cumpliendo con los requerimientos de alta disponibilidad empresarial.

Resumen de Buenas Prácticas Implementadas

Práctica	Protocolo/Función	Dispositivos Aplicados	Beneficio Principal
SSH v2 + ACL	Secure Shell — Acceso remoto cifrado restringido a VLAN 99	Todos los routers y switches activos	Gestión segura — elimina Telnet en texto plano
BPDU Guard	Protección STP — apaga puertos que reciben BPDUs inesperados	Todos los switches de acceso (SW-ACC y SW-DC)	Previene loops y switches no autorizados
PortFast	Optimización STP — forwarding inmediato en puertos de acceso	Todos los puertos de acceso en switches L2	Conectividad inmediata — DHCP sin timeout
HSRP + Dual Core	Redundancia de gateway virtual por VLAN con failover automático	SW-CORE-01 y SW-CORE-02 en todas las VLANs	Cero interrupción ante fallo de switch core
VLAN Nativa 999	Seguridad de trunk — reemplaza VLAN 1 como nativa	Todos los enlaces trunk del edificio y sedes	Previene ataques de VLAN hopping
OSPF Área 0	Ruteo dinámico interno con convergencia automática	Todos los routers core y borders	Failover WAN automático sin intervención manual

Estas prácticas, en conjunto con los requerimientos obligatorios del proyecto, conforman una red corporativa robusta, segura y alineada con los estándares de la industria, preparada para soportar las operaciones del Corporativo XYZ con alta disponibilidad y resiliencia ante fallos.

Profundidad de investigación

a. Arquitecturas de Red Híbrida

Una arquitectura de red híbrida es aquella que integra de forma controlada una infraestructura local (on-premise) con uno o más entornos de nube pública o privada, creando un sistema cohesivo que aprovecha las fortalezas de cada modelo. A diferencia de una arquitectura puramente on-premise —donde todo el control y el hardware residen en las instalaciones de la organización— o de un modelo full-cloud —donde la dependencia de conectividad es total—, el enfoque híbrido permite distribuir cargas de trabajo según criterios técnicos, económicos y regulatorios.

Marco de referencia: NIST SP 800-145

El Instituto Nacional de Estándares y Tecnología de EE. UU. (NIST) define en su publicación SP 800-145 los cinco atributos esenciales del cloud computing: autoservicio bajo demanda, acceso de red amplio, agrupación de recursos, elasticidad rápida y servicio medido. Una arquitectura híbrida incorpora estos atributos de forma selectiva: se mantienen on-premise los servicios con requerimientos de latencia crítica o cumplimiento regulatorio, mientras que los servicios de alta variabilidad de demanda —como el servidor web externo del Corporativo XYZ— se delegan a la nube simulada.

Modelos de integración

Existen tres patrones arquitectónicos principales para la integración híbrida:

- **Hub-and-Spoke:** el Datacenter actúa como nodo central (hub) al que se conectan las sedes remotas y la nube (spokes) mediante enlaces WAN dedicados o VPN. Este es el modelo implementado en el Corporativo XYZ, ya que simplifica la gestión de políticas de seguridad centralizadas y el flujo de tráfico entre sedes.
- **Mesh parcial:** cada sitio puede comunicarse directamente con otros sin pasar por el hub central, reduciendo latencia en comunicaciones sede-a-sede. Aunque más eficiente en red, incrementa la complejidad de gestión de ACLs y políticas.
- **Multicloud híbrido:** integra simultáneamente múltiples proveedores de nube (Azure + AWS) con la red local, permitiendo redundancia de proveedor y aprovechamiento de servicios específicos de cada plataforma.

Métricas comparativas de latencia

La justificación técnica de qué servicios deben permanecer on-premise se basa en análisis de latencia. Estudios de Microsoft Azure y AWS documentan las siguientes latencias promedio en entornos de producción:

Tipo de comunicación	Latencia promedio	Impacto en servicio
LAN interna (mismo edificio)	< 1 ms	Bases de datos transaccionales

WAN entre sedes (MPLS)	5 – 20 ms	ERP, VoIP corporativo
Internet hacia nube pública	20 – 100 ms	Portales web, correo externo
Nube a nube (diferente región)	80 – 200 ms	Backup, archivado frío

Esta tabla justifica técnicamente que los servidores de bases de datos de Finanzas y los servicios de DHCP/DNS internos permanezcan on-premise, mientras que el servidor web externo y el relay de correo se ubiquen en la nube simulada del proyecto.

b. Tecnologías de Datacenter On-Premise

El diseño del Datacenter del Corporativo XYZ sigue el modelo jerárquico de tres capas definido por Cisco en su Enterprise Network Architecture, que es el estándar de facto en redes empresariales de mediana y gran escala. Este modelo no solo organiza los dispositivos físicamente, sino que define responsabilidades funcionales claras para cada capa.

Modelo jerárquico: ratios de sobresuscripción

Un concepto fundamental que el diseño debe documentar son los ratios de oversubscription (sobresuscripción), que determinan cuántos puertos de acceso compiten por el ancho de banda de uplink hacia la distribución:

Capa	Ratio típico	Ancho de banda uplink	Función principal
Acceso → Distribución	20:1	1 GbE típico	Conectividad dispositivos finales
Distribución → Core	4:1	10 GbE recomendado	Ruteo inter-VLAN y políticas
Core → Datacenter	2:1	10/40 GbE	Transporte de alta velocidad

VLANs: segmentación avanzada y hardening de Capa 2

La segmentación mediante VLANs va más allá de separar departamentos. Desde una perspectiva de seguridad, cada VLAN representa un dominio de broadcast independiente que limita el alcance de ataques de red. Sin embargo, una configuración incorrecta puede crear vectores de ataque conocidos como VLAN Hopping. Las contramedidas técnicas esenciales son:

- VLAN nativa: nunca utilizar VLAN 1 como nativa en troncales. Se debe designar una VLAN sin usuarios (por ejemplo, VLAN 999) como nativa y asegurarse de que no haya hosts asignados a ella.
- BPDU Guard: habilitado en todos los puertos de acceso para evitar que un dispositivo no autorizado participe en el proceso de elección del árbol de expansión (STP), lo que podría permitir a un atacante redirigir tráfico.
- Dynamic ARP Inspection (DAI): valida que las solicitudes y respuestas ARP correspondan a asignaciones DHCP legítimas, previniendo ataques de ARP poisoning dentro de cada VLAN.
- Port Security: limitar el número de direcciones MAC por puerto (típicamente 1 en acceso, 2 en uplinks de AP) activa un mecanismo de shutdown automático ante violaciones, reduciendo el riesgo de conexiones no autorizadas.

c. Conectividad WAN y Tecnologías de Túneles

La conectividad entre el Datacenter del Corporativo XYZ y sus tres sedes remotas requiere tecnologías WAN que combinen seguridad, rendimiento y costo-efectividad. La elección entre MPLS, SD-WAN, IPSec e implementaciones de GRE tiene implicaciones técnicas y económicas directas sobre el diseño.

Comparativa técnica detallada

Tecnología	Cifrado	QoS nativo	Soporte multicast / ruteo dinámico	Costo relativo
MPLS	No (requiere capa adicional)	Sí (nativo)	Sí (OSPF/BGP sin encapsulamiento)	Alto
IPSec solo	Sí (AES-256)	Limitado	No soporta multicast → OSPF no funciona directamente	Bajo
GRE + IPSec	Sí (AES-256 via IPSec)	Parcial	Sí (GRE encapsula multicast, IPSec cifra el túnel GRE)	Bajo-Medio
SD-WAN	Sí (integrado)	Sí (dinámico)	Sí (orquestración centralizada)	Medio (OPEX)

GRE sobre IPSec: la solución correcta para OSPF sobre WAN

Un punto técnico crítico que el proyecto debe documentar es la razón por la cual se requiere la combinación GRE+IPSec cuando se ejecuta OSPF entre sedes remotas. OSPF depende del envío de paquetes multicast (dirección 224.0.0.5 para All-OSPF-Routers y 224.0.0.6 para DR/BDR) para el descubrimiento de vecinos y la sincronización de la base de datos de estado de enlace. Sin embargo, IPSec solo puede cifrar tráfico unicast de forma nativa; el multicast es descartado por el proceso IPSec.

La solución técnica es encapsular el tráfico OSPF multicast dentro de un túnel GRE (que lo convierte en tráfico unicast punto a punto), y luego aplicar IPSec sobre el túnel GRE para garantizar confidencialidad e integridad. Esta arquitectura GRE over IPSec es el estándar de la industria para VPN site-to-site con ruteo dinámico y está contemplada en el RFC 2784 (GRE) y en los RFC 4301-4309 (IPSec Architecture).

f. Gobernabilidad y Gestión de Redes

La gobernabilidad de red no se limita a la configuración de protocolos de acceso; abarca un conjunto de políticas, procedimientos y controles técnicos que garantizan la operación segura, trazable y auditable de la infraestructura. El marco ISO/IEC 27001 (Sistemas de Gestión de Seguridad de la Información) y el framework ITIL v4 establecen las mejores prácticas para la gestión de infraestructura de red en entornos corporativos.

SNMP: versiones y criticidad de seguridad

El proyecto menciona SNMP para monitoreo, pero es crucial documentar la diferencia de seguridad entre versiones, ya que representa una brecha de seguridad crítica en muchas organizaciones:

Versión	Autenticación	Cifrado de datos	Recomendación
SNMPv1	Community string en texto plano	Ninguno	PROHIBIDO en redes corporativas
SNMPv2c	Community string en texto plano	Ninguno	Evitar; solo para entornos aislados
SNMPv3	Usuario + SHA-256	AES-128 / AES-256	REQUERIDO en producción

SSH: requerimientos técnicos para implementación correcta

El proyecto documenta la dificultad de implementar SSH en Cisco Packet Tracer con ciertos modelos de switch. En un entorno de producción, los requerimientos mínimos para SSH v2 son: nombre de dominio configurado (ip domain-name), par de claves RSA de mínimo 2048 bits (las claves de 1024 bits ya son consideradas insuficientes según NIST SP 800-131A

desde 2015), y la deshabilitación explícita de SSHv1 mediante el comando `ip ssh version 2`.

Herramientas de visibilidad y análisis de tráfico

Más allá de SSH y SNMP, una gestión de red madura requiere herramientas de visibilidad de flujos de tráfico. NetFlow/IPFIX (RFC 7011) permite exportar metadatos de cada flujo de red —IP origen/destino, puertos, protocolo, volumen— hacia un colector centralizado. Esta capacidad es crítica para:

- Detección de anomalías: tráfico inusual desde la VLAN de CCTV hacia internet podría indicar una cámara comprometida siendo utilizada en una botnet.
- Planificación de capacidad: identificar qué aplicaciones o departamentos consumen mayor ancho de banda y proyectar necesidades futuras.
- Investigación forense: ante un incidente de seguridad, los registros de NetFlow permiten reconstruir el flujo de comunicaciones para determinar el alcance de la brecha.
- Cumplimiento regulatorio: en industrias reguladas, la retención de registros de flujo de red es un requerimiento de auditoría.

g. Servicios de Red: DHCP, DNS, Correo y Web

Los servicios de red son la capa de aplicación que sustenta la operación diaria del Corporativo XYZ. Su diseño debe contemplar no solo la funcionalidad básica, sino también alta disponibilidad, seguridad y escalabilidad.

DHCP: diseño jerárquico y consideraciones de seguridad

En la arquitectura del Corporativo XYZ, el servicio DHCP centralizado en el Datacenter debe llegar a 51 VLANs distribuidas en 7 pisos y 3 sedes remotas. El mecanismo `ip helper-address` en cada SVI de los switches de distribución reenvía el broadcast DHCP del cliente como unicast hacia el servidor centralizado. El dimensionamiento del servidor debe contemplar:

- Lease time apropiado por segmento: dispositivos fijos de escritorio pueden tener leases de 8-24 horas, mientras que segmentos de WiFi de visitantes deben configurarse con leases cortos de 1-2 horas para evitar el agotamiento del pool de direcciones.
- Reservas de IP estáticas: servidores, impresoras y cámaras CCTV deben tener asignaciones estáticas por MAC address dentro del servidor DHCP para garantizar direccionamiento predecible sin perder la gestión centralizada.
- DHCP Snooping: habilitado en todos los switches para prevenir la aparición de servidores DHCP no autorizados (rogue DHCP servers) que podrían asignar direcciones falsas de gateway a los usuarios, facilitando ataques man-in-the-middle.

DNS: arquitectura de resolución y seguridad

El diseño de DNS para el Corporativo XYZ debe implementar una arquitectura de split-DNS (DNS dividido): el servidor DNS interno resuelve los nombres del dominio corporativo

(corp.xyz.com) con direcciones IP privadas para usuarios internos, mientras que el DNS público expuesto en la nube simulada resuelve únicamente los registros públicos del portal web externo. Esta separación previene que usuarios externos puedan enumerar la topología interna de la red.

Adicionalmente, DNSSEC (DNS Security Extensions, RFC 4033-4035) debe contemplarse para firmar criptográficamente las zonas DNS, previniendo ataques de envenenamiento de caché DNS (DNS cache poisoning), donde un atacante podría redirigir a los usuarios a sitios maliciosos resolviendo nombres legítimos con IPs falsas.

Implementación de topología

SW-CORE-01 y 02

SW-CORE-01

```
SW-CORE-01#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Gig1/0/3, Gig1/0/4, Gig1/0/6, Gig1/0/7 Gig1/0/8, Gig1/0/9, Gig1/0/10, Gig1/0/12 Gig1/0/14, Gig1/0/16, Gig1/0/18, Gig1/0/20 Gig1/0/22, Gig1/0/24, Gig1/1/1, Gig1/1/2 Gig1/1/3, Gig1/1/4
10	TRANSPORT-CORE	active	
11	VLAN0011	active	
20	SERV-BACKOFFICE	active	
21	SERV-DNS	active	
22	SERV-DHCP	active	
23	SERV-MAIL	active	
24	SERV-NTP	active	
25	SERV-WEB	active	
30	SRV-DESARROLLO	active	
31	SRV-QA	active	
32	SRV-PREPROD	active	
33	SRV-PRODUCCION	active	
40	WIFI-CORE	active	
50	WIFI-VISIT	active	
99	MGMT-COB	active	
110	Dept-Showroom	active	
111	CCTV-Showroom	active	
112	WIFI-C-SHOWROOM	active	
113	WIFI-V-SHOWROOM	active	
120	DEPT-VENTAS	active	
121	CCTV-VENTAS	active	
122	WIFI-C-VENTAS	active	
123	WIFI-V-VENTAS	active	
130	DEPT-AUDITORIO	active	
131	CCTV-AUDITORIO	active	
132	WIFI-C-AUDITORIO	active	
133	WIFI-V-AUDITORIO	active	
140	DEPT-IT	active	
141	CCTV-IT	active	
142	WIFI-C-IT	active	
143	WIFI-V-IT	active	
150	DEPT-RRHH	active	
151	CCTV-RRHH	active	
152	WIFI-C-RRHH	active	
153	WIFI-V-RRHH	active	
160	DEPT-MARKETING	active	
161	CCTV-MARKETING	active	
162	WIFI-C-MKT	active	
163	WIFI-V-MKT	active	
170	DEPT-AUDITORIA	active	
171	CCTV-AUDITORIA	active	
172	WIFI-C-AUDITORIA	active	
173	WIFI-V-AUDITORIA	active	
180	DEPT-FINANZAS	active	
181	CCTV-FINANZAS	active	
182	WIFI-C-FINANZAS	active	
183	WIFI-V-FINANZAS	active	
999	NATIVE-UNUSED	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
SW-CORE-01#show vtp status
```

```
VTP Version capable      : 1 to 2
VTP version running      : 1
VTP Domain Name          : EmpresaXYZ
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                : 0060.3EC4.4100
Configuration last modified by 0.0.0.0 at 3-1-93 00:10:00
Local updater ID is 192.168.10.2 on interface Vl110 (lowest numbered VLAN interface found)

Feature VLAN :
-----
VTP Operating Mode      : Server
Maximum VLANs supported locally : 1005
Number of existing VLANs : 53
Configuration Revision  : 2441
MD5 digest              : 0xE4 0x42 0xBD 0x7C 0x16 0x3E 0xC0 0xF7
                        : 0xA5 0x42 0xFE 0x6E 0x6A 0x6B 0xC8 0x9F

SW-CORE-01#
SW-CORE-01#
SW-CORE-01#
```

```

Switch is in rapid-pvst mode
Root bridge for: TRANSPORT-CORE SERV-BACKOFFICE SERV-DNS SERV-DHCP SERV-MAIL SERV-NTP SERV-WEB SRV-
DESARROLLO SRV-QA SRV-PREPROD SRV-PRODUCCION MGMT-OOB Dept-Showroom CCTV-Showroom WIFI-C-SHOWROOM WIFI-V-
SHOWROOM DEPT-VENTAS CCTV-VENTAS WIFI-C-VENTAS WIFI-V-VENTAS DEPT-AUDITORIO CCTV-AUDITORIO WIFI-C-AUDITORIO
WIFI-V-AUDITORIO DEPT-IT CCTV-IT WIFI-C-IT WIFI-V-IT DEPT-RRHH CCTV-RRHH WIFI-C-RRHH WIFI-V-RRHH DEPT-
MARKETING CCTV-MARKETING WIFI-C-MKT WIFI-V-MKT DEPT-AUDITORIA CCTV-AUDITORIA WIFI-C-AUDITORIA WIFI-V-
AUDITORIA DEPT-FINANZAS CCTV-FINANZAS WIFI-C-FINANZAS WIFI-V-FINANZAS NATIVE-UNUSED
Extended system ID is enabled
Portfast Default is disabled
PortFast BPDU Guard Default is disabled
Portfast BPDU Filter Default is disabled
Loopguard Default is disabled
EtherChannel misconfig guard is disabled
UplinkFast is disabled
BackboneFast is disabled
Configured Pathcost method used is short

```

Name	Blocking	Listening	Learning	Forwarding	STP Active
VLAN0010	3	0	0	8	11
VLAN0020	3	0	0	8	11
VLAN0021	3	0	0	8	11
VLAN0022	3	0	0	8	11
VLAN0023	3	0	0	8	11
VLAN0024	3	0	0	8	11
VLAN0025	3	0	0	8	11
VLAN0030	3	0	0	8	11
VLAN0031	3	0	0	8	11
VLAN0032	3	0	0	8	11
VLAN0033	3	0	0	8	11
VLAN0099	3	0	0	8	11
VLAN0110	3	0	0	8	11
VLAN0111	3	0	0	8	11
VLAN0112	3	0	0	8	11
VLAN0113	3	0	0	8	11
VLAN0120	3	0	0	8	11
VLAN0121	3	0	0	8	11
VLAN0122	3	0	0	8	11
VLAN0123	3	0	0	8	11
VLAN0130	3	0	0	8	11
VLAN0131	3	0	0	8	11
VLAN0132	3	0	0	8	11
VLAN0133	3	0	0	8	11
VLAN0140	3	0	0	8	11
VLAN0141	3	0	0	8	11
VLAN0142	3	0	0	8	11
VLAN0143	3	0	0	8	11
VLAN0150	3	0	0	8	11
VLAN0151	3	0	0	8	11
VLAN0152	3	0	0	8	11
VLAN0153	3	0	0	8	11
VLAN0160	3	0	0	8	11
VLAN0161	3	0	0	8	11
VLAN0162	3	0	0	8	11
VLAN0163	3	0	0	8	11
VLAN0170	3	0	0	8	11
VLAN0171	3	0	0	8	11
VLAN0172	3	0	0	8	11
VLAN0173	3	0	0	8	11
VLAN0180	3	0	0	8	11
VLAN0181	3	0	0	8	11
VLAN0182	3	0	0	8	11
VLAN0183	3	0	0	8	11
VLAN0999	3	0	0	8	11

49 vlans	135	0	0	360	495

```

SW-CORE-01#show standby brief
P indicates configured to preempt.
|
Interface  Grp  Pri  P State   Active      Standby      Virtual IP
Vl110      10   110  P Active local      192.168.10.3 192.168.10.1
Vl120      20   110  P Active local      192.168.20.3 192.168.20.1
Vl121      21   110  P Active local      192.168.21.4 192.168.21.1
Vl122      22   110  P Active local      192.168.22.3 192.168.22.1
Vl123      23   110  P Active local      192.168.23.3 192.168.23.1
Vl124      24   110  P Active local      192.168.24.3 192.168.24.1
Vl125      25   110  P Active local      192.168.25.3 192.168.25.1
Vl130      30   110  P Active local      192.168.30.3 192.168.30.1
Vl131      31   110  P Active local      192.168.31.3 192.168.31.1
Vl132      32   110  P Active local      192.168.32.3 192.168.32.1
Vl133      33   110  P Active local      192.168.33.3 192.168.33.1
Vl199      99   110  P Active local      192.168.99.3 192.168.99.1
Vl1110     110   110  P Active local      10.10.10.3   10.10.10.1
Vl1111     111   110  P Active local      10.10.11.3   10.10.11.1
Vl1120     120   110  P Active local      10.10.20.3   10.10.20.1
Vl1121     121   110  P Active local      10.10.21.3   10.10.21.1
Vl1122     122   110  P Active local      10.10.22.3   10.10.22.1
Vl1123     123   110  P Active local      10.10.23.3   10.10.23.1
Vl1130     130   110  P Active local      10.10.30.3   10.10.30.1
Vl1131     131   110  P Active local      10.10.31.3   10.10.31.1
Vl1132     132   110  P Active local      10.10.32.3   10.10.32.1
Vl1133     133   110  P Active local      10.10.33.3   10.10.33.1
Vl1140     140   110  P Active local      10.10.40.3   10.10.40.1
Vl1141     141   110  P Active local      10.10.41.3   10.10.41.1
Vl1142     142   110  P Active local      10.10.42.3   10.10.42.1
Vl1143     143   110  P Active local      10.10.43.3   10.10.43.1
Vl1150     150   110  P Active local      10.10.50.3   10.10.50.1
Vl1151     151   110  P Active local      10.10.51.3   10.10.51.1
Vl1152     152   110  P Active local      10.10.52.3   10.10.52.1
Vl1153     153   110  P Active local      10.10.53.3   10.10.53.1
Vl1160     160   110  P Active local      10.10.60.3   10.10.60.1
Vl1161     161   110  P Active local      10.10.61.3   10.10.61.1
Vl1162     162   110  P Active local      10.10.62.3   10.10.62.1
Vl1163     163   110  P Active local      10.10.63.3   10.10.63.1
Vl1170     170   110  P Active local      10.10.70.3   10.10.70.1
Vl1171     171   110  P Active local      10.10.71.3   10.10.71.1
Vl1172     172   110  P Active local      10.10.72.3   10.10.72.1
Vl1173     173   110  P Active local      10.10.73.3   10.10.73.1
Vl1180     180   110  P Active local      10.10.80.3   10.10.80.1
Vl1181     181   110  P Active local      10.10.81.3   10.10.81.1
Vl1182     182   110  P Active local      10.10.82.3   10.10.82.1
Vl1183     183   110  P Active local      10.10.83.3   10.10.83.1
SW-CORE-01#
SW-CORE-01#

```

```

SW-CORE-01#show etherchannel summary
Flags: D - down        P - in port-channel
       I - stand-alone s - suspended
       H - Hot-standby (LACP only)
       R - Layer3       S - Layer2
       U - in use       f - failed to allocate aggregator
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1(SU)        LACP        Gig1/0/1(P) Gig1/0/2(P)
SW-CORE-01#
SW-CORE-01#
SW-CORE-01#
SW-CORE-01#

```

```
SW-CORE-01#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

```
Gateway of last resort is 10.0.0.1 to network 0.0.0.0
```

```
10.0.0.0/8 is variably subnetted, 62 subnets, 3 masks
C    10.0.0.0/30 is directly connected, GigabitEthernet1/0/5
L    10.0.0.2/32 is directly connected, GigabitEthernet1/0/5
C    10.10.10.0/24 is directly connected, Vlan110
L    10.10.10.2/32 is directly connected, Vlan110
C    10.10.11.0/24 is directly connected, Vlan111
L    10.10.11.2/32 is directly connected, Vlan111
C    10.10.20.0/24 is directly connected, Vlan120
L    10.10.20.2/32 is directly connected, Vlan120
C    10.10.21.0/24 is directly connected, Vlan121
L    10.10.21.2/32 is directly connected, Vlan121
C    10.10.22.0/24 is directly connected, Vlan122
L    10.10.22.2/32 is directly connected, Vlan122
C    10.10.23.0/24 is directly connected, Vlan123
L    10.10.23.2/32 is directly connected, Vlan123
C    10.10.30.0/24 is directly connected, Vlan130
L    10.10.30.2/32 is directly connected, Vlan130
C    10.10.31.0/24 is directly connected, Vlan131
L    10.10.31.2/32 is directly connected, Vlan131
C    10.10.32.0/24 is directly connected, Vlan132
L    10.10.32.2/32 is directly connected, Vlan132
C    10.10.33.0/24 is directly connected, Vlan133
L    10.10.33.2/32 is directly connected, Vlan133
C    10.10.40.0/24 is directly connected, Vlan140
L    10.10.40.2/32 is directly connected, Vlan140
C    10.10.41.0/24 is directly connected, Vlan141
L    10.10.41.2/32 is directly connected, Vlan141
C    10.10.42.0/24 is directly connected, Vlan142
L    10.10.42.2/32 is directly connected, Vlan142
C    10.10.43.0/24 is directly connected, Vlan143
L    10.10.43.2/32 is directly connected, Vlan143
C    10.10.50.0/24 is directly connected, Vlan150
L    10.10.50.2/32 is directly connected, Vlan150
C    10.10.51.0/24 is directly connected, Vlan151
L    10.10.51.2/32 is directly connected, Vlan151
C    10.10.52.0/24 is directly connected, Vlan152
L    10.10.52.2/32 is directly connected, Vlan152
C    10.10.53.0/24 is directly connected, Vlan153
L    10.10.53.2/32 is directly connected, Vlan153
C    10.10.60.0/24 is directly connected, Vlan160
L    10.10.60.2/32 is directly connected, Vlan160
C    10.10.61.0/24 is directly connected, Vlan161
L    10.10.61.2/32 is directly connected, Vlan161
C    10.10.62.0/24 is directly connected, Vlan162
L    10.10.62.2/32 is directly connected, Vlan162
C    10.10.63.0/24 is directly connected, Vlan163
L    10.10.63.2/32 is directly connected, Vlan163
C    10.10.70.0/24 is directly connected, Vlan170
L    10.10.70.2/32 is directly connected, Vlan170
C    10.10.71.0/24 is directly connected, Vlan171
L    10.10.71.2/32 is directly connected, Vlan171
C    10.10.72.0/24 is directly connected, Vlan172
L    10.10.72.2/32 is directly connected, Vlan172
```

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C 10.10.73.0/24 is directly connected, Vlan173
L 10.10.73.2/32 is directly connected, Vlan173
C 10.10.80.0/24 is directly connected, Vlan180
L 10.10.80.2/32 is directly connected, Vlan180
C 10.10.81.0/24 is directly connected, Vlan181
L 10.10.81.2/32 is directly connected, Vlan181
C 10.10.82.0/24 is directly connected, Vlan182
L 10.10.82.2/32 is directly connected, Vlan182
C 10.10.83.0/24 is directly connected, Vlan183
L 10.10.83.2/32 is directly connected, Vlan183
L 172.16.0.0/24 is subnetted, 18 subnets
O E2 172.16.10.0 [110/20] via 10.0.0.1, 00:46:15, GigabitEthernet1/0/5
O E2 172.16.11.0 [110/20] via 10.0.0.1, 00:46:15, GigabitEthernet1/0/5
O E2 172.16.12.0 [110/20] via 10.0.0.1, 00:46:15, GigabitEthernet1/0/5
O E2 172.16.13.0 [110/20] via 10.0.0.1, 00:46:15, GigabitEthernet1/0/5
O E2 172.16.14.0 [110/20] via 10.0.0.1, 00:46:15, GigabitEthernet1/0/5
O E2 172.16.15.0 [110/20] via 10.0.0.1, 00:46:15, GigabitEthernet1/0/5
O E2 172.16.20.0 [110/20] via 10.0.0.1, 00:45:03, GigabitEthernet1/0/5
O E2 172.16.21.0 [110/20] via 10.0.0.1, 00:45:03, GigabitEthernet1/0/5
O E2 172.16.22.0 [110/20] via 10.0.0.1, 00:45:03, GigabitEthernet1/0/5
O E2 172.16.23.0 [110/20] via 10.0.0.1, 00:45:03, GigabitEthernet1/0/5
O E2 172.16.24.0 [110/20] via 10.0.0.1, 00:45:03, GigabitEthernet1/0/5
O E2 172.16.25.0 [110/20] via 10.0.0.1, 00:45:03, GigabitEthernet1/0/5
O E2 172.16.30.0 [110/20] via 10.0.0.1, 00:46:15, GigabitEthernet1/0/5
O E2 172.16.31.0 [110/20] via 10.0.0.1, 00:46:15, GigabitEthernet1/0/5
O E2 172.16.32.0 [110/20] via 10.0.0.1, 00:46:15, GigabitEthernet1/0/5
O E2 172.16.33.0 [110/20] via 10.0.0.1, 00:46:15, GigabitEthernet1/0/5
O E2 172.16.34.0 [110/20] via 10.0.0.1, 00:46:15, GigabitEthernet1/0/5
O E2 172.16.35.0 [110/20] via 10.0.0.1, 00:46:15, GigabitEthernet1/0/5
C 192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.10.0/24 is directly connected, Vlan10
L 192.168.10.2/32 is directly connected, Vlan10
C 192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.20.0/24 is directly connected, Vlan20
L 192.168.20.2/32 is directly connected, Vlan20
C 192.168.21.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.21.0/24 is directly connected, Vlan21
L 192.168.21.3/32 is directly connected, Vlan21
C 192.168.22.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.22.0/24 is directly connected, Vlan22
L 192.168.22.2/32 is directly connected, Vlan22
C 192.168.23.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.23.0/24 is directly connected, Vlan23
L 192.168.23.2/32 is directly connected, Vlan23
C 192.168.24.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.24.0/24 is directly connected, Vlan24
L 192.168.24.2/32 is directly connected, Vlan24
C 192.168.25.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.25.0/24 is directly connected, Vlan25
L 192.168.25.2/32 is directly connected, Vlan25
C 192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.30.0/24 is directly connected, Vlan30
L 192.168.30.2/32 is directly connected, Vlan30
C 192.168.31.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.31.0/24 is directly connected, Vlan31
L 192.168.31.2/32 is directly connected, Vlan31
C 192.168.32.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.32.0/24 is directly connected, Vlan32
L 192.168.32.2/32 is directly connected, Vlan32
C 192.168.33.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.33.0/24 is directly connected, Vlan33
L 192.168.33.2/32 is directly connected, Vlan33
C 192.168.99.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.99.0/24 is directly connected, Vlan99
L 192.168.99.2/32 is directly connected, Vlan99
O*E2 0.0.0.0/0 [110/1] via 10.0.0.1, 00:46:15, GigabitEthernet1/0/5

```

Neighbor ID	Pri	State	Dead Time	Address	Interface
200.0.0.1	1	FULL/DR	00:00:31	10.0.0.1	GigabitEthernet1/0/5
192.168.99.3	200	FULL/SDR	00:00:31	192.168.10.3	Vlan10
5.5.5.5	1	FULL/DROTHER	00:00:30	192.168.10.15	Vlan10
7.7.7.7	1	FULL/DROTHER	00:00:30	192.168.10.17	Vlan10
11.11.11.11	1	FULL/DROTHER	00:00:30	192.168.10.11	Vlan10
3.3.3.3	1	FULL/DROTHER	00:00:30	192.168.10.13	Vlan10
4.4.4.4	1	FULL/DROTHER	00:00:30	192.168.10.6	Vlan10
2.2.2.2	1	FULL/DROTHER	00:00:30	192.168.10.12	Vlan10
6.6.6.6	1	FULL/DROTHER	00:00:38	192.168.10.16	Vlan10

SW-CORE-01#
 SW-CORE-01#

SW-CORE-01#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
Port-channel1	unassigned	YES	unset	up	up
GigabitEthernet1/0/1	unassigned	YES	unset	up	up
GigabitEthernet1/0/2	unassigned	YES	unset	up	up
GigabitEthernet1/0/3	unassigned	YES	unset	down	down
GigabitEthernet1/0/4	unassigned	YES	unset	down	down
GigabitEthernet1/0/5	10.0.0.2	YES	NVRAM	up	up
GigabitEthernet1/0/6	unassigned	YES	unset	down	down
GigabitEthernet1/0/7	unassigned	YES	unset	down	down
GigabitEthernet1/0/8	unassigned	YES	unset	down	down
GigabitEthernet1/0/9	unassigned	YES	unset	down	down
GigabitEthernet1/0/10	unassigned	YES	unset	down	down
GigabitEthernet1/0/11	unassigned	YES	unset	up	up
GigabitEthernet1/0/12	unassigned	YES	unset	down	down
GigabitEthernet1/0/13	unassigned	YES	unset	up	up
GigabitEthernet1/0/14	unassigned	YES	unset	down	down
GigabitEthernet1/0/15	unassigned	YES	unset	up	up
GigabitEthernet1/0/16	unassigned	YES	unset	down	down
GigabitEthernet1/0/17	unassigned	YES	unset	up	up
GigabitEthernet1/0/18	unassigned	YES	unset	down	down
GigabitEthernet1/0/19	unassigned	YES	unset	up	up
GigabitEthernet1/0/20	unassigned	YES	unset	down	down
GigabitEthernet1/0/21	unassigned	YES	unset	up	up
GigabitEthernet1/0/22	unassigned	YES	unset	down	down
GigabitEthernet1/0/23	unassigned	YES	unset	up	up
GigabitEthernet1/0/24	unassigned	YES	unset	down	down
GigabitEthernet1/1/1	unassigned	YES	unset	down	down
GigabitEthernet1/1/2	unassigned	YES	unset	down	down
GigabitEthernet1/1/3	unassigned	YES	unset	down	down
GigabitEthernet1/1/4	unassigned	YES	unset	down	down
Vlan1	unassigned	YES	unset	administratively down	down
Vlan10	192.168.10.2	YES	manual	up	up
Vlan11	unassigned	YES	unset	up	down
Vlan20	192.168.20.2	YES	manual	up	up
Vlan21	192.168.21.3	YES	manual	up	up
Vlan22	192.168.22.2	YES	manual	up	up
Vlan23	192.168.23.2	YES	manual	up	up
Vlan24	192.168.24.2	YES	manual	up	up
Vlan25	192.168.25.2	YES	manual	up	up
Vlan30	192.168.30.2	YES	manual	up	up
Vlan31	192.168.31.2	YES	manual	up	up
Vlan32	192.168.32.2	YES	manual	up	up
Vlan33	192.168.33.2	YES	manual	up	up
Vlan99	192.168.99.2	YES	manual	up	up
Vlan110	10.10.10.2	YES	manual	up	up
Vlan111	10.10.11.2	YES	manual	up	up
Vlan112	unassigned	YES	unset	up	up
Vlan113	unassigned	YES	unset	up	up
Vlan120	10.10.20.2	YES	manual	up	up
Vlan121	10.10.21.2	YES	manual	up	up
Vlan122	10.10.22.2	YES	manual	up	up
Vlan123	10.10.23.2	YES	manual	up	up
Vlan130	10.10.30.2	YES	manual	up	up
Vlan131	10.10.31.2	YES	manual	up	up
Vlan132	10.10.32.2	YES	manual	up	up
Vlan133	10.10.33.2	YES	manual	up	up
Vlan140	10.10.40.2	YES	manual	up	up
Vlan141	10.10.41.2	YES	manual	up	up
Vlan142	10.10.42.2	YES	manual	up	up
Vlan143	10.10.43.2	YES	manual	up	up
Vlan150	10.10.50.2	YES	manual	up	up
Vlan151	10.10.51.2	YES	manual	up	up
Vlan152	10.10.52.2	YES	manual	up	up
Vlan153	10.10.53.2	YES	manual	up	up
Vlan160	10.10.60.2	YES	manual	up	up

Vlan163	10.10.63.2	YES	manual	up	up
Vlan170	10.10.70.2	YES	manual	up	up
Vlan171	10.10.71.2	YES	manual	up	up
Vlan172	10.10.72.2	YES	manual	up	up
Vlan173	10.10.73.2	YES	manual	up	up
Vlan180	10.10.80.2	YES	manual	up	up
Vlan181	10.10.81.2	YES	manual	up	up
Vlan182	10.10.82.2	YES	manual	up	up
Vlan183	10.10.83.2	YES	manual	up	up

SW-CORE-01#

SW-CORE-02

```
SW-CORE-02#show vlan br
SW-CORE-02#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Gig1/0/3, Gig1/0/4, Gig1/0/5, Gig1/0/6 Gig1/0/7, Gig1/0/8, Gig1/0/9, Gig1/0/11 Gig1/0/13, Gig1/0/15, Gig1/0/17, Gig1/0/19 Gig1/0/21, Gig1/0/23, Gig1/0/24, Gig1/1/1 Gig1/1/2, Gig1/1/3, Gig1/1/4
10	TRANSPORT-CORE	active	
11	VLAN0011	active	
20	SERV-BACKOFFICE	active	
21	SERV-DNS	active	
22	SERV-DHCP	active	
23	SERV-MAIL	active	
24	SERV-NTP	active	
25	SERV-WEB	active	
30	SRV-DESARROLLO	active	
31	SRV-QA	active	
32	SRV-PREPROD	active	
33	SRV-PRODUCCION	active	
40	WIFI-CORP	active	
50	WIFI-VISIT	active	
99	MGMT-OOB	active	
110	Dept-Showroom	active	
111	CCTV-Showroom	active	
112	WIFI-C-SHOWROOM	active	
113	WIFI-V-SHOWROOM	active	
120	DEPT-VENTAS	active	
121	CCTV-VENTAS	active	
122	WIFI-C-VENTAS	active	
123	WIFI-V-VENTAS	active	
130	DEPT-AUDITORIO	active	
131	CCTV-AUDITORIO	active	
132	WIFI-C-AUDITORIO	active	
133	WIFI-V-AUDITORIO	active	
140	DEPT-IT	active	
141	CCTV-IT	active	
142	WIFI-C-IT	active	
143	WIFI-V-IT	active	
150	DEPT-RRHH	active	
151	CCTV-RRHH	active	
152	WIFI-C-RRHH	active	
153	WIFI-V-RRHH	active	
160	DEPT-MARKETING	active	
161	CCTV-MARKETING	active	
162	WIFI-C-MKT	active	
163	WIFI-V-MKT	active	
170	DEPT-AUDITORIA	active	
171	CCTV-AUDITORIA	active	
172	WIFI-C-AUDITORIA	active	
173	WIFI-V-AUDITORIA	active	
180	DEPT-FINANZAS	active	
181	CCTV-FINANZAS	active	
182	WIFI-C-FINANZAS	active	
183	WIFI-V-FINANZAS	active	
999	NATIVE-UNUSED	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
SW-CORE-02#
SW-CORE-02#
SW-CORE-02#
SW-CORE-02#
```

```
SW-CORE-02#show vtp status
VTP Version capable      : 1 to 2
VTP version running      : 1
VTP Domain Name          : EmpresaXYZ
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                : 0001.C701.BAC0
Configuration last modified by 0.0.0.0 at 3-6-93 02:13:33
Local updater ID is 192.168.10.3 on interface Vl10 (lowest numbered VLAN interface found)

Feature VLAN :
-----
VTP Operating Mode       : Server
Maximum VLANs supported locally : 1005
Number of existing VLANs : 53
Configuration Revision   : 2441
MD5 digest               : 0xAD 0x01 0xA0 0xBC 0x06 0x22 0x53 0xDA
                          : 0x7F 0xFD 0x75 0xA8 0x51 0x9C 0xE0 0xA3

SW-CORE-02#
```

```

SW-CORE-02#show spanning-tree summary
Switch is in rapid-pvst mode
Root bridge for: default SRV-QA
Extended system ID is enabled
PortFast Default is disabled
PortFast BPDU Guard Default is disabled
PortFast BPDU Filter Default is disabled
Loopguard Default is disabled
EtherChannel misconfig guard is disabled
UplinkFast is disabled
BackboneFast is disabled
Configured Pathcost method used is short

```

Name	Blocking	Listening	Learning	Forwarding	STP Active
VLAN0001	10	0	0	1	11
VLAN0010	7	0	0	4	11
VLAN0020	5	0	0	6	11
VLAN0021	6	0	0	5	11
VLAN0022	7	0	0	4	11
VLAN0023	6	0	0	5	11
VLAN0024	4	0	0	7	11
VLAN0025	9	0	0	2	11
VLAN0030	6	0	0	5	11
VLAN0031	3	0	0	8	11
VLAN0032	5	0	0	6	11
VLAN0033	4	0	0	7	11
VLAN0099	7	0	0	4	11
VLAN0110	4	0	0	7	11
VLAN0111	4	0	0	7	11
VLAN0112	6	0	0	5	11
VLAN0113	5	0	0	6	11
VLAN0120	5	0	0	6	11
VLAN0121	7	0	0	4	11
VLAN0122	6	0	0	5	11
VLAN0123	6	0	0	5	11
VLAN0130	6	0	0	5	11
VLAN0131	4	0	0	7	11
VLAN0132	5	0	0	6	11
VLAN0133	5	0	0	6	11
VLAN0140	9	0	0	2	11
VLAN0141	8	0	0	3	11
VLAN0142	9	0	0	2	11
VLAN0143	5	0	0	6	11
VLAN0150	5	0	0	6	11
VLAN0151	7	0	0	4	11
VLAN0152	6	0	0	5	11
VLAN0153	4	0	0	7	11
VLAN0160	8	0	0	3	11
VLAN0161	9	0	0	2	11
VLAN0162	4	0	0	7	11
VLAN0163	7	0	0	4	11
VLAN0170	5	0	0	6	11
VLAN0171	4	0	0	7	11
VLAN0172	4	0	0	7	11
VLAN0173	9	0	0	2	11
VLAN0180	6	0	0	5	11
VLAN0181	8	0	0	3	11
VLAN0182	5	0	0	6	11
VLAN0183	4	0	0	7	11
VLAN0999	5	0	0	6	11

49 vlans	273	0	0	233	506

```

SW-CORE-02#show standby brief
          P indicates configured to preempt.
          |
Interface  Grp  Pri  P  State  Active  Standby  Virtual IP
Vl110      10   100  Standby  192.168.10.2  local  192.168.10.1
Vl120      20   100  Standby  192.168.20.2  local  192.168.20.1
Vl121      21   100  Standby  192.168.21.3  local  192.168.21.1
Vl122      22   100  Standby  192.168.22.2  local  192.168.22.1
Vl123      23   100  Standby  192.168.23.2  local  192.168.23.1
Vl124      24   100  Standby  192.168.24.2  local  192.168.24.1
Vl125      25   100  Standby  192.168.25.2  local  192.168.25.1
Vl130      30   100  Standby  192.168.30.2  local  192.168.30.1
Vl131      31   100  Standby  192.168.31.2  local  192.168.31.1
Vl132      32   100  Standby  192.168.32.2  local  192.168.32.1
Vl133      33   100  Standby  192.168.33.2  local  192.168.33.1
Vl199      99   100  Standby  192.168.99.2  local  192.168.99.1
Vl1110     110   100  Standby  10.10.10.2    local  10.10.10.1
Vl1111     111   100  Standby  10.10.11.2    local  10.10.11.1
Vl1120     120   100  Standby  10.10.20.2    local  10.10.20.1
Vl1121     121   100  Standby  10.10.21.2    local  10.10.21.1
Vl1122     122   100  Standby  10.10.22.2    local  10.10.22.1
Vl1123     123   100  Standby  10.10.23.2    local  10.10.23.1
Vl1130     130   100  Standby  10.10.30.2    local  10.10.30.1
Vl1131     131   100  Standby  10.10.31.2    local  10.10.31.1
Vl1132     132   100  Standby  10.10.32.2    local  10.10.32.1
Vl1133     133   100  Standby  10.10.33.2    local  10.10.33.1
Vl1140     140   100  Standby  10.10.40.2    local  10.10.40.1
Vl1141     141   100  Standby  10.10.41.2    local  10.10.41.1
Vl1142     142   100  Standby  10.10.42.2    local  10.10.42.1
Vl1143     143   100  Standby  10.10.43.2    local  10.10.43.1
Vl1150     150   100  Standby  10.10.50.2    local  10.10.50.1
Vl1151     151   100  Standby  10.10.51.2    local  10.10.51.1
Vl1152     152   100  Standby  10.10.52.2    local  10.10.52.1
Vl1153     153   100  Standby  10.10.53.2    local  10.10.53.1
Vl1160     160   100  Standby  10.10.60.2    local  10.10.60.1
Vl1161     161   100  Standby  10.10.61.2    local  10.10.61.1
Vl1162     162   100  Standby  10.10.62.2    local  10.10.62.1
Vl1163     163   100  Standby  10.10.63.2    local  10.10.63.1
Vl1170     170   100  Standby  10.10.70.2    local  10.10.70.1
Vl1171     171   100  Standby  10.10.71.2    local  10.10.71.1
Vl1172     172   100  Standby  10.10.72.2    local  10.10.72.1
Vl1173     173   100  Standby  10.10.73.2    local  10.10.73.1
Vl1180     180   100  Standby  10.10.80.2    local  10.10.80.1
Vl1181     181   100  Standby  10.10.81.2    local  10.10.81.1
Vl1182     182   100  Standby  10.10.82.2    local  10.10.82.1
Vl1183     183   100  Standby  10.10.83.2    local  10.10.83.1

```

```

SW-CORE-02#show etherchannel summary
Flags: D - down        P - in port-channel
       I - stand-alone s - suspended
       H - Hot-standby (LACP only)
       R - Layer3       S - Layer2
       U - in use       f - failed to allocate aggregator
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1(SU)        LACP        Gig1/0/1(P) Gig1/0/2(P)
SW-CORE-02#
SW-CORE-02#
SW-CORE-02#

```

```
SW-CORE-02#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

```
Gateway of last resort is 192.168.10.2 to network 0.0.0.0
```

```
10.0.0.0/8 is variably subnetted, 61 subnets, 3 masks
O    10.0.0.0/30 [110/2] via 192.168.10.2, 00:55:20, Vlan10
C    10.10.10.0/24 is directly connected, Vlan110
L    10.10.10.3/32 is directly connected, Vlan110
C    10.10.11.0/24 is directly connected, Vlan111
L    10.10.11.3/32 is directly connected, Vlan111
C    10.10.20.0/24 is directly connected, Vlan120
L    10.10.20.3/32 is directly connected, Vlan120
C    10.10.21.0/24 is directly connected, Vlan121
L    10.10.21.3/32 is directly connected, Vlan121
C    10.10.22.0/24 is directly connected, Vlan122
L    10.10.22.3/32 is directly connected, Vlan122
C    10.10.23.0/24 is directly connected, Vlan123
L    10.10.23.3/32 is directly connected, Vlan123
C    10.10.30.0/24 is directly connected, Vlan130
L    10.10.30.3/32 is directly connected, Vlan130
C    10.10.31.0/24 is directly connected, Vlan131
L    10.10.31.3/32 is directly connected, Vlan131
C    10.10.32.0/24 is directly connected, Vlan132
L    10.10.32.3/32 is directly connected, Vlan132
C    10.10.33.0/24 is directly connected, Vlan133
L    10.10.33.3/32 is directly connected, Vlan133
C    10.10.40.0/24 is directly connected, Vlan140
L    10.10.40.3/32 is directly connected, Vlan140
C    10.10.41.0/24 is directly connected, Vlan141
L    10.10.41.3/32 is directly connected, Vlan141
C    10.10.42.0/24 is directly connected, Vlan142
L    10.10.42.3/32 is directly connected, Vlan142
C    10.10.43.0/24 is directly connected, Vlan143
L    10.10.43.3/32 is directly connected, Vlan143
C    10.10.50.0/24 is directly connected, Vlan150
L    10.10.50.3/32 is directly connected, Vlan150
C    10.10.51.0/24 is directly connected, Vlan151
L    10.10.51.3/32 is directly connected, Vlan151
C    10.10.52.0/24 is directly connected, Vlan152
L    10.10.52.3/32 is directly connected, Vlan152
C    10.10.53.0/24 is directly connected, Vlan153
L    10.10.53.3/32 is directly connected, Vlan153
C    10.10.60.0/24 is directly connected, Vlan160
L    10.10.60.3/32 is directly connected, Vlan160
C    10.10.61.0/24 is directly connected, Vlan161
L    10.10.61.3/32 is directly connected, Vlan161
C    10.10.62.0/24 is directly connected, Vlan162
L    10.10.62.3/32 is directly connected, Vlan162
C    10.10.63.0/24 is directly connected, Vlan163
L    10.10.63.3/32 is directly connected, Vlan163
C    10.10.70.0/24 is directly connected, Vlan170
L    10.10.70.3/32 is directly connected, Vlan170
C    10.10.71.0/24 is directly connected, Vlan171
L    10.10.71.3/32 is directly connected, Vlan171
C    10.10.72.0/24 is directly connected, Vlan172
L    10.10.72.3/32 is directly connected, Vlan172
C    10.10.73.0/24 is directly connected, Vlan173
L    10.10.73.3/32 is directly connected, Vlan173
```

```

L    10.10.73.3/32 is directly connected, Vlan173
C    10.10.80.0/24 is directly connected, Vlan180
L    10.10.80.3/32 is directly connected, Vlan180
C    10.10.81.0/24 is directly connected, Vlan181
L    10.10.81.3/32 is directly connected, Vlan181
C    10.10.82.0/24 is directly connected, Vlan182
L    10.10.82.3/32 is directly connected, Vlan182
C    10.10.83.0/24 is directly connected, Vlan183
L    10.10.83.3/32 is directly connected, Vlan183
L    172.16.0.0/24 is subnetted, 18 subnets
O E2 172.16.10.0 [110/20] via 192.168.10.2, 00:55:20, Vlan10
O E2 172.16.11.0 [110/20] via 192.168.10.2, 00:55:20, Vlan10
O E2 172.16.12.0 [110/20] via 192.168.10.2, 00:55:20, Vlan10
O E2 172.16.13.0 [110/20] via 192.168.10.2, 00:55:20, Vlan10
O E2 172.16.14.0 [110/20] via 192.168.10.2, 00:55:20, Vlan10
O E2 172.16.15.0 [110/20] via 192.168.10.2, 00:55:20, Vlan10
O E2 172.16.20.0 [110/20] via 192.168.10.2, 00:54:12, Vlan10
O E2 172.16.21.0 [110/20] via 192.168.10.2, 00:54:12, Vlan10
O E2 172.16.22.0 [110/20] via 192.168.10.2, 00:54:12, Vlan10
O E2 172.16.23.0 [110/20] via 192.168.10.2, 00:54:12, Vlan10
O E2 172.16.24.0 [110/20] via 192.168.10.2, 00:54:12, Vlan10
O E2 172.16.25.0 [110/20] via 192.168.10.2, 00:54:12, Vlan10
O E2 172.16.30.0 [110/20] via 192.168.10.2, 00:55:20, Vlan10
O E2 172.16.31.0 [110/20] via 192.168.10.2, 00:55:20, Vlan10
O E2 172.16.32.0 [110/20] via 192.168.10.2, 00:55:20, Vlan10
O E2 172.16.33.0 [110/20] via 192.168.10.2, 00:55:20, Vlan10
O E2 172.16.34.0 [110/20] via 192.168.10.2, 00:55:20, Vlan10
O E2 172.16.35.0 [110/20] via 192.168.10.2, 00:55:20, Vlan10
L    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.10.0/24 is directly connected, Vlan10
L    192.168.10.3/32 is directly connected, Vlan10
L    192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.20.0/24 is directly connected, Vlan20
L    192.168.20.3/32 is directly connected, Vlan20
L    192.168.21.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.21.0/24 is directly connected, Vlan21
L    192.168.21.4/32 is directly connected, Vlan21
L    192.168.22.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.22.0/24 is directly connected, Vlan22
L    192.168.22.3/32 is directly connected, Vlan22
L    192.168.23.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.23.0/24 is directly connected, Vlan23
L    192.168.23.3/32 is directly connected, Vlan23
L    192.168.24.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.24.0/24 is directly connected, Vlan24
L    192.168.24.3/32 is directly connected, Vlan24
L    192.168.25.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.25.0/24 is directly connected, Vlan25
L    192.168.25.3/32 is directly connected, Vlan25
L    192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.30.0/24 is directly connected, Vlan30
L    192.168.30.3/32 is directly connected, Vlan30
L    192.168.31.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.31.0/24 is directly connected, Vlan31
L    192.168.31.3/32 is directly connected, Vlan31
L    192.168.32.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.32.0/24 is directly connected, Vlan32
L    192.168.32.3/32 is directly connected, Vlan32
L    192.168.33.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.33.0/24 is directly connected, Vlan33
L    192.168.33.3/32 is directly connected, Vlan33
L    192.168.99.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.99.0/24 is directly connected, Vlan99
L    192.168.99.3/32 is directly connected, Vlan99
O*E2 0.0.0.0/0 [110/1] via 192.168.10.2, 00:55:20, Vlan10

```

```
SW-CORE-02#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
5.5.5.5	1	FULL/DROTHER	00:00:31	192.168.10.15	Vlan10
11.11.11.11	1	FULL/DROTHER	00:00:32	192.168.10.11	Vlan10
192.168.99.2	255	FULL/DR	00:00:31	192.168.10.2	Vlan10
3.3.3.3	1	FULL/DROTHER	00:00:32	192.168.10.13	Vlan10
4.4.4.4	1	FULL/DROTHER	00:00:31	192.168.10.6	Vlan10
2.2.2.2	1	FULL/DROTHER	00:00:32	192.168.10.12	Vlan10
6.6.6.6	1	FULL/DROTHER	00:00:30	192.168.10.16	Vlan10
7.7.7.7	1	FULL/DROTHER	00:00:32	192.168.10.17	Vlan10

```
SW-CORE-02#
```

```
SW-CORE-02#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
5.5.5.5	1	FULL/DROTHER	00:00:31	192.168.10.15	Vlan10
11.11.11.11	1	FULL/DROTHER	00:00:32	192.168.10.11	Vlan10
192.168.99.2	255	FULL/DR	00:00:31	192.168.10.2	Vlan10
3.3.3.3	1	FULL/DROTHER	00:00:32	192.168.10.13	Vlan10
4.4.4.4	1	FULL/DROTHER	00:00:31	192.168.10.6	Vlan10
2.2.2.2	1	FULL/DROTHER	00:00:32	192.168.10.12	Vlan10
6.6.6.6	1	FULL/DROTHER	00:00:30	192.168.10.16	Vlan10
7.7.7.7	1	FULL/DROTHER	00:00:32	192.168.10.17	Vlan10

```
SW-CORE-02#show ip in
```

```
SW-CORE-02#show ip inte
```

```
SW-CORE-02#show ip interface bri
```

```
SW-CORE-02#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
Port-channel1	unassigned	YES	unset	up	up
GigabitEthernet1/0/1	unassigned	YES	unset	up	up
GigabitEthernet1/0/2	unassigned	YES	unset	up	up
GigabitEthernet1/0/3	unassigned	YES	unset	down	down
GigabitEthernet1/0/4	unassigned	YES	unset	down	down
GigabitEthernet1/0/5	unassigned	YES	unset	up	up
GigabitEthernet1/0/6	unassigned	YES	unset	down	down
GigabitEthernet1/0/7	unassigned	YES	unset	down	down
GigabitEthernet1/0/8	unassigned	YES	unset	down	down
GigabitEthernet1/0/9	unassigned	YES	unset	down	down
GigabitEthernet1/0/10	unassigned	YES	unset	up	up
GigabitEthernet1/0/11	unassigned	YES	unset	down	down
GigabitEthernet1/0/12	unassigned	YES	unset	up	up
GigabitEthernet1/0/13	unassigned	YES	unset	down	down
GigabitEthernet1/0/14	unassigned	YES	unset	up	up
GigabitEthernet1/0/15	unassigned	YES	unset	down	down
GigabitEthernet1/0/16	unassigned	YES	unset	up	up
GigabitEthernet1/0/17	unassigned	YES	unset	down	down
GigabitEthernet1/0/18	unassigned	YES	unset	up	up
GigabitEthernet1/0/19	unassigned	YES	unset	down	down
GigabitEthernet1/0/20	unassigned	YES	unset	up	up
GigabitEthernet1/0/21	unassigned	YES	unset	down	down
GigabitEthernet1/0/22	unassigned	YES	unset	up	up
GigabitEthernet1/0/23	unassigned	YES	unset	down	down
GigabitEthernet1/0/24	unassigned	YES	unset	down	down
GigabitEthernet1/1/1	unassigned	YES	unset	down	down
GigabitEthernet1/1/2	unassigned	YES	unset	down	down
GigabitEthernet1/1/3	unassigned	YES	unset	down	down
GigabitEthernet1/1/4	unassigned	YES	unset	down	down
Vlan1	unassigned	YES	unset	administratively down	down
Vlan10	192.168.10.3	YES	manual	up	up
Vlan20	192.168.20.3	YES	manual	up	up
Vlan21	192.168.21.4	YES	manual	up	up
Vlan22	192.168.22.3	YES	manual	up	up
Vlan23	192.168.23.3	YES	manual	up	up
Vlan24	192.168.24.3	YES	manual	up	up
Vlan25	192.168.25.3	YES	manual	up	up
Vlan30	192.168.30.3	YES	manual	up	up
Vlan31	192.168.31.3	YES	manual	up	up
Vlan32	192.168.32.3	YES	manual	up	up
Vlan33	192.168.33.3	YES	manual	up	up
Vlan40	192.168.40.1	YES	manual	up	down
Vlan50	192.168.50.1	YES	manual	up	down
Vlan99	192.168.99.3	YES	manual	up	up
Vlan110	10.10.10.3	YES	manual	up	up
Vlan111	10.10.11.3	YES	manual	up	up
Vlan112	unassigned	YES	unset	up	up
Vlan113	unassigned	YES	unset	up	up

Vlan120	10.10.20.3	YES	manual	up	up
Vlan121	10.10.21.3	YES	manual	up	up
Vlan122	10.10.22.3	YES	manual	up	up
Vlan123	10.10.23.3	YES	manual	up	up
Vlan130	10.10.30.3	YES	manual	up	up
Vlan131	10.10.31.3	YES	manual	up	up
Vlan132	10.10.32.3	YES	manual	up	up
Vlan133	10.10.33.3	YES	manual	up	up
Vlan140	10.10.40.3	YES	manual	up	up
Vlan141	10.10.41.3	YES	manual	up	up
Vlan142	10.10.42.3	YES	manual	up	up
Vlan143	10.10.43.3	YES	manual	up	up
Vlan150	10.10.50.3	YES	manual	up	up
Vlan151	10.10.51.3	YES	manual	up	up
Vlan152	10.10.52.3	YES	manual	up	up
Vlan153	10.10.53.3	YES	manual	up	up
Vlan160	10.10.60.3	YES	manual	up	up
Vlan161	10.10.61.3	YES	manual	up	up
Vlan162	10.10.62.3	YES	manual	up	up
Vlan163	10.10.63.3	YES	manual	up	up
Vlan170	10.10.70.3	YES	manual	up	up
Vlan171	10.10.71.3	YES	manual	up	up
Vlan172	10.10.72.3	YES	manual	up	up
Vlan173	10.10.73.3	YES	manual	up	up
Vlan180	10.10.80.3	YES	manual	up	up
Vlan181	10.10.81.3	YES	manual	up	up
Vlan182	10.10.82.3	YES	manual	up	up
Vlan183	10.10.83.3	YES	manual	up	up

```
SW-CORE-02#
```

SW-CORE-02#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
Port-channel1	unassigned	YES	unset	up	up
GigabitEthernet1/0/1	unassigned	YES	unset	up	up
GigabitEthernet1/0/2	unassigned	YES	unset	up	up
GigabitEthernet1/0/3	unassigned	YES	unset	down	down
GigabitEthernet1/0/4	unassigned	YES	unset	down	down
GigabitEthernet1/0/5	unassigned	YES	unset	up	up
GigabitEthernet1/0/6	unassigned	YES	unset	down	down
GigabitEthernet1/0/7	unassigned	YES	unset	down	down
GigabitEthernet1/0/8	unassigned	YES	unset	down	down
GigabitEthernet1/0/9	unassigned	YES	unset	down	down
GigabitEthernet1/0/10	unassigned	YES	unset	up	up
GigabitEthernet1/0/11	unassigned	YES	unset	down	down
GigabitEthernet1/0/12	unassigned	YES	unset	up	up
GigabitEthernet1/0/13	unassigned	YES	unset	down	down
GigabitEthernet1/0/14	unassigned	YES	unset	up	up
GigabitEthernet1/0/15	unassigned	YES	unset	down	down
GigabitEthernet1/0/16	unassigned	YES	unset	up	up
GigabitEthernet1/0/17	unassigned	YES	unset	down	down
GigabitEthernet1/0/18	unassigned	YES	unset	up	up
GigabitEthernet1/0/19	unassigned	YES	unset	down	down
GigabitEthernet1/0/20	unassigned	YES	unset	up	up
GigabitEthernet1/0/21	unassigned	YES	unset	down	down
GigabitEthernet1/0/22	unassigned	YES	unset	up	up
GigabitEthernet1/0/23	unassigned	YES	unset	down	down
GigabitEthernet1/0/24	unassigned	YES	unset	down	down
GigabitEthernet1/1/1	unassigned	YES	unset	down	down
GigabitEthernet1/1/2	unassigned	YES	unset	down	down
GigabitEthernet1/1/3	unassigned	YES	unset	down	down
GigabitEthernet1/1/4	unassigned	YES	unset	down	down
Vlan1	unassigned	YES	unset	administratively down	down
Vlan10	192.168.10.3	YES	manual	up	up
Vlan20	192.168.20.3	YES	manual	up	up
Vlan21	192.168.21.4	YES	manual	up	up
Vlan22	192.168.22.3	YES	manual	up	up
Vlan23	192.168.23.3	YES	manual	up	up
Vlan24	192.168.24.3	YES	manual	up	up
Vlan25	192.168.25.3	YES	manual	up	up
Vlan30	192.168.30.3	YES	manual	up	up
Vlan31	192.168.31.3	YES	manual	up	up
Vlan32	192.168.32.3	YES	manual	up	up
Vlan33	192.168.33.3	YES	manual	up	up
Vlan40	192.168.40.1	YES	manual	up	down
Vlan50	192.168.50.1	YES	manual	up	down
Vlan99	192.168.99.3	YES	manual	up	up
Vlan110	10.10.10.3	YES	manual	up	up
Vlan111	10.10.11.3	YES	manual	up	up
Vlan112	unassigned	YES	unset	up	up
Vlan113	unassigned	YES	unset	up	up
Vlan120	10.10.20.3	YES	manual	up	up
Vlan121	10.10.21.3	YES	manual	up	up
Vlan122	10.10.22.3	YES	manual	up	up
Vlan123	10.10.23.3	YES	manual	up	up
Vlan130	10.10.30.3	YES	manual	up	up
Vlan131	10.10.31.3	YES	manual	up	up
Vlan132	10.10.32.3	YES	manual	up	up
Vlan133	10.10.33.3	YES	manual	up	up
Vlan140	10.10.40.3	YES	manual	up	up
Vlan141	10.10.41.3	YES	manual	up	up
Vlan142	10.10.42.3	YES	manual	up	up
Vlan143	10.10.43.3	YES	manual	up	up
Vlan150	10.10.50.3	YES	manual	up	up
Vlan151	10.10.51.3	YES	manual	up	up
Vlan152	10.10.52.3	YES	manual	up	up

Vlan153	10.10.53.3	YES	manual	up	up
Vlan160	10.10.60.3	YES	manual	up	up
Vlan161	10.10.61.3	YES	manual	up	up
Vlan162	10.10.62.3	YES	manual	up	up
Vlan163	10.10.63.3	YES	manual	up	up
Vlan170	10.10.70.3	YES	manual	up	up
Vlan171	10.10.71.3	YES	manual	up	up
Vlan172	10.10.72.3	YES	manual	up	up
Vlan173	10.10.73.3	YES	manual	up	up
Vlan180	10.10.80.3	YES	manual	up	up
Vlan181	10.10.81.3	YES	manual	up	up
Vlan182	10.10.82.3	YES	manual	up	up
Vlan183	10.10.83.3	YES	manual	up	up

SW-DIST-L1 al L7

SW-DIST-L1

```
SW-DIST-L1#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24
10	TRANSPORT-CORE	active	
11	VLAN0011	active	
20	SERV-BACKOFFICE	active	
21	SERV-DNS	active	
22	SERV-DHCP	active	
23	SERV-MAIL	active	
24	SERV-NTP	active	
25	SERV-WEB	active	
30	SRV-DESARROLLO	active	
31	SRV-QA	active	
32	SRV-PREPROD	active	
33	SRV-PRODUCCION	active	
40	WIFI-CORP	active	
50	WIFI-VISIT	active	
99	MGMT-OOB	active	
110	Dept-Showroom	active	
111	CCTV-Showroom	active	
112	WIFI-C-SHOWROOM	active	
113	WIFI-V-SHOWROOM	active	
120	DEPT-VENTAS	active	
121	CCTV-VENTAS	active	
122	WIFI-C-VENTAS	active	
123	WIFI-V-VENTAS	active	
130	DEPT-AUDITORIO	active	
131	CCTV-AUDITORIO	active	
132	WIFI-C-AUDITORIO	active	
133	WIFI-V-AUDITORIO	active	
140	DEPT-IT	active	
141	CCTV-IT	active	
142	WIFI-C-IT	active	
143	WIFI-V-IT	active	
150	DEPT-RRHH	active	
151	CCTV-RRHH	active	
152	WIFI-C-RRHH	active	
153	WIFI-V-RRHH	active	
160	DEPT-MARKETING	active	
161	CCTV-MARKETING	active	
162	WIFI-C-MKT	active	
163	WIFI-V-MKT	active	
170	DEPT-AUDITORIA	active	
171	CCTV-AUDITORIA	active	
172	WIFI-C-AUDITORIA	active	
173	WIFI-V-AUDITORIA	active	
180	DEPT-FINANZAS	active	
181	CCTV-FINANZAS	active	
182	WIFI-C-FINANZAS	active	
183	WIFI-V-FINANZAS	active	
999	NATIVE-UNUSED	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
SW-DIST-L1#  
SW-DIST-L1#  
SW-DIST-L1#
```

```
SW-DIST-L1#show stan
```

```
SW-DIST-L1#show standby brief
```

P indicates configured to preempt.

Interface	Grp	Pri	P	State	Active	Standby	Virtual IP
SW-DIST-L1#							

SW-DIST-L1#

```
SW-DIST-L1#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
5.5.5.5	1	2WAY/DROTHER	00:00:36	192.168.10.15	Vlan10
7.7.7.7	1	2WAY/DROTHER	00:00:36	192.168.10.17	Vlan10
192.168.99.2	255	FULL/DR	00:00:34	192.168.10.2	Vlan10
3.3.3.3	1	2WAY/DROTHER	00:00:36	192.168.10.13	Vlan10
4.4.4.4	1	2WAY/DROTHER	00:00:36	192.168.10.6	Vlan10
1.1.1.2	1	2WAY/DROTHER	00:00:36	192.168.10.2	Vlan10
6.6.6.6	1	2WAY/DROTHER	00:00:34	192.168.10.16	Vlan10
192.168.99.3	200	FULL/BRP	00:00:36	192.168.10.3	Vlan10

```

SW-DIST-L1#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     on        802.1q         trunking    999
Gig0/1    on        802.1q         trunking    999
Gig0/2    on        802.1q         trunking    999

Port      Vlans allowed on trunk
Fa0/1     1-1005
Gig0/1    1-1005
Gig0/2    1-1005

Port      Vlans allowed and active in management domain
Fa0/1
1,10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999
Gig0/1
1,10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999
Gig0/2
1,10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1
10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999
Gig0/1
1,10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999
Gig0/2
10,11,21,22,23,25,31,40,50,121,123,130,140,141,142,152,160,161,163,171,173,999

```

CW DIST 114

```
SW-DIST-L1#show ip interface brief
Interface      IP-Address      OK? Method Status  Protocol
FastEthernet0/1 unassigned      YES unset  up      up
FastEthernet0/2 unassigned      YES unset  down    down
FastEthernet0/3 unassigned      YES unset  down    down
FastEthernet0/4 unassigned      YES unset  down    down
FastEthernet0/5 unassigned      YES unset  down    down
FastEthernet0/6 unassigned      YES unset  down    down
FastEthernet0/7 unassigned      YES unset  down    down
FastEthernet0/8 unassigned      YES unset  down    down
FastEthernet0/9 unassigned      YES unset  down    down
FastEthernet0/10 unassigned      YES unset  down    down
FastEthernet0/11 unassigned      YES unset  down    down
FastEthernet0/12 unassigned      YES unset  down    down
FastEthernet0/13 unassigned      YES unset  down    down
FastEthernet0/14 unassigned      YES unset  down    down
FastEthernet0/15 unassigned      YES unset  down    down
FastEthernet0/16 unassigned      YES unset  down    down
FastEthernet0/17 unassigned      YES unset  down    down
FastEthernet0/18 unassigned      YES unset  down    down
FastEthernet0/19 unassigned      YES unset  down    down
FastEthernet0/20 unassigned      YES unset  down    down
FastEthernet0/21 unassigned      YES unset  down    down
FastEthernet0/22 unassigned      YES unset  down    down
FastEthernet0/23 unassigned      YES unset  down    down
FastEthernet0/24 unassigned      YES unset  down    down
GigabitEthernet0/1 unassigned      YES unset  up       up
GigabitEthernet0/2 unassigned      YES unset  up       up
Vlan1          192.168.10.11  YES unset  administratively down down
Vlan10         192.168.10.11  YES manual up        up
```

SW-DIST-L2

```
SW-DIST-L2#show vlan brief
VLAN Name                Status Ports
-----
1    default            active Fa0/2, Fa0/3, Fa0/4, Fa0/5,
    Fa0/6, Fa0/7, Fa0/8, Fa0/9,
    Fa0/10, Fa0/11, Fa0/12, Fa0/13,
    Fa0/14, Fa0/15, Fa0/16, Fa0/17,
    Fa0/18, Fa0/19, Fa0/20, Fa0/21,
    Fa0/22, Fa0/23, Fa0/24
10   TRANSPORT-CORE      active
11   VLAN0011            active
20   SERV-BACKOFFICE     active
21   SERV-DNS            active
22   SERV-DHCP           active
23   SERV-MAIL           active
24   SERV-NTP            active
25   SERV-WEB            active
30   SRV-DESARROLLO      active
31   SRV-QA              active
32   SRV-PREPROD         active
33   SRV-PRODUCCION      active
40   WIFI-CORP           active
50   WIFI-VISIT          active
99   MGMT-OOB            active
110  Dept-Showroom        active
111  CCTV-Showroom      active
112  WIFI-C-SHOWROOM      active
113  WIFI-V-SHOWROOM      active
120  DEPT-VENTAS          active
121  CCTV-VENTAS        active
122  WIFI-C-VENTAS        active
123  WIFI-V-VENTAS        active
130  DEPT-AUDITORIO       active
131  CCTV-AUDITORIO     active
132  WIFI-C-AUDITORIO     active
133  WIFI-V-AUDITORIO     active
140  DEPT-IT              active
141  CCTV-IT            active
142  WIFI-C-IT            active
143  WIFI-V-IT            active
150  DEPT-RRHH            active
151  CCTV-RRHH          active
152  WIFI-C-RRHH          active
153  WIFI-V-RRHH          active
160  DEPT-MARKETING       active
161  CCTV-MARKETING     active
162  WIFI-C-MKT           active
163  WIFI-V-MKT           active
170  DEPT-AUDITORIA       active
171  CCTV-AUDITORIA     active
172  WIFI-C-AUDITORIA     active
173  WIFI-V-AUDITORIA     active
180  DEPT-FINANZAS        active
181  CCTV-FINANZAS       active
182  WIFI-C-FINANZAS       active
183  WIFI-V-FINANZAS       active
999  NATIVE-UNUSED        active
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default       active
1005 trnet-default         active
SW-DIST-L2#
```

```

SW-DIST-L2#show standby brief
                P indicates configured to preempt.
                |
Interface  Grp  Pri  P State      Active      Standby      Virtual IP
SW-DIST-L2#

```

```

SW-DIST-L2#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

```

Gateway of last resort is 192.168.10.1 to network 0.0.0.0

```

    10.0.0.0/8 is variably subnetted, 31 subnets, 2 masks
O    10.0.0.0/30 [110/2] via 192.168.10.2, 01:05:06, Vlan10
O    10.10.10.0/24 [110/2] via 192.168.10.3, 01:05:06, Vlan10
        [110/2] via 192.168.10.2, 01:05:06, Vlan10
O    10.10.11.0/24 [110/2] via 192.168.10.3, 01:05:06, Vlan10
        [110/2] via 192.168.10.2, 01:05:06, Vlan10
O    10.10.20.0/24 [110/2] via 192.168.10.3, 01:05:06, Vlan10
        [110/2] via 192.168.10.2, 01:05:06, Vlan10
O    10.10.21.0/24 [110/2] via 192.168.10.3, 01:05:06, Vlan10
        [110/2] via 192.168.10.2, 01:05:06, Vlan10
O    10.10.22.0/24 [110/2] via 192.168.10.3, 01:05:06, Vlan10
        [110/2] via 192.168.10.2, 01:05:06, Vlan10
O    10.10.23.0/24 [110/2] via 192.168.10.3, 01:05:06, Vlan10
        [110/2] via 192.168.10.2, 01:05:06, Vlan10
O    10.10.30.0/24 [110/2] via 192.168.10.3, 01:05:06, Vlan10
        [110/2] via 192.168.10.2, 01:05:06, Vlan10
O    10.10.31.0/24 [110/2] via 192.168.10.3, 01:05:06, Vlan10
        [110/2] via 192.168.10.2, 01:05:06, Vlan10
O    10.10.32.0/24 [110/2] via 192.168.10.3, 01:05:06, Vlan10
        [110/2] via 192.168.10.2, 01:05:06, Vlan10
O    10.10.33.0/24 [110/2] via 192.168.10.3, 01:05:06, Vlan10
        [110/2] via 192.168.10.2, 01:05:06, Vlan10
O    10.10.40.0/24 [110/2] via 192.168.10.3, 01:05:06, Vlan10
        [110/2] via 192.168.10.2, 01:05:06, Vlan10
O    10.10.41.0/24 [110/2] via 192.168.10.3, 01:05:06, Vlan10
        [110/2] via 192.168.10.2, 01:05:06, Vlan10
O    10.10.42.0/24 [110/2] via 192.168.10.3, 01:05:06, Vlan10
        [110/2] via 192.168.10.2, 01:05:06, Vlan10
O    10.10.43.0/24 [110/2] via 192.168.10.3, 01:05:06, Vlan10
        [110/2] via 192.168.10.2, 01:05:06, Vlan10
O    10.10.50.0/24 [110/2] via 192.168.10.3, 01:05:06, Vlan10
        [110/2] via 192.168.10.2, 01:05:06, Vlan10
O    10.10.51.0/24 [110/2] via 192.168.10.3, 01:05:06, Vlan10
        [110/2] via 192.168.10.2, 01:05:06, Vlan10
O    10.10.52.0/24 [110/2] via 192.168.10.3, 01:05:06, Vlan10
        [110/2] via 192.168.10.2, 01:05:06, Vlan10
O    10.10.53.0/24 [110/2] via 192.168.10.3, 01:05:06, Vlan10
        [110/2] via 192.168.10.2, 01:05:06, Vlan10
O    10.10.60.0/24 [110/2] via 192.168.10.3, 01:05:06, Vlan10
        [110/2] via 192.168.10.2, 01:05:06, Vlan10
O    10.10.61.0/24 [110/2] via 192.168.10.3, 01:05:06, Vlan10
        [110/2] via 192.168.10.2, 01:05:06, Vlan10
O    10.10.62.0/24 [110/2] via 192.168.10.3, 01:05:06, Vlan10
        [110/2] via 192.168.10.2, 01:05:06, Vlan10
O    10.10.63.0/24 [110/2] via 192.168.10.3, 01:05:06, Vlan10
        [110/2] via 192.168.10.2, 01:05:06, Vlan10
O    10.10.70.0/24 [110/2] via 192.168.10.3, 01:05:06, Vlan10
        [110/2] via 192.168.10.2, 01:05:06, Vlan10
O    10.10.71.0/24 [110/2] via 192.168.10.3, 01:05:06, Vlan10
        [110/2] via 192.168.10.2, 01:05:06, Vlan10
O    10.10.72.0/24 [110/2] via 192.168.10.3, 01:05:06, Vlan10
        [110/2] via 192.168.10.2, 01:05:06, Vlan10
O    10.10.73.0/24 [110/2] via 192.168.10.3, 01:05:06, Vlan10
        [110/2] via 192.168.10.2, 01:05:06, Vlan10
O    10.10.80.0/24 [110/2] via 192.168.10.3, 01:05:06, Vlan10

```

```

O      10.10.81.0/24 [110/2] via 192.168.10.2, 01:05:06, Vlan10
O      10.10.82.0/24 [110/2] via 192.168.10.2, 01:05:06, Vlan10
O      10.10.83.0/24 [110/2] via 192.168.10.3, 01:05:06, Vlan10
O      10.10.83.0/24 [110/2] via 192.168.10.2, 01:05:06, Vlan10
172.16.0.0/24 is summarized, 18 subnets
O E2 172.16.10.0 [110/20] via 192.168.10.2, 00:02:56, Vlan10
O E2 172.16.11.0 [110/20] via 192.168.10.2, 00:02:56, Vlan10
O E2 172.16.12.0 [110/20] via 192.168.10.2, 00:02:56, Vlan10
O E2 172.16.13.0 [110/20] via 192.168.10.2, 00:02:56, Vlan10
O E2 172.16.14.0 [110/20] via 192.168.10.2, 00:02:56, Vlan10
O E2 172.16.15.0 [110/20] via 192.168.10.2, 00:02:56, Vlan10
O E2 172.16.20.0 [110/20] via 192.168.10.2, 01:03:59, Vlan10
O E2 172.16.21.0 [110/20] via 192.168.10.2, 01:03:59, Vlan10
O E2 172.16.22.0 [110/20] via 192.168.10.2, 01:03:59, Vlan10
O E2 172.16.23.0 [110/20] via 192.168.10.2, 01:03:59, Vlan10
O E2 172.16.24.0 [110/20] via 192.168.10.2, 01:03:59, Vlan10
O E2 172.16.25.0 [110/20] via 192.168.10.2, 01:03:59, Vlan10
O E2 172.16.30.0 [110/20] via 192.168.10.2, 01:05:06, Vlan10
O E2 172.16.31.0 [110/20] via 192.168.10.2, 01:05:06, Vlan10
O E2 172.16.32.0 [110/20] via 192.168.10.2, 01:05:06, Vlan10
O E2 172.16.33.0 [110/20] via 192.168.10.2, 01:05:06, Vlan10
O E2 172.16.34.0 [110/20] via 192.168.10.2, 01:05:06, Vlan10
O E2 172.16.35.0 [110/20] via 192.168.10.2, 01:05:06, Vlan10
C      192.168.10.0/24 is directly connected, Vlan10
O      192.168.20.0/24 [110/2] via 192.168.10.2, 01:05:06, Vlan10
O      192.168.21.0/24 [110/2] via 192.168.10.2, 01:05:06, Vlan10
O      192.168.22.0/24 [110/2] via 192.168.10.2, 01:05:06, Vlan10
O      192.168.23.0/24 [110/2] via 192.168.10.2, 01:05:06, Vlan10
O      192.168.24.0/24 [110/2] via 192.168.10.2, 01:05:06, Vlan10
O      192.168.25.0/24 [110/2] via 192.168.10.2, 01:05:06, Vlan10
O      192.168.30.0/24 [110/2] via 192.168.10.2, 01:05:06, Vlan10
O      192.168.31.0/24 [110/2] via 192.168.10.2, 01:05:06, Vlan10
O      192.168.32.0/24 [110/2] via 192.168.10.2, 01:05:06, Vlan10
O      192.168.33.0/24 [110/2] via 192.168.10.2, 01:05:06, Vlan10
O      192.168.99.0/24 [110/2] via 192.168.10.2, 01:05:06, Vlan10
S*    0.0.0.0/0 [1/0] via 192.168.10.1

```

```
SW-DIST-L2#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
5.5.5.5	1	2WAY/DROTHER	00:00:38	192.168.10.15	Vlan10
7.7.7.7	1	2WAY/DROTHER	00:00:38	192.168.10.17	Vlan10
11.11.11.11	1	2WAY/DROTHER	00:00:38	192.168.10.11	Vlan10
192.168.99.2	255	FULL/DR	00:00:37	192.168.10.2	Vlan10
6.6.6.6	1	2WAY/DROTHER	00:00:36	192.168.10.16	Vlan10
4.4.4.4	1	2WAY/DROTHER	00:00:38	192.168.10.6	Vlan10
192.168.99.3	200	FULL/BDR	00:00:39	192.168.10.3	Vlan10
3.3.3.3	1	2WAY/DROTHER	00:00:39	192.168.10.13	Vlan10

```
SW-DIST-L2#
```

```
SW-DIST-L2#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Fa0/1	on	802.1q	trunking	999
Gig0/1	on	802.1q	trunking	999
Gig0/2	on	802.1q	trunking	999

```
Port Vlan allowed on trunk
```

```

Fa0/1 1-1005
Gig0/1 1-1005
Gig0/2 1-1005

```

```
Port Vlan allowed and active in management domain
```

```

Fa0/1
1,10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999
Gig0/1
1,10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999
Gig0/2
1,10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999

```

```
Port Vlan in spanning tree forwarding state and not pruned
```

```

Fa0/1
10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999
Gig0/1
1,10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999
Gig0/2
11,23,25,31,32,40,50,99,112,113,120,133,140,141,142,151,160,161,163,170,173,181

```

```
SW-DIST-L2#
```

```
SW-DIST-L2#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/1	unassigned	YES	unset	up	up
FastEthernet0/2	unassigned	YES	unset	down	down
FastEthernet0/3	unassigned	YES	unset	down	down
FastEthernet0/4	unassigned	YES	unset	down	down
FastEthernet0/5	unassigned	YES	unset	down	down
FastEthernet0/6	unassigned	YES	unset	down	down
FastEthernet0/7	unassigned	YES	unset	down	down
FastEthernet0/8	unassigned	YES	unset	down	down
FastEthernet0/9	unassigned	YES	unset	down	down
FastEthernet0/10	unassigned	YES	unset	down	down
FastEthernet0/11	unassigned	YES	unset	down	down
FastEthernet0/12	unassigned	YES	unset	down	down
FastEthernet0/13	unassigned	YES	unset	down	down
FastEthernet0/14	unassigned	YES	unset	down	down
FastEthernet0/15	unassigned	YES	unset	down	down
FastEthernet0/16	unassigned	YES	unset	down	down
FastEthernet0/17	unassigned	YES	unset	down	down
FastEthernet0/18	unassigned	YES	unset	down	down
FastEthernet0/19	unassigned	YES	unset	down	down
FastEthernet0/20	unassigned	YES	unset	down	down
FastEthernet0/21	unassigned	YES	unset	down	down
FastEthernet0/22	unassigned	YES	unset	down	down
FastEthernet0/23	unassigned	YES	unset	down	down
FastEthernet0/24	unassigned	YES	unset	down	down
GigabitEthernet0/1	unassigned	YES	unset	up	up
GigabitEthernet0/2	unassigned	YES	unset	up	up
Vlan1	unassigned	YES	unset	administratively down	down
Vlan10	192.168.10.12	YES	manual	up	up

SW-DIST-L3

```
SW-DIST-L3#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24
10	TRANSPORT-CORE	active	
11	VLAN0011	active	
20	SERV-BACKOFFICE	active	
21	SERV-DNS	active	
22	SERV-DHCP	active	
23	SERV-MAIL	active	
24	SERV-NTF	active	
25	SERV-WEB	active	
30	SRV-DESARROLLO	active	
31	SRV-QA	active	
32	SRV-PREPROD	active	
33	SRV-PRODUCCION	active	
40	WIFI-CORP	active	
50	WIFI-VISIT	active	
99	MGMT-OOB	active	
110	Dept-Showroom	active	
111	CCTV-Showroom	active	
112	WIFI-C-SHOWROOM	active	
113	WIFI-V-SHOWROOM	active	
120	DEPT-VENTAS	active	
121	CCTV-VENTAS	active	
122	WIFI-C-VENTAS	active	
123	WIFI-V-VENTAS	active	
130	DEPT-AUDITORIO	active	
131	CCTV-AUDITORIO	active	
132	WIFI-C-AUDITORIO	active	
133	WIFI-V-AUDITORIO	active	
140	DEPT-IT	active	
141	CCTV-IT	active	
142	WIFI-C-IT	active	
143	WIFI-V-IT	active	
150	DEPT-RRHH	active	
151	CCTV-RRHH	active	
152	WIFI-C-RRHH	active	
153	WIFI-V-RRHH	active	
160	DEPT-MARKETING	active	
161	CCTV-MARKETING	active	
162	WIFI-C-MKT	active	
163	WIFI-V-MKT	active	
170	DEPT-AUDITORIA	active	
171	CCTV-AUDITORIA	active	
172	WIFI-C-AUDITORIA	active	
173	WIFI-V-AUDITORIA	active	
180	DEPT-FINANZAS	active	
181	CCTV-FINANZAS	active	
182	WIFI-C-FINANZAS	active	
183	WIFI-V-FINANZAS	active	
999	NATIVE-UNUSED	active	
1002	fdi-default	active	
1003	token-ring-default	active	
1004	fdinet-default	active	
1005	trnet-default	active	

```
SW-DIST-L3#
SW-DIST-L3#
```

```

SW-DIST-L3#show stan
SW-DIST-L3#show standby brief
          F indicates configured to preempt.
          |
Interface  Grp  Pri  P State      Active      Standby      Virtual IP
SW-DIST-L3#

```

```

SW-DIST-L3#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

```

Gateway of last resort is 192.168.10.2 to network 0.0.0.0

```

10.0.0.0/8 is variably subnetted, 31 subnets, 2 masks
O    10.0.0.0/30 [110/2] via 192.168.10.2, 01:06:29, Vlan10
O    10.10.10.0/24 [110/2] via 192.168.10.3, 01:06:19, Vlan10
      [110/2] via 192.168.10.2, 01:06:19, Vlan10
O    10.10.11.0/24 [110/2] via 192.168.10.3, 01:06:19, Vlan10
      [110/2] via 192.168.10.2, 01:06:19, Vlan10
O    10.10.20.0/24 [110/2] via 192.168.10.3, 01:06:19, Vlan10
      [110/2] via 192.168.10.2, 01:06:19, Vlan10
O    10.10.21.0/24 [110/2] via 192.168.10.3, 01:06:19, Vlan10
      [110/2] via 192.168.10.2, 01:06:19, Vlan10
O    10.10.22.0/24 [110/2] via 192.168.10.3, 01:06:19, Vlan10
      [110/2] via 192.168.10.2, 01:06:19, Vlan10
O    10.10.23.0/24 [110/2] via 192.168.10.3, 01:06:19, Vlan10
      [110/2] via 192.168.10.2, 01:06:19, Vlan10
O    10.10.30.0/24 [110/2] via 192.168.10.3, 01:06:19, Vlan10
      [110/2] via 192.168.10.2, 01:06:19, Vlan10
O    10.10.31.0/24 [110/2] via 192.168.10.3, 01:06:19, Vlan10
      [110/2] via 192.168.10.2, 01:06:19, Vlan10
O    10.10.32.0/24 [110/2] via 192.168.10.3, 01:06:19, Vlan10
      [110/2] via 192.168.10.2, 01:06:19, Vlan10
O    10.10.33.0/24 [110/2] via 192.168.10.3, 01:06:19, Vlan10
      [110/2] via 192.168.10.2, 01:06:19, Vlan10
O    10.10.40.0/24 [110/2] via 192.168.10.3, 01:06:19, Vlan10
      [110/2] via 192.168.10.2, 01:06:19, Vlan10
O    10.10.41.0/24 [110/2] via 192.168.10.3, 01:06:19, Vlan10
      [110/2] via 192.168.10.2, 01:06:19, Vlan10
O    10.10.42.0/24 [110/2] via 192.168.10.3, 01:06:19, Vlan10
      [110/2] via 192.168.10.2, 01:06:19, Vlan10
O    10.10.43.0/24 [110/2] via 192.168.10.3, 01:06:19, Vlan10
      [110/2] via 192.168.10.2, 01:06:19, Vlan10
O    10.10.50.0/24 [110/2] via 192.168.10.3, 01:06:19, Vlan10
      [110/2] via 192.168.10.2, 01:06:19, Vlan10
O    10.10.51.0/24 [110/2] via 192.168.10.3, 01:06:19, Vlan10
      [110/2] via 192.168.10.2, 01:06:19, Vlan10
O    10.10.52.0/24 [110/2] via 192.168.10.3, 01:06:19, Vlan10
      [110/2] via 192.168.10.2, 01:06:19, Vlan10
O    10.10.53.0/24 [110/2] via 192.168.10.3, 01:06:19, Vlan10
      [110/2] via 192.168.10.2, 01:06:19, Vlan10
O    10.10.60.0/24 [110/2] via 192.168.10.3, 01:06:19, Vlan10
      [110/2] via 192.168.10.2, 01:06:19, Vlan10
O    10.10.61.0/24 [110/2] via 192.168.10.3, 01:06:19, Vlan10
      [110/2] via 192.168.10.2, 01:06:19, Vlan10
O    10.10.62.0/24 [110/2] via 192.168.10.3, 01:06:19, Vlan10
      [110/2] via 192.168.10.2, 01:06:19, Vlan10
O    10.10.63.0/24 [110/2] via 192.168.10.3, 01:06:19, Vlan10
      [110/2] via 192.168.10.2, 01:06:19, Vlan10
O    10.10.70.0/24 [110/2] via 192.168.10.3, 01:06:19, Vlan10
      [110/2] via 192.168.10.2, 01:06:19, Vlan10
O    10.10.71.0/24 [110/2] via 192.168.10.3, 01:06:19, Vlan10
      [110/2] via 192.168.10.2, 01:06:19, Vlan10
O    10.10.72.0/24 [110/2] via 192.168.10.3, 01:06:19, Vlan10
      [110/2] via 192.168.10.2, 01:06:19, Vlan10
O    10.10.73.0/24 [110/2] via 192.168.10.3, 01:06:19, Vlan10
      [110/2] via 192.168.10.2, 01:06:19, Vlan10
O    10.10.80.0/24 [110/2] via 192.168.10.3, 01:06:19, Vlan10

```

```

O      10.10.81.0/24 [110/2] via 192.168.10.2, 01:06:19, Vlan10
O      10.10.82.0/24 [110/2] via 192.168.10.3, 01:06:19, Vlan10
O      10.10.83.0/24 [110/2] via 192.168.10.2, 01:06:19, Vlan10
O      10.10.83.0/24 [110/2] via 192.168.10.3, 01:06:19, Vlan10
O      10.10.83.0/24 [110/2] via 192.168.10.2, 01:06:19, Vlan10
O      172.16.0.0/24 is subnetted, 18 subnets
O E2   172.16.10.0 [110/20] via 192.168.10.2, 00:04:14, Vlan10
O E2   172.16.11.0 [110/20] via 192.168.10.2, 00:04:14, Vlan10
O E2   172.16.12.0 [110/20] via 192.168.10.2, 00:04:14, Vlan10
O E2   172.16.13.0 [110/20] via 192.168.10.2, 00:04:14, Vlan10
O E2   172.16.14.0 [110/20] via 192.168.10.2, 00:04:14, Vlan10
O E2   172.16.15.0 [110/20] via 192.168.10.2, 00:04:14, Vlan10
O E2   172.16.20.0 [110/20] via 192.168.10.2, 01:05:17, Vlan10
O E2   172.16.21.0 [110/20] via 192.168.10.2, 01:05:17, Vlan10
O E2   172.16.22.0 [110/20] via 192.168.10.2, 01:05:17, Vlan10
O E2   172.16.23.0 [110/20] via 192.168.10.2, 01:05:17, Vlan10
O E2   172.16.24.0 [110/20] via 192.168.10.2, 01:05:17, Vlan10
O E2   172.16.25.0 [110/20] via 192.168.10.2, 01:05:17, Vlan10
O E2   172.16.30.0 [110/20] via 192.168.10.2, 01:06:29, Vlan10
O E2   172.16.31.0 [110/20] via 192.168.10.2, 01:06:29, Vlan10
O E2   172.16.32.0 [110/20] via 192.168.10.2, 01:06:29, Vlan10
O E2   172.16.33.0 [110/20] via 192.168.10.2, 01:06:29, Vlan10
O E2   172.16.34.0 [110/20] via 192.168.10.2, 01:06:29, Vlan10
O E2   172.16.35.0 [110/20] via 192.168.10.2, 01:06:29, Vlan10
C      192.168.10.0/24 is directly connected, Vlan10
O      192.168.20.0/24 [110/2] via 192.168.10.2, 01:06:29, Vlan10
O      192.168.21.0/24 [110/2] via 192.168.10.2, 01:06:29, Vlan10
O      192.168.22.0/24 [110/2] via 192.168.10.2, 01:06:29, Vlan10
O      192.168.23.0/24 [110/2] via 192.168.10.2, 01:06:29, Vlan10
O      192.168.24.0/24 [110/2] via 192.168.10.2, 01:06:29, Vlan10
O      192.168.25.0/24 [110/2] via 192.168.10.2, 01:06:29, Vlan10
O      192.168.30.0/24 [110/2] via 192.168.10.2, 01:06:29, Vlan10
O      192.168.31.0/24 [110/2] via 192.168.10.2, 01:06:29, Vlan10
O      192.168.32.0/24 [110/2] via 192.168.10.2, 01:06:29, Vlan10
O      192.168.33.0/24 [110/2] via 192.168.10.2, 01:06:29, Vlan10
O      192.168.39.0/24 [110/2] via 192.168.10.2, 01:06:29, Vlan10
O=E2  0.0.0.0/0 [110/1] via 192.168.10.2, 01:06:29, Vlan10

```

```

SW-DIST-L3#show ip ospf ne
SW-DIST-L3#show ip os
SW-DIST-L3#show ip ospf ne
SW-DIST-L3#show ip ospf neighbor

```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.99.3	200	FULL/DR	00:00:34	192.168.10.3	Vlan10
5.5.5.5	1	2WAY/DROTHER	00:00:33	192.168.10.15	Vlan10
7.7.7.7	1	2WAY/DROTHER	00:00:33	192.168.10.17	Vlan10
11.11.11.11	1	2WAY/DROTHER	00:00:33	192.168.10.11	Vlan10
192.168.99.2	255	FULL/DR	00:00:32	192.168.10.2	Vlan10
4.4.4.4	1	2WAY/DROTHER	00:00:33	192.168.10.6	Vlan10
6.6.6.6	1	2WAY/DROTHER	00:00:31	192.168.10.16	Vlan10
2.2.2.2	1	2WAY/DROTHER	00:00:33	192.168.10.12	Vlan10

```

SW-DIST-L3#

```

```

SW-DIST-L3#show interfaces trunk

```

Port	Mode	Encapsulation	Status	Native vlan
Fa0/1	on	802.1q	trunking	999
Gig0/1	on	802.1q	trunking	999
Gig0/2	on	802.1q	trunking	999

Port	Vlans allowed on trunk
Fa0/1	1-1005
Gig0/1	1-1005
Gig0/2	1-1005

Port	Vlans allowed and active in management domain
------	---

Fa0/1	1,10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999
-------	--

Gig0/1	1,10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999
--------	--

Gig0/2	1,10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999
--------	--

Port	Vlans in spanning tree forwarding state and not pruned
------	--

Fa0/1	10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999
-------	--

Gig0/1	1,10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999
--------	--

Gig0/2	10,11,20,21,22,23,25,31,32,33,40,50,99,121,122,123,130,132,140,141,142,150,151,152,160,161,162,163,173,180,181,182,183,999
--------	--

```

SW-DIST-L3#

```

```

SW-DIST-L3#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/1 unassigned      YES unset  up          up
FastEthernet0/2 unassigned      YES unset  down        down
FastEthernet0/3 unassigned      YES unset  down        down
FastEthernet0/4 unassigned      YES unset  down        down
FastEthernet0/5 unassigned      YES unset  down        down
FastEthernet0/6 unassigned      YES unset  down        down
FastEthernet0/7 unassigned      YES unset  down        down
FastEthernet0/8 unassigned      YES unset  down        down
FastEthernet0/9 unassigned      YES unset  down        down
FastEthernet0/10 unassigned      YES unset  down        down
FastEthernet0/11 unassigned      YES unset  down        down
FastEthernet0/12 unassigned      YES unset  down        down
FastEthernet0/13 unassigned      YES unset  down        down
FastEthernet0/14 unassigned      YES unset  down        down
FastEthernet0/15 unassigned      YES unset  down        down
FastEthernet0/16 unassigned      YES unset  down        down
FastEthernet0/17 unassigned      YES unset  down        down
FastEthernet0/18 unassigned      YES unset  down        down
FastEthernet0/19 unassigned      YES unset  down        down
FastEthernet0/20 unassigned      YES unset  down        down
FastEthernet0/21 unassigned      YES unset  down        down
FastEthernet0/22 unassigned      YES unset  down        down
FastEthernet0/23 unassigned      YES unset  down        down
FastEthernet0/24 unassigned      YES unset  down        down
GigabitEthernet0/1 unassigned      YES unset  up          up
GigabitEthernet0/2 unassigned      YES unset  up          up
Vlan1          unassigned      YES unset  administratively down down
Vlan10         192.168.10.13  YES manual up          up

```

SW-DIST-L4

```

SW-DIST-L4#show vlan brief
VLAN Name                Status      Ports
-----
1    default              active      Fa0/2, Fa0/4, Fa0/5, Fa0/6
                                           Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                           Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                           Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                           Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                           Fa0/23, Fa0/24
10   TRANSPORT-CORE       active
11   VLAN0011              active
20   SERV-BACKOFFICE       active
21   SERV-DNS               active
22   SERV-DHCP              active
23   SERV-MAIL              active
24   SERV-NTP               active
25   SERV-WEB               active
30   SRV-DESARROLLO        active
31   SRV-QA                 active
32   SRV-FREPROD           active
33   SRV-PRODUCCION        active
40   WIFI-CORP              active
50   WIFI-VISIT            active
99   MGMT-COB               active
110  Dept-Showroom          active
111  CCTV-Showroom        active
112  WIFI-C-SHOWROOM        active
113  WIFI-V-SHOWROOM        active
120  DEPT-VENTAS            active
121  CCTV-VENTAS          active
122  WIFI-C-VENTAS          active
123  WIFI-V-VENTAS          active
130  DEPT-AUDITORIO         active
131  CCTV-AUDITORIO       active
132  WIFI-C-AUDITORIO       active
133  WIFI-V-AUDITORIO       active
140  DEPT-IT                active
141  CCTV-IT              active
142  WIFI-C-IT              active
143  WIFI-V-IT              active
150  DEPT-RRHH              active
151  CCTV-RRHH            active
152  WIFI-C-RRHH            active
153  WIFI-V-RRHH            active
160  DEPT-MARKETING          active
161  CCTV-MARKETING        active
162  WIFI-C-MKT             active
163  WIFI-V-MKT             active
170  DEPT-AUDITORIA         active
171  CCTV-AUDITORIA       active
172  WIFI-C-AUDITORIA       active
173  WIFI-V-AUDITORIA       active
180  DEPT-FINANZAS          active
181  CCTV-FINANZAS        active
182  WIFI-C-FINANZAS        active
183  WIFI-V-FINANZAS        active
999  NATIVE-UNUSED          active
1002 fddi-default           active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default          active
SW-DIST-L4#

```

SW-DIST-L4#show standby brief

P indicates configured to preempt.

Interface	Grp	Pri	P	State	Active	Standby	Virtual IP
SW-DIST-L4#							

IOS Command Line Interface

SW-DIST-L4#show ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is 192.168.10.2 to network 0.0.0.0

10.0.0.0/8 is variably subnetted, 31 subnets, 2 masks

O	10.0.0.0/30	[110/2]	via 192.168.10.2, 01:19:19, Vlan10
O	10.10.10.0/24	[110/2]	via 192.168.10.3, 01:19:19, Vlan10
		[110/2]	via 192.168.10.2, 01:19:19, Vlan10
O	10.10.11.0/24	[110/2]	via 192.168.10.3, 01:19:19, Vlan10
		[110/2]	via 192.168.10.2, 01:19:19, Vlan10
O	10.10.20.0/24	[110/2]	via 192.168.10.3, 01:19:19, Vlan10
		[110/2]	via 192.168.10.2, 01:19:19, Vlan10
O	10.10.21.0/24	[110/2]	via 192.168.10.3, 01:19:19, Vlan10
		[110/2]	via 192.168.10.2, 01:19:19, Vlan10
O	10.10.22.0/24	[110/2]	via 192.168.10.3, 01:19:19, Vlan10
		[110/2]	via 192.168.10.2, 01:19:19, Vlan10
O	10.10.23.0/24	[110/2]	via 192.168.10.3, 01:19:19, Vlan10
		[110/2]	via 192.168.10.2, 01:19:19, Vlan10
O	10.10.30.0/24	[110/2]	via 192.168.10.3, 01:19:19, Vlan10
		[110/2]	via 192.168.10.2, 01:19:19, Vlan10
O	10.10.31.0/24	[110/2]	via 192.168.10.3, 01:19:19, Vlan10
		[110/2]	via 192.168.10.2, 01:19:19, Vlan10
O	10.10.32.0/24	[110/2]	via 192.168.10.3, 01:19:19, Vlan10
		[110/2]	via 192.168.10.2, 01:19:19, Vlan10
O	10.10.33.0/24	[110/2]	via 192.168.10.3, 01:19:19, Vlan10
		[110/2]	via 192.168.10.2, 01:19:19, Vlan10
O	10.10.40.0/24	[110/2]	via 192.168.10.3, 01:19:19, Vlan10
		[110/2]	via 192.168.10.2, 01:19:19, Vlan10
O	10.10.41.0/24	[110/2]	via 192.168.10.3, 01:19:19, Vlan10
		[110/2]	via 192.168.10.2, 01:19:19, Vlan10
O	10.10.42.0/24	[110/2]	via 192.168.10.3, 01:19:19, Vlan10
		[110/2]	via 192.168.10.2, 01:19:19, Vlan10
O	10.10.43.0/24	[110/2]	via 192.168.10.3, 01:19:19, Vlan10
		[110/2]	via 192.168.10.2, 01:19:19, Vlan10
O	10.10.50.0/24	[110/2]	via 192.168.10.3, 01:19:19, Vlan10
		[110/2]	via 192.168.10.2, 01:19:19, Vlan10
O	10.10.51.0/24	[110/2]	via 192.168.10.3, 01:19:19, Vlan10
		[110/2]	via 192.168.10.2, 01:19:19, Vlan10
O	10.10.52.0/24	[110/2]	via 192.168.10.3, 01:19:19, Vlan10
		[110/2]	via 192.168.10.2, 01:19:19, Vlan10
O	10.10.53.0/24	[110/2]	via 192.168.10.3, 01:19:19, Vlan10
		[110/2]	via 192.168.10.2, 01:19:19, Vlan10
O	10.10.60.0/24	[110/2]	via 192.168.10.3, 01:19:19, Vlan10
		[110/2]	via 192.168.10.2, 01:19:19, Vlan10
O	10.10.61.0/24	[110/2]	via 192.168.10.3, 01:19:19, Vlan10
		[110/2]	via 192.168.10.2, 01:19:19, Vlan10
O	10.10.62.0/24	[110/2]	via 192.168.10.3, 01:19:19, Vlan10
		[110/2]	via 192.168.10.2, 01:19:19, Vlan10
O	10.10.63.0/24	[110/2]	via 192.168.10.3, 01:19:19, Vlan10
		[110/2]	via 192.168.10.2, 01:19:19, Vlan10
O	10.10.70.0/24	[110/2]	via 192.168.10.3, 01:19:19, Vlan10
		[110/2]	via 192.168.10.2, 01:19:19, Vlan10
O	10.10.71.0/24	[110/2]	via 192.168.10.3, 01:19:19, Vlan10
		[110/2]	via 192.168.10.2, 01:19:19, Vlan10
O	10.10.72.0/24	[110/2]	via 192.168.10.3, 01:19:19, Vlan10
		[110/2]	via 192.168.10.2, 01:19:19, Vlan10
O	10.10.73.0/24	[110/2]	via 192.168.10.3, 01:19:19, Vlan10
		[110/2]	via 192.168.10.2, 01:19:19, Vlan10

```

O    10.10.80.0/24 [110/2] via 192.168.10.3, 01:19:19, Vlan10
    [110/2] via 192.168.10.2, 01:19:19, Vlan10
O    10.10.81.0/24 [110/2] via 192.168.10.3, 01:19:19, Vlan10
    [110/2] via 192.168.10.2, 01:19:19, Vlan10
O    10.10.82.0/24 [110/2] via 192.168.10.3, 01:19:19, Vlan10
    [110/2] via 192.168.10.2, 01:19:19, Vlan10
O    10.10.83.0/24 [110/2] via 192.168.10.3, 01:19:19, Vlan10
    [110/2] via 192.168.10.2, 01:19:19, Vlan10
172.16.0.0/24 is subnetted, 18 subnets
O E2 172.16.10.0 [110/20] via 192.168.10.2, 00:17:20, Vlan10
O E2 172.16.11.0 [110/20] via 192.168.10.2, 00:17:20, Vlan10
O E2 172.16.12.0 [110/20] via 192.168.10.2, 00:17:20, Vlan10
O E2 172.16.13.0 [110/20] via 192.168.10.2, 00:17:20, Vlan10
O E2 172.16.14.0 [110/20] via 192.168.10.2, 00:17:20, Vlan10
O E2 172.16.15.0 [110/20] via 192.168.10.2, 00:17:20, Vlan10
O E2 172.16.20.0 [110/20] via 192.168.10.2, 01:18:23, Vlan10
O E2 172.16.21.0 [110/20] via 192.168.10.2, 01:18:23, Vlan10
O E2 172.16.22.0 [110/20] via 192.168.10.2, 01:18:23, Vlan10
O E2 172.16.23.0 [110/20] via 192.168.10.2, 01:18:23, Vlan10
O E2 172.16.24.0 [110/20] via 192.168.10.2, 01:18:23, Vlan10
O E2 172.16.25.0 [110/20] via 192.168.10.2, 01:18:23, Vlan10
O E2 172.16.30.0 [110/20] via 192.168.10.2, 01:19:19, Vlan10
O E2 172.16.31.0 [110/20] via 192.168.10.2, 01:19:19, Vlan10
O E2 172.16.32.0 [110/20] via 192.168.10.2, 01:19:19, Vlan10
O E2 172.16.33.0 [110/20] via 192.168.10.2, 01:19:19, Vlan10
O E2 172.16.34.0 [110/20] via 192.168.10.2, 01:19:19, Vlan10
O E2 172.16.35.0 [110/20] via 192.168.10.2, 01:19:19, Vlan10
C    192.168.10.0/24 is directly connected, Vlan10
O    192.168.20.0/24 [110/2] via 192.168.10.2, 01:19:19, Vlan10
O    192.168.21.0/24 [110/2] via 192.168.10.2, 01:19:19, Vlan10
O    192.168.22.0/24 [110/2] via 192.168.10.2, 01:19:19, Vlan10
O    192.168.23.0/24 [110/2] via 192.168.10.2, 01:19:19, Vlan10
O    192.168.24.0/24 [110/2] via 192.168.10.2, 01:19:19, Vlan10
O    192.168.25.0/24 [110/2] via 192.168.10.2, 01:19:19, Vlan10
O    192.168.30.0/24 [110/2] via 192.168.10.2, 01:19:19, Vlan10
O    192.168.31.0/24 [110/2] via 192.168.10.2, 01:19:19, Vlan10
O    192.168.32.0/24 [110/2] via 192.168.10.2, 01:19:19, Vlan10
O    192.168.33.0/24 [110/2] via 192.168.10.2, 01:19:19, Vlan10
O    192.168.99.0/24 [110/2] via 192.168.10.2, 01:19:19, Vlan10
O*E2 0.0.0.0/0 [110/1] via 192.168.10.2, 01:19:19, Vlan10

```

SW-DIST-L4#

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.99.3	200	FULL/BDP	00:00:36	192.168.10.3	Vlan10
5.5.5.5	1	2WAY/DROTHER	00:00:35	192.168.10.15	Vlan10
7.7.7.7	1	2WAY/DROTHER	00:00:35	192.168.10.17	Vlan10
11.11.11.11	1	2WAY/DROTHER	00:00:35	192.168.10.11	Vlan10
192.168.99.2	255	FULL/DR	00:00:35	192.168.10.2	Vlan10
6.6.6.6	1	2WAY/DROTHER	00:00:33	192.168.10.16	Vlan10
3.3.3.3	1	2WAY/DROTHER	00:00:36	192.168.10.13	Vlan10
2.2.2.2	1	2WAY/DROTHER	00:00:34	192.168.10.12	Vlan10

SW-DIST-L4#show interfaces trunk

Port	Mode	Encapsulation	Status	Native vlan
Fa0/1	on	802.1q	trunking	999
Fa0/3	auto	n-802.1q	trunking	999
Gig0/1	on	802.1q	trunking	999
Gig0/2	on	802.1q	trunking	999

Port Vlans allowed on trunk

Fa0/1	1-1005
Fa0/3	1-1005
Gig0/1	1-1005
Gig0/2	1-1005

Port Vlans allowed and active in management domain

Fa0/1	1,10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999
Fa0/3	1,10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999
Gig0/1	1,10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999
Gig0/2	1,10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999

Port Vlans in spanning tree forwarding state and not pruned

Fa0/1	10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999
Fa0/3	10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999
Gig0/1	1,10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999
Gig0/2	11,22,25,30,31,40,50,99,112,121,132,133,140,142,151,161,173,180,181

```

SW-DIST-L4#show ip interface brief
Interface                IP-Address      OK? Method Status  Protocol
FastEthernet0/1          unassigned      YES unset  up      up
FastEthernet0/2          unassigned      YES unset  down    down
FastEthernet0/3          unassigned      YES unset  up      up
FastEthernet0/4          unassigned      YES unset  down    down
FastEthernet0/5          unassigned      YES unset  down    down
FastEthernet0/6          unassigned      YES unset  down    down
FastEthernet0/7          unassigned      YES unset  down    down
FastEthernet0/8          unassigned      YES unset  down    down
FastEthernet0/9          unassigned      YES unset  down    down
FastEthernet0/10         unassigned      YES unset  down    down
FastEthernet0/11         unassigned      YES unset  down    down
FastEthernet0/12         unassigned      YES unset  down    down
FastEthernet0/13         unassigned      YES unset  down    down
FastEthernet0/14         unassigned      YES unset  down    down
FastEthernet0/15         unassigned      YES unset  down    down
FastEthernet0/16         unassigned      YES unset  down    down
FastEthernet0/17         unassigned      YES unset  down    down
FastEthernet0/18         unassigned      YES unset  down    down
FastEthernet0/19         unassigned      YES unset  down    down
FastEthernet0/20         unassigned      YES unset  down    down
FastEthernet0/21         unassigned      YES unset  down    down
FastEthernet0/22         unassigned      YES unset  down    down
FastEthernet0/23         unassigned      YES unset  down    down
FastEthernet0/24         unassigned      YES unset  down    down
GigabitEthernet0/1       unassigned      YES unset  up      up
GigabitEthernet0/2       unassigned      YES unset  up      up
Vlan1                    unassigned      YES unset  administratively down down
Vlan10                    192.168.10.6   YES manual up       up
SW-DIST-L4#

```

SW-DIST-L5

```

SW-DIST-L5#show vlan brief

VLAN Name                Status    Ports
-----
1    default              active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                           Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                           Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                           Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                           Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                           Fa0/23, Fa0/24
10   TRANSPORT-CORE       active
11   VLAN0011              active
20   SERV-BACKOFFICE       active
21   SERV-DNS              active
22   SERV-DHCP             active
23   SERV-MAIL             active
24   SERV-NTP              active
25   SERV-WEB              active
30   SRV-DESARROLLO        active
31   SRV-QA                active
32   SRV-PREPROD           active
33   SRV-PRODUCCION        active
40   WIFI-CORP             active
50   WIFI-VISIT            active
99   MGMT-OOB              active
110  Dept-Showroom         active
111  CCTV-Showroom       active
112  WIFI-C-SHOWROOM       active
113  WIFI-V-SHOWROOM       active
120  DEPT-VENTAS           active
121  CCTV-VENTAS         active
122  WIFI-C-VENTAS         active
123  WIFI-V-VENTAS         active
130  DEPT-AUDITORIO        active
131  CCTV-AUDITORIO      active
132  WIFI-C-AUDITORIO      active
133  WIFI-V-AUDITORIO      active
140  DEPT-II               active
141  CCTV-II             active
142  WIFI-C-II             active
143  WIFI-V-II             active
150  DEPT-RRHH             active
151  CCTV-RRHH           active
152  WIFI-C-RRHH           active
153  WIFI-V-RRHH           active
160  DEPT-MARKETING         active
161  CCTV-MARKETING       active
162  WIFI-C-MKT            active
163  WIFI-V-MKT            active
170  DEPT-AUDITORIA        active
171  CCTV-AUDITORIA      active
172  WIFI-C-AUDITORIA      active
173  WIFI-V-AUDITORIA      active
180  DEPT-FINANZAS         active
181  CCTV-FINANZAS       active
182  WIFI-C-FINANZAS       active
183  WIFI-V-FINANZAS       active
999  NATIVE-UNUSED         active
1002 fddi-default          active
1003 token-ring-default   active
1004 fddinet-default       active
1005 trnet-default        active
SW-DIST-L5#
SW-DIST-L5#

```

```

SW-DIST-L5#show standby brief
                P indicates configured to preempt.
                |
Interface  Grp  Pri P State      Active      Standby      Virtual IP
sw-NTST-15+

```

```

SW-DIST-L5#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

```

Gateway of last resort is 192.168.10.2 to network 0.0.0.0

```

10.0.0.0/8 is variably subnetted, 31 subnets, 2 masks
O    10.0.0.0/30 [110/2] via 192.168.10.2, 01:20:46, Vlan10
O    10.10.10.0/24 [110/2] via 192.168.10.3, 01:20:46, Vlan10
      [110/2] via 192.168.10.2, 01:20:46, Vlan10
O    10.10.11.0/24 [110/2] via 192.168.10.3, 01:20:46, Vlan10
      [110/2] via 192.168.10.2, 01:20:46, Vlan10
O    10.10.20.0/24 [110/2] via 192.168.10.3, 01:20:46, Vlan10
      [110/2] via 192.168.10.2, 01:20:46, Vlan10
O    10.10.21.0/24 [110/2] via 192.168.10.3, 01:20:46, Vlan10
      [110/2] via 192.168.10.2, 01:20:46, Vlan10
O    10.10.22.0/24 [110/2] via 192.168.10.3, 01:20:46, Vlan10
      [110/2] via 192.168.10.2, 01:20:46, Vlan10
O    10.10.23.0/24 [110/2] via 192.168.10.3, 01:20:46, Vlan10
      [110/2] via 192.168.10.2, 01:20:46, Vlan10
O    10.10.30.0/24 [110/2] via 192.168.10.3, 01:20:46, Vlan10
      [110/2] via 192.168.10.2, 01:20:46, Vlan10
O    10.10.31.0/24 [110/2] via 192.168.10.3, 01:20:46, Vlan10
      [110/2] via 192.168.10.2, 01:20:46, Vlan10
O    10.10.32.0/24 [110/2] via 192.168.10.3, 01:20:46, Vlan10
      [110/2] via 192.168.10.2, 01:20:46, Vlan10
O    10.10.33.0/24 [110/2] via 192.168.10.3, 01:20:46, Vlan10
      [110/2] via 192.168.10.2, 01:20:46, Vlan10
O    10.10.40.0/24 [110/2] via 192.168.10.3, 01:20:46, Vlan10
      [110/2] via 192.168.10.2, 01:20:46, Vlan10
O    10.10.41.0/24 [110/2] via 192.168.10.3, 01:20:46, Vlan10
      [110/2] via 192.168.10.2, 01:20:46, Vlan10
O    10.10.42.0/24 [110/2] via 192.168.10.3, 01:20:46, Vlan10
      [110/2] via 192.168.10.2, 01:20:46, Vlan10
O    10.10.43.0/24 [110/2] via 192.168.10.3, 01:20:46, Vlan10
      [110/2] via 192.168.10.2, 01:20:46, Vlan10
O    10.10.50.0/24 [110/2] via 192.168.10.3, 01:20:46, Vlan10
      [110/2] via 192.168.10.2, 01:20:46, Vlan10
O    10.10.51.0/24 [110/2] via 192.168.10.3, 01:20:46, Vlan10
      [110/2] via 192.168.10.2, 01:20:46, Vlan10
O    10.10.52.0/24 [110/2] via 192.168.10.3, 01:20:46, Vlan10
      [110/2] via 192.168.10.2, 01:20:46, Vlan10
O    10.10.53.0/24 [110/2] via 192.168.10.3, 01:20:46, Vlan10
      [110/2] via 192.168.10.2, 01:20:46, Vlan10
O    10.10.60.0/24 [110/2] via 192.168.10.3, 01:20:46, Vlan10
      [110/2] via 192.168.10.2, 01:20:46, Vlan10
O    10.10.61.0/24 [110/2] via 192.168.10.3, 01:20:46, Vlan10
      [110/2] via 192.168.10.2, 01:20:46, Vlan10
O    10.10.62.0/24 [110/2] via 192.168.10.3, 01:20:46, Vlan10
      [110/2] via 192.168.10.2, 01:20:46, Vlan10
O    10.10.63.0/24 [110/2] via 192.168.10.3, 01:20:46, Vlan10
      [110/2] via 192.168.10.2, 01:20:46, Vlan10
O    10.10.70.0/24 [110/2] via 192.168.10.3, 01:20:46, Vlan10
      [110/2] via 192.168.10.2, 01:20:46, Vlan10
O    10.10.71.0/24 [110/2] via 192.168.10.3, 01:20:46, Vlan10
      [110/2] via 192.168.10.2, 01:20:46, Vlan10
O    10.10.72.0/24 [110/2] via 192.168.10.3, 01:20:46, Vlan10
      [110/2] via 192.168.10.2, 01:20:46, Vlan10
O    10.10.73.0/24 [110/2] via 192.168.10.3, 01:20:46, Vlan10
      [110/2] via 192.168.10.2, 01:20:46, Vlan10
O    10.10.80.0/24 [110/2] via 192.168.10.3, 01:20:46, Vlan10

```

```

O      10.10.81.0/24 [110/2] via 192.168.10.2, 01:20:46, Vlan10
O      10.10.82.0/24 [110/2] via 192.168.10.3, 01:20:46, Vlan10
O      10.10.83.0/24 [110/2] via 192.168.10.2, 01:20:46, Vlan10
O      10.10.83.0/24 [110/2] via 192.168.10.3, 01:20:46, Vlan10
O      10.10.83.0/24 [110/2] via 192.168.10.2, 01:20:46, Vlan10
O E2 172.16.10.0 [110/20] via 192.168.10.2, 00:18:41, Vlan10
O E2 172.16.11.0 [110/20] via 192.168.10.2, 00:18:41, Vlan10
O E2 172.16.12.0 [110/20] via 192.168.10.2, 00:18:41, Vlan10
O E2 172.16.13.0 [110/20] via 192.168.10.2, 00:18:41, Vlan10
O E2 172.16.14.0 [110/20] via 192.168.10.2, 00:18:41, Vlan10
O E2 172.16.15.0 [110/20] via 192.168.10.2, 00:18:41, Vlan10
O E2 172.16.20.0 [110/20] via 192.168.10.2, 01:19:44, Vlan10
O E2 172.16.21.0 [110/20] via 192.168.10.2, 01:19:44, Vlan10
O E2 172.16.22.0 [110/20] via 192.168.10.2, 01:19:44, Vlan10
O E2 172.16.23.0 [110/20] via 192.168.10.2, 01:19:44, Vlan10
O E2 172.16.24.0 [110/20] via 192.168.10.2, 01:19:44, Vlan10
O E2 172.16.25.0 [110/20] via 192.168.10.2, 01:19:44, Vlan10
O E2 172.16.30.0 [110/20] via 192.168.10.2, 01:20:46, Vlan10
O E2 172.16.31.0 [110/20] via 192.168.10.2, 01:20:46, Vlan10
O E2 172.16.32.0 [110/20] via 192.168.10.2, 01:20:46, Vlan10
O E2 172.16.33.0 [110/20] via 192.168.10.2, 01:20:46, Vlan10
O E2 172.16.34.0 [110/20] via 192.168.10.2, 01:20:46, Vlan10
O E2 172.16.35.0 [110/20] via 192.168.10.2, 01:20:46, Vlan10
C 192.168.10.0/24 is directly connected, Vlan10
O 192.168.20.0/24 [110/2] via 192.168.10.2, 01:20:46, Vlan10
O 192.168.21.0/24 [110/2] via 192.168.10.2, 01:20:46, Vlan10
O 192.168.22.0/24 [110/2] via 192.168.10.2, 01:20:46, Vlan10
O 192.168.23.0/24 [110/2] via 192.168.10.2, 01:20:46, Vlan10
O 192.168.24.0/24 [110/2] via 192.168.10.2, 01:20:46, Vlan10
O 192.168.25.0/24 [110/2] via 192.168.10.2, 01:20:46, Vlan10
O 192.168.30.0/24 [110/2] via 192.168.10.2, 01:20:46, Vlan10
O 192.168.31.0/24 [110/2] via 192.168.10.2, 01:20:46, Vlan10
O 192.168.32.0/24 [110/2] via 192.168.10.2, 01:20:46, Vlan10
O 192.168.33.0/24 [110/2] via 192.168.10.2, 01:20:46, Vlan10
O 192.168.99.0/24 [110/2] via 192.168.10.2, 01:20:46, Vlan10
O*E2 0.0.0.0/0 [110/1] via 192.168.10.2, 01:20:46, Vlan10

```

SW-DIST-L5#show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
7.7.7.7	1	2WAY/DROTHER	00:00:32	192.168.10.17	Vlan10
11.11.11.11	1	2WAY/DROTHER	00:00:31	192.168.10.11	Vlan10
192.168.99.2	255	FULL/DR	00:00:31	192.168.10.2	Vlan10
3.3.3.3	1	2WAY/DROTHER	00:00:32	192.168.10.13	Vlan10
2.2.2.2	1	2WAY/DROTHER	00:00:31	192.168.10.12	Vlan10
6.6.6.6	1	2WAY/DROTHER	00:00:39	192.168.10.16	Vlan10
4.4.4.4	1	2WAY/DROTHER	00:00:31	192.168.10.6	Vlan10
192.168.99.3	200	FULL/BDR	00:00:32	192.168.10.3	Vlan10

SW-DIST-L5#

SW-DIST-L5#show interfaces trunk

Port	Mode	Encapsulation	Status	Native vlan
Fa0/1	on	802.1q	trunking	999
Fa0/2	on	802.1q	trunking	999
Gig0/1	on	802.1q	trunking	999
Gig0/2	on	802.1q	trunking	999

Port Vlans allowed on trunk

Port	Vlans allowed on trunk
Fa0/1	1-1005
Fa0/2	10,20,30,40,50,99,150-153,999
Gig0/1	1-1005
Gig0/2	1-1005

Port Vlans allowed and active in management domain

Port	Vlans allowed and active in management domain
Fa0/1	1,10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999
Fa0/2	10,20,30,40,50,99,150,151,152,153,999
Gig0/1	1,10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999
Gig0/2	1,10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999

Port Vlans in spanning tree forwarding state and not pruned

Port	Vlans in spanning tree forwarding state and not pruned
Fa0/1	10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999
Fa0/2	10,20,30,40,50,99,150,151,152,153,999
Gig0/1	1,10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999
Gig0/2	10,11,22,25,30,31,40,50,110,111,121,122,123,131,140,141,142,143,153,160,161,172,173,180,181,182

SW-DIST-L5#

```

SW-DIST-L5#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/1	unassigned	YES	unset	up	up
FastEthernet0/2	unassigned	YES	unset	up	up
FastEthernet0/3	unassigned	YES	unset	down	down
FastEthernet0/4	unassigned	YES	unset	down	down
FastEthernet0/5	unassigned	YES	unset	down	down
FastEthernet0/6	unassigned	YES	unset	down	down
FastEthernet0/7	unassigned	YES	unset	down	down
FastEthernet0/8	unassigned	YES	unset	down	down
FastEthernet0/9	unassigned	YES	unset	down	down
FastEthernet0/10	unassigned	YES	unset	down	down
FastEthernet0/11	unassigned	YES	unset	down	down
FastEthernet0/12	unassigned	YES	unset	down	down
FastEthernet0/13	unassigned	YES	unset	down	down
FastEthernet0/14	unassigned	YES	unset	down	down
FastEthernet0/15	unassigned	YES	unset	down	down
FastEthernet0/16	unassigned	YES	unset	down	down
FastEthernet0/17	unassigned	YES	unset	down	down
FastEthernet0/18	unassigned	YES	unset	down	down
FastEthernet0/19	unassigned	YES	unset	down	down
FastEthernet0/20	unassigned	YES	unset	down	down
FastEthernet0/21	unassigned	YES	unset	down	down
FastEthernet0/22	unassigned	YES	unset	down	down
FastEthernet0/23	unassigned	YES	unset	down	down
FastEthernet0/24	unassigned	YES	unset	down	down
GigabitEthernet0/1	unassigned	YES	unset	up	up
GigabitEthernet0/2	unassigned	YES	unset	up	up
Vlan1	unassigned	YES	unset	administratively down	down
Vlan10	192.168.10.15	YES	manual	up	up

```

SW-DIST-L5#

```

SW-DIST-L6

```

SW-DIST-L6#show vlan brief

```

VLAN Name	Status	Ports
1 default	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24
10 TRANSPORT-CORE	active	
11 VLAN0011	active	
20 SERV-BACKOFFICE	active	
21 SERV-DNS	active	
22 SERV-DHCP	active	
23 SERV-MAIL	active	
24 SERV-NTP	active	
25 SERV-WEB	active	
30 SRV-DESARROLLO	active	
31 SRV-QA	active	
32 SRV-PREPROD	active	
33 SRV-PRODUCCION	active	
40 WIFI-CORP	active	
50 WIFI-VISIT	active	
99 MGMT-OOB	active	
110 Dept-Showroom	active	
111 CCTV-Showroom	active	
112 WIFI-C-SHOWROOM	active	
113 WIFI-V-SHOWROOM	active	
120 DEPT-VENTAS	active	
121 CCTV-VENTAS	active	
122 WIFI-C-VENTAS	active	
123 WIFI-V-VENTAS	active	
130 DEPT-AUDITORIO	active	
131 CCTV-AUDITORIO	active	
132 WIFI-C-AUDITORIO	active	
133 WIFI-V-AUDITORIO	active	
140 DEPT-IT	active	
141 CCTV-IT	active	
142 WIFI-C-IT	active	
143 WIFI-V-IT	active	
150 DEPT-RRHH	active	
151 CCTV-RRHH	active	
152 WIFI-C-RRHH	active	
153 WIFI-V-RRHH	active	
160 DEPT-MARKETING	active	
161 CCTV-MARKETING	active	
162 WIFI-C-MKT	active	
163 WIFI-V-MKT	active	
170 DEPT-AUDITORIA	active	
171 CCTV-AUDITORIA	active	
172 WIFI-C-AUDITORIA	active	
173 WIFI-V-AUDITORIA	active	
180 DEPT-FINANZAS	active	
181 CCTV-FINANZAS	active	
182 WIFI-C-FINANZAS	active	
183 WIFI-V-FINANZAS	active	
999 NATIVE-UNUSED	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```

SW-DIST-L6#
SW-DIST-L6#

```

```

SW-DIST-L6#show standby brief
                P indicates configured to preempt.
                |
Interface  Grp  Pri  P State      Active      Standby      Virtual IP
SW-DIST-L6#
SW-DIST-L6#
SW-DIST-L6#

```

```

SW-DIST-L6#show standby brief
                P indicates configured to preempt.
                |
Interface  Grp  Pri  P State      Active      Standby      Virtual IP
SW-DIST-L6#
SW-DIST-L6#
SW-DIST-L6#
SW-DIST-L6#show ip ro
SW-DIST-L6#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

```

Gateway of last resort is 192.168.10.2 to network 0.0.0.0

```

10.0.0.0/8 is variably subnetted, 31 subnets, 2 masks
O    10.0.0.0/30 [110/2] via 192.168.10.2, 01:22:20, Vlan10
O    10.10.10.0/24 [110/2] via 192.168.10.3, 01:22:10, Vlan10
      [110/2] via 192.168.10.2, 01:22:10, Vlan10
O    10.10.11.0/24 [110/2] via 192.168.10.3, 01:22:10, Vlan10
      [110/2] via 192.168.10.2, 01:22:10, Vlan10
O    10.10.20.0/24 [110/2] via 192.168.10.3, 01:22:10, Vlan10
      [110/2] via 192.168.10.2, 01:22:10, Vlan10
O    10.10.21.0/24 [110/2] via 192.168.10.3, 01:22:10, Vlan10
      [110/2] via 192.168.10.2, 01:22:10, Vlan10
O    10.10.22.0/24 [110/2] via 192.168.10.3, 01:22:10, Vlan10
      [110/2] via 192.168.10.2, 01:22:10, Vlan10
O    10.10.23.0/24 [110/2] via 192.168.10.3, 01:22:10, Vlan10
      [110/2] via 192.168.10.2, 01:22:10, Vlan10
O    10.10.30.0/24 [110/2] via 192.168.10.3, 01:22:10, Vlan10
      [110/2] via 192.168.10.2, 01:22:10, Vlan10
O    10.10.31.0/24 [110/2] via 192.168.10.3, 01:22:10, Vlan10
      [110/2] via 192.168.10.2, 01:22:10, Vlan10
O    10.10.32.0/24 [110/2] via 192.168.10.3, 01:22:10, Vlan10
      [110/2] via 192.168.10.2, 01:22:10, Vlan10
O    10.10.33.0/24 [110/2] via 192.168.10.3, 01:22:10, Vlan10
      [110/2] via 192.168.10.2, 01:22:10, Vlan10
O    10.10.40.0/24 [110/2] via 192.168.10.3, 01:22:10, Vlan10
      [110/2] via 192.168.10.2, 01:22:10, Vlan10
O    10.10.41.0/24 [110/2] via 192.168.10.3, 01:22:10, Vlan10
      [110/2] via 192.168.10.2, 01:22:10, Vlan10
O    10.10.42.0/24 [110/2] via 192.168.10.3, 01:22:10, Vlan10
      [110/2] via 192.168.10.2, 01:22:10, Vlan10
O    10.10.43.0/24 [110/2] via 192.168.10.3, 01:22:10, Vlan10
      [110/2] via 192.168.10.2, 01:22:10, Vlan10
O    10.10.50.0/24 [110/2] via 192.168.10.3, 01:22:10, Vlan10
      [110/2] via 192.168.10.2, 01:22:10, Vlan10
O    10.10.51.0/24 [110/2] via 192.168.10.3, 01:22:10, Vlan10
      [110/2] via 192.168.10.2, 01:22:10, Vlan10
O    10.10.52.0/24 [110/2] via 192.168.10.3, 01:22:10, Vlan10
      [110/2] via 192.168.10.2, 01:22:10, Vlan10
O    10.10.53.0/24 [110/2] via 192.168.10.3, 01:22:10, Vlan10
      [110/2] via 192.168.10.2, 01:22:10, Vlan10
O    10.10.60.0/24 [110/2] via 192.168.10.3, 01:22:10, Vlan10
      [110/2] via 192.168.10.2, 01:22:10, Vlan10
O    10.10.61.0/24 [110/2] via 192.168.10.3, 01:22:10, Vlan10
      [110/2] via 192.168.10.2, 01:22:10, Vlan10
O    10.10.62.0/24 [110/2] via 192.168.10.3, 01:22:10, Vlan10
      [110/2] via 192.168.10.2, 01:22:10, Vlan10
O    10.10.63.0/24 [110/2] via 192.168.10.3, 01:22:10, Vlan10
      [110/2] via 192.168.10.2, 01:22:10, Vlan10
O    10.10.70.0/24 [110/2] via 192.168.10.3, 01:22:10, Vlan10
      [110/2] via 192.168.10.2, 01:22:10, Vlan10

```

```

O      10.10.71.0/24 [110/2] via 192.168.10.3, 01:22:10, Vlan10
      [110/2] via 192.168.10.2, 01:22:10, Vlan10
O      10.10.72.0/24 [110/2] via 192.168.10.3, 01:22:10, Vlan10
      [110/2] via 192.168.10.2, 01:22:10, Vlan10
O      10.10.73.0/24 [110/2] via 192.168.10.3, 01:22:10, Vlan10
      [110/2] via 192.168.10.2, 01:22:10, Vlan10
O      10.10.80.0/24 [110/2] via 192.168.10.3, 01:22:10, Vlan10
      [110/2] via 192.168.10.2, 01:22:10, Vlan10
O      10.10.81.0/24 [110/2] via 192.168.10.3, 01:22:10, Vlan10
      [110/2] via 192.168.10.2, 01:22:10, Vlan10
O      10.10.82.0/24 [110/2] via 192.168.10.3, 01:22:10, Vlan10
      [110/2] via 192.168.10.2, 01:22:10, Vlan10
O      10.10.83.0/24 [110/2] via 192.168.10.3, 01:22:10, Vlan10
      [110/2] via 192.168.10.2, 01:22:10, Vlan10
172.16.0.0/24 is subnetted, 18 subnets
O E2   172.16.10.0 [110/20] via 192.168.10.2, 00:20:11, Vlan10
O E2   172.16.11.0 [110/20] via 192.168.10.2, 00:20:11, Vlan10
O E2   172.16.12.0 [110/20] via 192.168.10.2, 00:20:11, Vlan10
O E2   172.16.13.0 [110/20] via 192.168.10.2, 00:20:11, Vlan10
O E2   172.16.14.0 [110/20] via 192.168.10.2, 00:20:11, Vlan10
O E2   172.16.15.0 [110/20] via 192.168.10.2, 00:20:11, Vlan10
O E2   172.16.20.0 [110/20] via 192.168.10.2, 01:21:14, Vlan10
O E2   172.16.21.0 [110/20] via 192.168.10.2, 01:21:14, Vlan10
O E2   172.16.22.0 [110/20] via 192.168.10.2, 01:21:14, Vlan10
O E2   172.16.23.0 [110/20] via 192.168.10.2, 01:21:14, Vlan10
O E2   172.16.24.0 [110/20] via 192.168.10.2, 01:21:14, Vlan10
O E2   172.16.25.0 [110/20] via 192.168.10.2, 01:21:14, Vlan10
O E2   172.16.30.0 [110/20] via 192.168.10.2, 01:22:20, Vlan10
O E2   172.16.31.0 [110/20] via 192.168.10.2, 01:22:20, Vlan10
O E2   172.16.32.0 [110/20] via 192.168.10.2, 01:22:20, Vlan10
O E2   172.16.33.0 [110/20] via 192.168.10.2, 01:22:20, Vlan10
O E2   172.16.34.0 [110/20] via 192.168.10.2, 01:22:20, Vlan10
O E2   172.16.35.0 [110/20] via 192.168.10.2, 01:22:20, Vlan10
C      192.168.10.0/24 is directly connected, Vlan10
O      192.168.20.0/24 [110/2] via 192.168.10.2, 01:22:20, Vlan10
O      192.168.21.0/24 [110/2] via 192.168.10.2, 01:22:20, Vlan10
O      192.168.22.0/24 [110/2] via 192.168.10.2, 01:22:20, Vlan10
O      192.168.23.0/24 [110/2] via 192.168.10.2, 01:22:20, Vlan10
O      192.168.24.0/24 [110/2] via 192.168.10.2, 01:22:20, Vlan10
O      192.168.25.0/24 [110/2] via 192.168.10.2, 01:22:20, Vlan10
O      192.168.30.0/24 [110/2] via 192.168.10.2, 01:22:20, Vlan10
O      192.168.31.0/24 [110/2] via 192.168.10.2, 01:22:20, Vlan10
O      192.168.32.0/24 [110/2] via 192.168.10.2, 01:22:20, Vlan10
O      192.168.33.0/24 [110/2] via 192.168.10.2, 01:22:20, Vlan10
O      192.168.99.0/24 [110/2] via 192.168.10.2, 01:22:20, Vlan10
O*E2   0.0.0.0/0 [110/1] via 192.168.10.2, 01:22:20, Vlan10

```

SW-DIST-L6#show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.99.3	200	FULL/BDR	00:00:37	192.168.10.3	Vlan10
5.5.5.5	1	2WAY/DROTHER	00:00:36	192.168.10.15	Vlan10
11.11.11.11	1	2WAY/DROTHER	00:00:36	192.168.10.11	Vlan10
192.168.99.2	255	FULL/DR	00:00:36	192.168.10.2	Vlan10
3.3.3.3	1	2WAY/DROTHER	00:00:37	192.168.10.13	Vlan10
4.4.4.4	1	2WAY/DROTHER	00:00:36	192.168.10.6	Vlan10
2.2.2.2	1	2WAY/DROTHER	00:00:35	192.168.10.12	Vlan10
7.7.7.7	1	2WAY/DROTHER	00:00:36	192.168.10.17	Vlan10

SW-DIST-L6#

SW-DIST-L6#show interfaces trunk

Port	Mode	Encapsulation	Status	Native vlan
Fa0/1	on	802.1q	trunking	999
Gig0/1	on	802.1q	trunking	999
Gig0/2	on	802.1q	trunking	999

Port Vlans allowed on trunk

Fa0/1	1-1005
Gig0/1	1-1005
Gig0/2	1-1005

Port Vlans allowed and active in management domain

Fa0/1	1,10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999
Gig0/1	1,10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999
Gig0/2	1,10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999

Port Vlans in spanning tree forwarding state and not pruned

Fa0/1	1,10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999
Gig0/1	1,10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999
Gig0/2	1,11,31,40,50,161

```

SW-DIST-L6#show ip interface brief
-----
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/1          unassigned      YES unset   up          up
FastEthernet0/2          unassigned      YES unset   down        down
FastEthernet0/3          unassigned      YES unset   down        down
FastEthernet0/4          unassigned      YES unset   down        down
FastEthernet0/5          unassigned      YES unset   down        down
FastEthernet0/6          unassigned      YES unset   down        down
FastEthernet0/7          unassigned      YES unset   down        down
FastEthernet0/8          unassigned      YES unset   down        down
FastEthernet0/9          unassigned      YES unset   down        down
FastEthernet0/10         unassigned      YES unset   down        down
FastEthernet0/11         unassigned      YES unset   down        down
FastEthernet0/12         unassigned      YES unset   down        down
FastEthernet0/13         unassigned      YES unset   down        down
FastEthernet0/14         unassigned      YES unset   down        down
FastEthernet0/15         unassigned      YES unset   down        down
FastEthernet0/16         unassigned      YES unset   down        down
FastEthernet0/17         unassigned      YES unset   down        down
FastEthernet0/18         unassigned      YES unset   down        down
FastEthernet0/19         unassigned      YES unset   down        down
FastEthernet0/20         unassigned      YES unset   down        down
FastEthernet0/21         unassigned      YES unset   down        down
FastEthernet0/22         unassigned      YES unset   down        down
FastEthernet0/23         unassigned      YES unset   down        down
FastEthernet0/24         unassigned      YES unset   down        down
GigabitEthernet0/1       unassigned      YES unset   up          up
GigabitEthernet0/2       unassigned      YES unset   up          up
Vlan1                    unassigned      YES unset   administratively down down
Vlan10                   192.168.10.16  YES manual  up          up
SW-DIST-L6#

```

SW-DIST-L7

```

SW-DIST-L7#show vlan brief
-----
VLAN Name                Status      Ports
-----
1    default              active      Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                           Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                           Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                           Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                           Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                           Fa0/22, Fa0/23, Fa0/24
10   TRANSPORT-CORE       active
11   VLAN0011              active
20   SERV-BACKOFFICE       active
21   SERV-DNS              active
22   SERV-DHCP             active
23   SERV-MAIL             active
24   SERV-NTP              active
25   SERV-WEB              active
30   SRV-DESARROLLO        active
31   SRV-QA                active
32   SRV-PREPROD           active
33   SRV-PRODUCCION        active
40   WIFI-CORP             active
50   WIFI-VISIT           active
99   MGMT-OOB              active
110  Dept-Showroom         active
111  CCTV-Showroom       active
112  WIFI-C-SHOWROOM       active
113  WIFI-V-SHOWROOM       active
120  DEPT-VENTAS           active
121  CCTV-VENTAS         active
122  WIFI-C-VENTAS         active
123  WIFI-V-VENTAS         active
130  DEPT-AUDITORIO        active
131  CCTV-AUDITORIO       active
132  WIFI-C-AUDITORIO       active
133  WIFI-V-AUDITORIO       active
140  DEPT-IT               active
141  CCTV-IT             active
142  WIFI-C-IT             active
143  WIFI-V-IT             active
150  DEPT-RRHH             active
151  CCTV-RRHH           active
152  WIFI-C-RRHH           active
153  WIFI-V-RRHH           active
160  DEPT-MARKETING         active
161  CCTV-MARKETING       active
162  WIFI-C-MKT            active
163  WIFI-V-MKT            active
170  DEPT-AUDITORIA         active
171  CCTV-AUDITORIA       active
172  WIFI-C-AUDITORIA       active
173  WIFI-V-AUDITORIA       active
180  DEPT-FINANZAS          active
181  CCTV-FINANZAS        active
182  WIFI-C-FINANZAS        active
183  WIFI-V-FINANZAS        active
999  NATIVE-UNUSED         active
1002 fddi-default           active
1003 token-ring-default     active
1004 fddinet-default        active
1005 trnet-default          active

```

```

SW-DIST-L7#show standby brief
                P indicates configured to preempt.
                |
Interface  Grp  Pri  P State      Active      Standby      Virtual IP
SW-DIST-L7#

```

```

SW-DIST-L7#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

```

Gateway of last resort is 192.168.10.2 to network 0.0.0.0

```

10.0.0.0/8 is variably subnetted, 31 subnets, 2 masks
O    10.0.0.0/30 [110/2] via 192.168.10.2, 01:23:45, Vlan10
O    10.10.10.0/24 [110/2] via 192.168.10.3, 01:23:45, Vlan10
O    [110/2] via 192.168.10.2, 01:23:45, Vlan10
O    10.10.11.0/24 [110/2] via 192.168.10.3, 01:23:45, Vlan10
O    [110/2] via 192.168.10.2, 01:23:45, Vlan10
O    10.10.20.0/24 [110/2] via 192.168.10.3, 01:23:45, Vlan10
O    [110/2] via 192.168.10.2, 01:23:45, Vlan10
O    10.10.21.0/24 [110/2] via 192.168.10.3, 01:23:45, Vlan10
O    [110/2] via 192.168.10.2, 01:23:45, Vlan10
O    10.10.22.0/24 [110/2] via 192.168.10.3, 01:23:45, Vlan10
O    [110/2] via 192.168.10.2, 01:23:45, Vlan10
O    10.10.23.0/24 [110/2] via 192.168.10.3, 01:23:45, Vlan10
O    [110/2] via 192.168.10.2, 01:23:45, Vlan10
O    10.10.30.0/24 [110/2] via 192.168.10.3, 01:23:45, Vlan10
O    [110/2] via 192.168.10.2, 01:23:45, Vlan10
O    10.10.31.0/24 [110/2] via 192.168.10.3, 01:23:45, Vlan10
O    [110/2] via 192.168.10.2, 01:23:45, Vlan10
O    10.10.32.0/24 [110/2] via 192.168.10.3, 01:23:45, Vlan10
O    [110/2] via 192.168.10.2, 01:23:45, Vlan10
O    10.10.33.0/24 [110/2] via 192.168.10.3, 01:23:45, Vlan10
O    [110/2] via 192.168.10.2, 01:23:45, Vlan10
O    10.10.40.0/24 [110/2] via 192.168.10.3, 01:23:45, Vlan10
O    [110/2] via 192.168.10.2, 01:23:45, Vlan10
O    10.10.41.0/24 [110/2] via 192.168.10.3, 01:23:45, Vlan10
O    [110/2] via 192.168.10.2, 01:23:45, Vlan10
O    10.10.42.0/24 [110/2] via 192.168.10.3, 01:23:45, Vlan10
O    [110/2] via 192.168.10.2, 01:23:45, Vlan10
O    10.10.43.0/24 [110/2] via 192.168.10.3, 01:23:45, Vlan10
O    [110/2] via 192.168.10.2, 01:23:45, Vlan10
O    10.10.50.0/24 [110/2] via 192.168.10.3, 01:23:45, Vlan10
O    [110/2] via 192.168.10.2, 01:23:45, Vlan10
O    10.10.51.0/24 [110/2] via 192.168.10.3, 01:23:45, Vlan10
O    [110/2] via 192.168.10.2, 01:23:45, Vlan10
O    10.10.52.0/24 [110/2] via 192.168.10.3, 01:23:45, Vlan10
O    [110/2] via 192.168.10.2, 01:23:45, Vlan10
O    10.10.53.0/24 [110/2] via 192.168.10.3, 01:23:45, Vlan10
O    [110/2] via 192.168.10.2, 01:23:45, Vlan10
O    10.10.60.0/24 [110/2] via 192.168.10.3, 01:23:45, Vlan10
O    [110/2] via 192.168.10.2, 01:23:45, Vlan10
O    10.10.61.0/24 [110/2] via 192.168.10.3, 01:23:45, Vlan10
O    [110/2] via 192.168.10.2, 01:23:45, Vlan10
O    10.10.62.0/24 [110/2] via 192.168.10.3, 01:23:45, Vlan10
O    [110/2] via 192.168.10.2, 01:23:45, Vlan10
O    10.10.63.0/24 [110/2] via 192.168.10.3, 01:23:45, Vlan10
O    [110/2] via 192.168.10.2, 01:23:45, Vlan10
O    10.10.70.0/24 [110/2] via 192.168.10.3, 01:23:45, Vlan10
O    [110/2] via 192.168.10.2, 01:23:45, Vlan10
O    10.10.71.0/24 [110/2] via 192.168.10.3, 01:23:45, Vlan10
O    [110/2] via 192.168.10.2, 01:23:45, Vlan10
O    10.10.72.0/24 [110/2] via 192.168.10.3, 01:23:45, Vlan10
O    [110/2] via 192.168.10.2, 01:23:45, Vlan10
O    10.10.73.0/24 [110/2] via 192.168.10.3, 01:23:45, Vlan10
O    [110/2] via 192.168.10.2, 01:23:45, Vlan10
O    10.10.80.0/24 [110/2] via 192.168.10.3, 01:23:45, Vlan10
O    [110/2] via 192.168.10.2, 01:23:45, Vlan10

```

```

O      10.10.81.0/24 [110/2] via 192.168.10.3, 01:23:45, Vlan10
      [110/2] via 192.168.10.2, 01:23:45, Vlan10
O      10.10.82.0/24 [110/2] via 192.168.10.3, 01:23:45, Vlan10
      [110/2] via 192.168.10.2, 01:23:45, Vlan10
O      10.10.83.0/24 [110/2] via 192.168.10.3, 01:23:45, Vlan10
      [110/2] via 192.168.10.2, 01:23:45, Vlan10
172.16.0.0/24 is subnetted, 18 subnets
O E2 172.16.10.0 [110/20] via 192.168.10.2, 00:21:30, Vlan10
O E2 172.16.11.0 [110/20] via 192.168.10.2, 00:21:30, Vlan10
O E2 172.16.12.0 [110/20] via 192.168.10.2, 00:21:30, Vlan10
O E2 172.16.13.0 [110/20] via 192.168.10.2, 00:21:30, Vlan10
O E2 172.16.14.0 [110/20] via 192.168.10.2, 00:21:30, Vlan10
O E2 172.16.15.0 [110/20] via 192.168.10.2, 00:21:30, Vlan10
O E2 172.16.20.0 [110/20] via 192.168.10.2, 01:22:33, Vlan10
O E2 172.16.21.0 [110/20] via 192.168.10.2, 01:22:33, Vlan10
O E2 172.16.22.0 [110/20] via 192.168.10.2, 01:22:33, Vlan10
O E2 172.16.23.0 [110/20] via 192.168.10.2, 01:22:33, Vlan10
O E2 172.16.24.0 [110/20] via 192.168.10.2, 01:22:33, Vlan10
O E2 172.16.25.0 [110/20] via 192.168.10.2, 01:22:33, Vlan10
O E2 172.16.30.0 [110/20] via 192.168.10.2, 01:23:45, Vlan10
O E2 172.16.31.0 [110/20] via 192.168.10.2, 01:23:45, Vlan10
O E2 172.16.32.0 [110/20] via 192.168.10.2, 01:23:45, Vlan10
O E2 172.16.33.0 [110/20] via 192.168.10.2, 01:23:45, Vlan10
O E2 172.16.34.0 [110/20] via 192.168.10.2, 01:23:45, Vlan10
O E2 172.16.35.0 [110/20] via 192.168.10.2, 01:23:45, Vlan10
C 192.168.10.0/24 is directly connected, Vlan10
O 192.168.20.0/24 [110/2] via 192.168.10.2, 01:23:45, Vlan10
O 192.168.21.0/24 [110/2] via 192.168.10.2, 01:23:45, Vlan10
O 192.168.22.0/24 [110/2] via 192.168.10.2, 01:23:45, Vlan10
O 192.168.23.0/24 [110/2] via 192.168.10.2, 01:23:45, Vlan10
O 192.168.24.0/24 [110/2] via 192.168.10.2, 01:23:45, Vlan10
O 192.168.25.0/24 [110/2] via 192.168.10.2, 01:23:45, Vlan10
O 192.168.30.0/24 [110/2] via 192.168.10.2, 01:23:45, Vlan10
O 192.168.31.0/24 [110/2] via 192.168.10.2, 01:23:45, Vlan10
O 192.168.32.0/24 [110/2] via 192.168.10.2, 01:23:45, Vlan10
O 192.168.33.0/24 [110/2] via 192.168.10.2, 01:23:45, Vlan10
O 192.168.99.0/24 [110/2] via 192.168.10.2, 01:23:45, Vlan10
O*E2 0.0.0.0/0 [110/1] via 192.168.10.2, 01:23:45, Vlan10

```

SW-DIST-L7#

SW-DIST-L7#show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.99.3	200	FULL/BDR	00:00:32	192.168.10.3	Vlan10
5.5.5.5	1	2WAY/DROTHER	00:00:32	192.168.10.15	Vlan10
11.11.11.11	1	2WAY/DROTHER	00:00:32	192.168.10.11	Vlan10
192.168.99.2	255	FULL/DR	00:00:31	192.168.10.2	Vlan10
3.3.3.3	1	2WAY/DROTHER	00:00:32	192.168.10.13	Vlan10
6.6.6.6	1	2WAY/DROTHER	00:00:39	192.168.10.16	Vlan10
2.2.2.2	1	2WAY/DROTHER	00:00:31	192.168.10.12	Vlan10
4.4.4.4	1	2WAY/DROTHER	00:00:31	192.168.10.6	Vlan10

SW-DIST-L7#

SW-DIST-L7#show interfaces trunk

Port	Mode	Encapsulation	Status	Native vlan
Fa0/1	on	802.1q	trunking	999
Gig0/1	on	802.1q	trunking	999
Gig0/2	on	802.1q	trunking	999

Port Vlans allowed on trunk

Fa0/1 1-1005
Gig0/1 1-1005
Gig0/2 1-1005

Port Vlans allowed and active in management domain

Fa0/1
1,10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999
Gig0/1
1,10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999
Gig0/2
1,10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999

Port Vlans in spanning tree forwarding state and not pruned

Fa0/1
10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999
Gig0/1
1,10,11,20,21,22,23,24,25,30,31,32,33,40,50,99,110,111,112,113,120,121,122,123,130,131,132,133,140,141,142,143,150,151,152,153,160,161,162,163,170,171,172,173,180,181,182,183,999
Gig0/2
10,11,20,21,24,25,30,31,40,50,99,112,113,120,122,130,140,141,142,143,150,151,152,160,161,163,170,173,181

SW-DIST-L7#

SW-DIST-L7#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/1	unassigned	YES	unset	up	up
FastEthernet0/2	unassigned	YES	unset	down	down
FastEthernet0/3	unassigned	YES	unset	down	down
FastEthernet0/4	unassigned	YES	unset	down	down
FastEthernet0/5	unassigned	YES	unset	down	down
FastEthernet0/6	unassigned	YES	unset	down	down
FastEthernet0/7	unassigned	YES	unset	down	down
FastEthernet0/8	unassigned	YES	unset	down	down
FastEthernet0/9	unassigned	YES	unset	down	down
FastEthernet0/10	unassigned	YES	unset	down	down
FastEthernet0/11	unassigned	YES	unset	down	down
FastEthernet0/12	unassigned	YES	unset	down	down
FastEthernet0/13	unassigned	YES	unset	down	down
FastEthernet0/14	unassigned	YES	unset	down	down
FastEthernet0/15	unassigned	YES	unset	down	down
FastEthernet0/16	unassigned	YES	unset	down	down
FastEthernet0/17	unassigned	YES	unset	down	down
FastEthernet0/18	unassigned	YES	unset	down	down
FastEthernet0/19	unassigned	YES	unset	down	down
FastEthernet0/20	unassigned	YES	unset	down	down
FastEthernet0/21	unassigned	YES	unset	down	down
FastEthernet0/22	unassigned	YES	unset	down	down
FastEthernet0/23	unassigned	YES	unset	down	down
FastEthernet0/24	unassigned	YES	unset	down	down
GigabitEthernet0/1	unassigned	YES	unset	up	up
GigabitEthernet0/2	unassigned	YES	unset	up	up
Vlan1	unassigned	YES	unset	administratively down	down
Vlan10	192.168.10.17	YES	manual	up	up

SW-ACC-SHW-01

SW-ACC-SHW-01#show vlan brief

VLAN	Name	Status	Ports
1	default	active	Gig0/1, Gig0/2
10	TRANSPORT-CORE	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24
11	VLAN0011	active	
20	SERV-BACKOFFICE	active	
21	SERV-DNS	active	
22	SERV-DHCP	active	
23	SERV-MAIL	active	
24	SERV-NTP	active	
25	SERV-WEB	active	
30	SRV-DESARROLLO	active	
31	SRV-QA	active	
32	SRV-PREPROD	active	
33	SRV-PRODUCCION	active	
40	WIFI-CORP	active	
50	WIFI-VISIT	active	
99	MGMT-OOB	active	
110	Dept-Showroom	active	Fa0/2
111	CCTV-Showroom	active	Fa0/3
112	WIFI-C-SHOWROOM	active	
113	WIFI-V-SHOWROOM	active	
120	DEPT-VENTAS	active	
121	CCTV-VENTAS	active	
122	WIFI-C-VENTAS	active	
123	WIFI-V-VENTAS	active	
130	DEPT-AUDITORIO	active	
131	CCTV-AUDITORIO	active	
132	WIFI-C-AUDITORIO	active	
133	WIFI-V-AUDITORIO	active	
140	DEPT-IT	active	
141	CCTV-IT	active	
142	WIFI-C-IT	active	
143	WIFI-V-IT	active	
150	DEPT-RRHH	active	
151	CCTV-RRHH	active	
152	WIFI-C-RRHH	active	
153	WIFI-V-RRHH	active	
160	DEPT-MARKETING	active	
161	CCTV-MARKETING	active	
162	WIFI-C-MKT	active	
163	WIFI-V-MKT	active	
170	DEPT-AUDITORIA	active	
171	CCTV-AUDITORIA	active	
172	WIFI-C-AUDITORIA	active	
173	WIFI-V-AUDITORIA	active	
180	DEPT-FINANZAS	active	
181	CCTV-FINANZAS	active	
182	WIFI-C-FINANZAS	active	
183	WIFI-V-FINANZAS	active	
999	NATIVE-UNUSED	active	
1002	fdi-default	active	
1003	token-ring-default	active	
1004	fdinet-default	active	
1005	trnet-default	active	

SW-ACC-SHW-01#
SW-ACC-SHW-01#
SW-ACC-SHW-01#
SW-ACC-SHW-01#
SW-ACC-SHW-01#

SW-ACC-SHW-01#show interfaces status

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1	Enlace_Trunk_Hacia	connected	trunk	a-full	a-100	10/100BaseTX
Fa0/2		connected	110	a-full	a-100	10/100BaseTX
Fa0/3		connected	111	a-full	a-100	10/100BaseTX
Fa0/4		notconnect	10	auto	auto	10/100BaseTX
Fa0/5		notconnect	10	auto	auto	10/100BaseTX
Fa0/6		notconnect	10	auto	auto	10/100BaseTX
Fa0/7		notconnect	10	auto	auto	10/100BaseTX
Fa0/8		notconnect	10	auto	auto	10/100BaseTX
Fa0/9		notconnect	10	auto	auto	10/100BaseTX
Fa0/10		notconnect	10	auto	auto	10/100BaseTX
Fa0/11		notconnect	10	auto	auto	10/100BaseTX
Fa0/12		notconnect	10	auto	auto	10/100BaseTX
Fa0/13		notconnect	10	auto	auto	10/100BaseTX
Fa0/14		notconnect	10	auto	auto	10/100BaseTX
Fa0/15		notconnect	10	auto	auto	10/100BaseTX
Fa0/16		notconnect	10	auto	auto	10/100BaseTX
Fa0/17		notconnect	10	auto	auto	10/100BaseTX
Fa0/18		notconnect	10	auto	auto	10/100BaseTX
Fa0/19		notconnect	10	auto	auto	10/100BaseTX
Fa0/20		notconnect	10	auto	auto	10/100BaseTX
Fa0/21		notconnect	10	auto	auto	10/100BaseTX
Fa0/22		notconnect	10	auto	auto	10/100BaseTX
Fa0/23		notconnect	10	auto	auto	10/100BaseTX
Fa0/24		notconnect	10	auto	auto	10/100BaseTX
Gig0/1		notconnect	1	auto	auto	10/100/1000BaseTX
Gig0/2		notconnect	1	auto	auto	10/100/1000BaseTX

SW-ACC-SHW-01#show spanning-tree

VLAN0099

Spanning tree enabled protocol rstp

Root ID Priority 4195
 Address 000A.F354.D2B3
 Cost 23
 Port 1(FastEthernet0/1)
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32867 (priority 32768 sys-id-ext 99)
 Address 0001.C738.139A
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
 Aging Time 20

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Root	FWD	19	128.1	P2p

VLAN0110

Spanning tree enabled protocol rstp

Root ID Priority 4206
 Address 000A.F354.D2B3
 Cost 23
 Port 1(FastEthernet0/1)
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32878 (priority 32768 sys-id-ext 110)
 Address 0001.C738.139A
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
 Aging Time 20

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/2	Desg	FWD	19	128.2	P2p
Fa0/1	Root	FWD	19	128.1	P2p

VLAN0111

Spanning tree enabled protocol rstp

Root ID Priority 4207
 Address 000A.F354.D2B3
 Cost 23
 Port 1(FastEthernet0/1)
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32879 (priority 32768 sys-id-ext 111)
 Address 0001.C738.139A
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
 Aging Time 20

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Root	FWD	19	128.1	P2p
Fa0/3	Desg	FWD	19	128.3	P2p

VLAN0112

Spanning tree enabled protocol rstp

Root ID Priority 4208
 Address 000A.F354.D2B3
 Cost 23
 Port 1(FastEthernet0/1)
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32880 (priority 32768 sys-id-ext 112)
 Address 0001.C738.139A
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
 Aging Time 20

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----------	------	-----	------	----------	------

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Root	FWD	19	128.1	P2p

VLAN0113

```

Spanning tree enabled protocol rstp
Root ID      Priority      4209
              Address      000A.F354.D2B3
              Cost         23
              Port         1(FastEthernet0/1)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID    Priority      32881 (priority 32768 sys-id-ext 113)
              Address      0001.C738.139A
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time   20

```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Root	FWD	19	128.1	P2p

VLAN0999

```

Spanning tree enabled protocol rstp
Root ID      Priority      5095
              Address      000A.F354.D2B3
              Cost         23
              Port         1(FastEthernet0/1)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID    Priority      33767 (priority 32768 sys-id-ext 999)
              Address      0001.C738.139A
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time   20

```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Root	FWD	19	128.1	P2p

SW-ACC-SHW-01#show mac address-table
Mac Address Table

Vlan	Mac Address	Type	Ports
99	0007.ec58.240e	DYNAMIC	Fa0/1
99	000a.f354.d20d	DYNAMIC	Fa0/1
99	0050.0f9c.7301	DYNAMIC	Fa0/1
110	0050.0f9c.7301	DYNAMIC	Fa0/1
111	0007.ec58.2410	DYNAMIC	Fa0/1
111	000a.f354.d20f	DYNAMIC	Fa0/1
111	0050.0f9c.7301	DYNAMIC	Fa0/1
112	0050.0f9c.7301	DYNAMIC	Fa0/1
113	0050.0f9c.7301	DYNAMIC	Fa0/1

SW-ACC-SHW-01#

SW-ACC-VEN-01

SW-ACC-VEN-01#show vlan brief

VLAN	Name	Status	Ports
1	default	active	Gig0/1, Gig0/2
10	TRANSPORT-CORE	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24
11	VLAN0011	active	
20	SERV-BACKOFFICE	active	
21	SERV-DNS	active	
22	SERV-DHCP	active	
23	SERV-MAIL	active	
24	SERV-NTP	active	
25	SERV-WEB	active	
30	SRV-DESARROLLO	active	
31	SRV-QA	active	
32	SRV-PREPROD	active	
33	SRV-PRODUCCION	active	
40	WIFI-CORP	active	
50	WIFI-VISIT	active	
99	MGMT-OOB	active	
110	Dept-Showroom	active	
111	CCTV-Showroom	active	
112	WIFI-C-SHOWROOM	active	
113	WIFI-V-SHOWROOM	active	
120	DEPT-VENTAS	active	
121	CCTV-VENTAS	active	Fa0/2
122	WIFI-C-VENTAS	active	Fa0/3
123	WIFI-V-VENTAS	active	
130	DEPT-AUDITORIO	active	
131	CCTV-AUDITORIO	active	
132	WIFI-C-AUDITORIO	active	
133	WIFI-V-AUDITORIO	active	
140	DEPT-IT	active	
141	CCTV-IT	active	
142	WIFI-C-IT	active	
143	WIFI-V-IT	active	
150	DEPT-RRHH	active	
151	CCTV-RRHH	active	
152	WIFI-C-RRHH	active	
153	WIFI-V-RRHH	active	
160	DEPT-MARKETING	active	
161	CCTV-MARKETING	active	
162	WIFI-C-MKT	active	
163	WIFI-V-MKT	active	
170	DEPT-AUDITORIA	active	
171	CCTV-AUDITORIA	active	
172	WIFI-C-AUDITORIA	active	
173	WIFI-V-AUDITORIA	active	
180	DEPT-FINANZAS	active	
181	CCTV-FINANZAS	active	
182	WIFI-C-FINANZAS	active	
183	WIFI-V-FINANZAS	active	
999	NATIVE-UNUSED	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

SW-ACC-VEN-01#
SW-ACC-VEN-01#

SW-ACC-VEN-01#show interfaces status

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1	Enlace_Trunk_Hacia	connected	trunk	a-full	a-100	10/100BaseTX
Fa0/2		connected	121	a-full	a-100	10/100BaseTX
Fa0/3		connected	122	a-full	a-100	10/100BaseTX
Fa0/4		notconnect	10	auto	auto	10/100BaseTX
Fa0/5		notconnect	10	auto	auto	10/100BaseTX
Fa0/6		notconnect	10	auto	auto	10/100BaseTX
Fa0/7		notconnect	10	auto	auto	10/100BaseTX
Fa0/8		notconnect	10	auto	auto	10/100BaseTX
Fa0/9		notconnect	10	auto	auto	10/100BaseTX
Fa0/10		notconnect	10	auto	auto	10/100BaseTX
Fa0/11		notconnect	10	auto	auto	10/100BaseTX
Fa0/12		notconnect	10	auto	auto	10/100BaseTX
Fa0/13		notconnect	10	auto	auto	10/100BaseTX
Fa0/14		notconnect	10	auto	auto	10/100BaseTX
Fa0/15		notconnect	10	auto	auto	10/100BaseTX
Fa0/16		notconnect	10	auto	auto	10/100BaseTX
Fa0/17		notconnect	10	auto	auto	10/100BaseTX
Fa0/18		notconnect	10	auto	auto	10/100BaseTX
Fa0/19		notconnect	10	auto	auto	10/100BaseTX
Fa0/20		notconnect	10	auto	auto	10/100BaseTX
Fa0/21		notconnect	10	auto	auto	10/100BaseTX
Fa0/22		notconnect	10	auto	auto	10/100BaseTX
Fa0/23		notconnect	10	auto	auto	10/100BaseTX
Fa0/24		notconnect	10	auto	auto	10/100BaseTX
Gig0/1		notconnect	trunk	auto	auto	10/100/1000BaseTX
Gig0/2		notconnect	1	auto	auto	10/100/1000BaseTX

```

SW-ACC-VE-N-01#show spanning-tree
VLAN0099
Spanning tree enabled protocol rstp
Root ID    Priority    4195
           Address    000A.F354.D2B3
           Cost       23
           Port       1(FastEthernet0/1)
           Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID   Priority    32867 (priority 32768 sys-id-ext 99)
           Address    00D0.FFB9.009E
           Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time 20

Interface   Role Sts Cost      Prio.Nbr Type
-----
Fa0/1       Root FWD 19        128.1    P2p

VLAN0110
Spanning tree enabled protocol rstp
Root ID    Priority    4206
           Address    000A.F354.D2B3
           Cost       23
           Port       1(FastEthernet0/1)
           Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID   Priority    32878 (priority 32768 sys-id-ext 110)
           Address    00D0.FFB9.009E
           Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time 20

Interface   Role Sts Cost      Prio.Nbr Type
-----
Fa0/1       Root FWD 19        128.1    P2p

VLAN0111
Spanning tree enabled protocol rstp
Root ID    Priority    4207
           Address    000A.F354.D2B3
           Cost       23
           Port       1(FastEthernet0/1)
           Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID   Priority    32879 (priority 32768 sys-id-ext 111)
           Address    00D0.FFB9.009E
           Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time 20

Interface   Role Sts Cost      Prio.Nbr Type
-----
Fa0/1       Root FWD 19        128.1    P2p

VLAN0112
Spanning tree enabled protocol rstp
Root ID    Priority    4208
           Address    000A.F354.D2B3
           Cost       23
           Port       1(FastEthernet0/1)
           Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID   Priority    32880 (priority 32768 sys-id-ext 112)
           Address    00D0.FFB9.009E
           Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time 20

```

```

VLAN0113
Spanning tree enabled protocol rstp
Root ID      Priority      4209
             Address      000A.F354.D2B3
             Cost          23
             Port          1(FastEthernet0/1)
             Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID    Priority      32881 (priority 32768 sys-id-ext 113)
             Address      00D0.FFB9.009E
             Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time    20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Root FWD 19        128.1    P2p

VLAN0121
Spanning tree enabled protocol rstp
Root ID      Priority      4217
             Address      000A.F354.D2B3
             Cost          23
             Port          1(FastEthernet0/1)
             Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID    Priority      32889 (priority 32768 sys-id-ext 121)
             Address      00D0.FFB9.009E
             Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time    20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Root FWD 19        128.1    P2p
Fa0/2          Desg FWD 19        128.2    P2p

VLAN0122
Spanning tree enabled protocol rstp
Root ID      Priority      4218
             Address      000A.F354.D2B3
             Cost          23
             Port          1(FastEthernet0/1)
             Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID    Priority      32890 (priority 32768 sys-id-ext 122)
             Address      00D0.FFB9.009E
             Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time    20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Root FWD 19        128.1    P2p
Fa0/3          Desg FWD 19        128.3    P2p

VLAN0999
Spanning tree enabled protocol rstp
Root ID      Priority      5095
             Address      000A.F354.D2B3
             Cost          23
             Port          1(FastEthernet0/1)
             Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID    Priority      33767 (priority 32768 sys-id-ext 999)
             Address      00D0.FFB9.009E
             Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time    20

Interface      Role Sts Cost      Prio.Nbr Type
-----

```

```

SW-ACC-VEN-01#show mac address-table
Mac Address Table

```

Vlan	Mac Address	Type	Ports
99	0000.0c07.ac63	DYNAMIC	Fa0/1
99	0007.ec58.240e	DYNAMIC	Fa0/1
99	000a.f354.d20d	DYNAMIC	Fa0/1
99	000c.85ac.bed1	DYNAMIC	Fa0/1
110	0000.0c07.ac6e	DYNAMIC	Fa0/1
110	0001.c9a9.d842	DYNAMIC	Fa0/1
110	0007.ec58.240f	DYNAMIC	Fa0/1
110	000a.f354.d20e	DYNAMIC	Fa0/1
110	000c.85ac.bed1	DYNAMIC	Fa0/1
111	0007.ec58.2410	DYNAMIC	Fa0/1
111	000a.f354.d20f	DYNAMIC	Fa0/1
111	000c.85ac.bed1	DYNAMIC	Fa0/1
112	000c.85ac.bed1	DYNAMIC	Fa0/1
113	000c.85ac.bed1	DYNAMIC	Fa0/1
121	0007.ec58.2414	DYNAMIC	Fa0/1
121	000a.f354.d213	DYNAMIC	Fa0/1
122	0007.ec58.2415	DYNAMIC	Fa0/1
122	000a.f354.d214	DYNAMIC	Fa0/1
122	000c.85ac.bed1	DYNAMIC	Fa0/1

SW-ACC-AUT-01

SW-ACC-AUT-01#show vlan brief

VLAN	Name	Status	Ports
1	default	active	Gig0/1, Gig0/2
10	TRANSPORT-CORE	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24
11	VLAN0011	active	
20	SERV-BACKOFFICE	active	
21	SERV-DNS	active	
22	SERV-DHCP	active	
23	SERV-MAIL	active	
24	SERV-NTP	active	
25	SERV-WEB	active	
30	SRV-DESARROLLO	active	
31	SRV-QA	active	
32	SRV-PREPROD	active	
33	SRV-PRODUCCION	active	
40	WIFI-CORP	active	
50	WIFI-VISIT	active	
99	MGMT-OOB	active	
110	Dept-Showroom	active	
111	CCTV-Showroom	active	
112	WIFI-C-SHOWROOM	active	
113	WIFI-V-SHOWROOM	active	
120	DEPT-VENTAS	active	
121	CCTV-VENTAS	active	
122	WIFI-C-VENTAS	active	
123	WIFI-V-VENTAS	active	
130	DEPT-AUDITORIO	active	Fa0/2
131	CCTV-AUDITORIO	active	
132	WIFI-C-AUDITORIO	active	
133	WIFI-V-AUDITORIO	active	
140	DEPT-IT	active	
141	CCTV-IT	active	
142	WIFI-C-IT	active	
143	WIFI-V-IT	active	
150	DEPT-RRHH	active	
151	CCTV-RRHH	active	
152	WIFI-C-RRHH	active	
153	WIFI-V-RRHH	active	
160	DEPT-MARKETING	active	
161	CCTV-MARKETING	active	
162	WIFI-C-MKT	active	
163	WIFI-V-MKT	active	
170	DEPT-AUDITORIA	active	
171	CCTV-AUDITORIA	active	
172	WIFI-C-AUDITORIA	active	
173	WIFI-V-AUDITORIA	active	
180	DEPT-FINANZAS	active	
181	CCTV-FINANZAS	active	
182	WIFI-C-FINANZAS	active	
183	WIFI-V-FINANZAS	active	
999	NATIVE-UNUSED	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	
SW-ACC-AUT-01#			
SW-ACC-AUT-01#			
SW-ACC-AUT-01#			
SW-ACC-AUT-01#			
SW-ACC-AUT-01#			

SW-ACC-AUT-01#show interfaces status

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1	Enlace_Trunk_Hacia	connected	trunk	a-full	a-100	10/100BaseTX
Fa0/2		connected	130	a-full	a-100	10/100BaseTX
Fa0/3		notconnect	10	auto	auto	10/100BaseTX
Fa0/4		notconnect	10	auto	auto	10/100BaseTX
Fa0/5		notconnect	10	auto	auto	10/100BaseTX
Fa0/6		notconnect	10	auto	auto	10/100BaseTX
Fa0/7		notconnect	10	auto	auto	10/100BaseTX
Fa0/8		notconnect	10	auto	auto	10/100BaseTX
Fa0/9		notconnect	10	auto	auto	10/100BaseTX
Fa0/10		notconnect	10	auto	auto	10/100BaseTX
Fa0/11		notconnect	10	auto	auto	10/100BaseTX
Fa0/12		notconnect	10	auto	auto	10/100BaseTX
Fa0/13		notconnect	10	auto	auto	10/100BaseTX
Fa0/14		notconnect	10	auto	auto	10/100BaseTX
Fa0/15		notconnect	10	auto	auto	10/100BaseTX
Fa0/16		notconnect	10	auto	auto	10/100BaseTX
Fa0/17		notconnect	10	auto	auto	10/100BaseTX
Fa0/18		notconnect	10	auto	auto	10/100BaseTX
Fa0/19		notconnect	10	auto	auto	10/100BaseTX
Fa0/20		notconnect	10	auto	auto	10/100BaseTX
Fa0/21		notconnect	10	auto	auto	10/100BaseTX
Fa0/22		notconnect	10	auto	auto	10/100BaseTX
Fa0/23		notconnect	10	auto	auto	10/100BaseTX
Fa0/24		notconnect	10	auto	auto	10/100BaseTX
Gig0/1		notconnect	trunk	auto	auto	10/100/1000BaseTX
Gig0/2		notconnect	1	auto	auto	10/100/1000BaseTX

```

SW-ACC-AUT-01#show spanning-tree
VLAN0099
  Spanning tree enabled protocol rstp
  Root ID    Priority    4195
            Address    000A.F354.D2B3
            Cost        23
            Port        1(FastEthernet0/1)
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32867 (priority 32768 sys-id-ext 99)
            Address    0009.7CBE.1EB8
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time  20

```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Root	FWD	19	128.1	P2p

```

VLAN0130
  Spanning tree enabled protocol rstp
  Root ID    Priority    4226
            Address    000A.F354.D2B3
            Cost        23
            Port        1(FastEthernet0/1)
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32898 (priority 32768 sys-id-ext 130)
            Address    0009.7CBE.1EB8
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time  20

```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Root	FWD	19	128.1	P2p
Fa0/2	Desg	FWD	19	128.2	P2p

```

VLAN0131
  Spanning tree enabled protocol rstp
  Root ID    Priority    4227
            Address    000A.F354.D2B3
            Cost        23
            Port        1(FastEthernet0/1)
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32899 (priority 32768 sys-id-ext 131)
            Address    0009.7CBE.1EB8
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time  20

```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Root	FWD	19	128.1	P2p

```

VLAN0132
  Spanning tree enabled protocol rstp
  Root ID    Priority    4228
            Address    000A.F354.D2B3
            Cost        23
            Port        1(FastEthernet0/1)
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32900 (priority 32768 sys-id-ext 132)
            Address    0009.7CBE.1EB8
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time  20

```

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----------	------	-----	------	----------	------

Fa0/1	Root	FWD	19	128.1	P2p
-------	------	-----	----	-------	-----

```

VLAN0133
  Spanning tree enabled protocol rstp
  Root ID    Priority    4229
            Address    000A.F354.D2B3
            Cost        23
            Port        1(FastEthernet0/1)
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32901 (priority 32768 sys-id-ext 133)
            Address    0009.7CBE.1EB8
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time  20

```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Root	FWD	19	128.1	P2p

```

VLAN0999
  Spanning tree enabled protocol rstp
  Root ID    Priority    5095
            Address    000A.F354.D2B3
            Cost        23
            Port        1(FastEthernet0/1)
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    33767 (priority 32768 sys-id-ext 999)
            Address    0009.7CBE.1EB8
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time  20

```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Root	FWD	19	128.1	P2p

```

SW-ACC-AUT-01#

```

```
SW-ACC-AUT-01#show mac address-table
Mac Address Table
```

Vlan	Mac Address	Type	Ports
99	0007.ec58.240e	DYNAMIC	Fa0/1
99	000a.f354.d20d	DYNAMIC	Fa0/1
99	0090.0ca0.2dac	DYNAMIC	Fa0/1
130	0007.ec58.2417	DYNAMIC	Fa0/1
130	000a.f354.d216	DYNAMIC	Fa0/1
130	0090.0ca0.2dac	DYNAMIC	Fa0/1
131	0007.ec58.2418	DYNAMIC	Fa0/1
131	000a.f354.d217	DYNAMIC	Fa0/1
131	0090.0ca0.2dac	DYNAMIC	Fa0/1
132	0007.ec58.2419	DYNAMIC	Fa0/1
132	000a.f354.d218	DYNAMIC	Fa0/1
132	0090.0ca0.2dac	DYNAMIC	Fa0/1
133	0007.ec58.241a	DYNAMIC	Fa0/1
133	000a.f354.d219	DYNAMIC	Fa0/1
133	0090.0ca0.2dac	DYNAMIC	Fa0/1

```
SW-ACC-AUT-01#
```

SW-DC-SRV-01

```
SW-DC-SRV-01#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Gig0/1, Gig0/2
10	TRANSPORT-CORE	active	Fa0/4, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24
11	VLAN0011	active	
20	SERV-BACKOFFICE	active	Fa0/2
21	SERV-DNS	active	Fa0/10, Fa0/11, Fa0/12
22	SERV-DHCP	active	Fa0/8, Fa0/9
23	SERV-MAIL	active	Fa0/3
24	SERV-NTP	active	
25	SERV-WEB	active	Fa0/5, Fa0/6, Fa0/7
30	SRV-DESARROLLO	active	
31	SRV-QA	active	
32	SRV-PREPROD	active	
33	SRV-PRODUCCION	active	
40	WIFI-CORP	active	
50	WIFI-VISIT	active	
99	MGMT-OOB	active	
110	Dept-Showroom	active	
111	CCTV-Showroom	active	
112	WIFI-C-SHOWROOM	active	
113	WIFI-V-SHOWROOM	active	
120	DEPT-VENTAS	active	
121	CCTV-VENTAS	active	
122	WIFI-C-VENTAS	active	
123	WIFI-V-VENTAS	active	
130	DEPT-AUDITORIO	active	
131	CCTV-AUDITORIO	active	
132	WIFI-C-AUDITORIO	active	
133	WIFI-V-AUDITORIO	active	
140	DEPT-IT	active	
141	CCTV-IT	active	
142	WIFI-C-IT	active	
143	WIFI-V-IT	active	
150	DEPT-RRHH	active	
151	CCTV-RRHH	active	
152	WIFI-C-RRHH	active	
153	WIFI-V-RRHH	active	
160	DEPT-MARKETING	active	
161	CCTV-MARKETING	active	
162	WIFI-C-MKT	active	
163	WIFI-V-MKT	active	
170	DEPT-AUDITORIA	active	
171	CCTV-AUDITORIA	active	
172	WIFI-C-AUDITORIA	active	
173	WIFI-V-AUDITORIA	active	
180	DEPT-FINANZAS	active	
181	CCTV-FINANZAS	active	
182	WIFI-C-FINANZAS	active	
183	WIFI-V-FINANZAS	active	
999	NATIVE-UNUSED	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
SW-DC-SRV-01#
```

```
SW-DC-SRV-01#
```

```

SW-DC-SRV-01#show interfaces status
Port      Name      Status      Vlan      Duplex  Speed Type
Fa0/1      Name      connected   trunk     a-full  a-100  10/100BaseTX
Fa0/2      notconnect 20          auto      auto    10/100BaseTX
Fa0/3      Conexion a Backoff connected 23          a-full    a-100  10/100BaseTX
Fa0/4      notconnect 10          auto      auto    10/100BaseTX
Fa0/5      notconnect 25          auto      auto    10/100BaseTX
Fa0/6      connected 25          a-full    a-100  10/100BaseTX
Fa0/7      notconnect 25          auto      auto    10/100BaseTX
Fa0/8      connected 22          a-full    a-100  10/100BaseTX
Fa0/9      notconnect 22          auto      auto    10/100BaseTX
Fa0/10     connected 21          a-full    a-100  10/100BaseTX
Fa0/11     notconnect 21          auto      auto    10/100BaseTX
Fa0/12     notconnect 21          auto      auto    10/100BaseTX
Fa0/13     notconnect 10          auto      auto    10/100BaseTX
Fa0/14     notconnect 10          auto      auto    10/100BaseTX
Fa0/15     notconnect 10          auto      auto    10/100BaseTX
Fa0/16     notconnect 10          auto      auto    10/100BaseTX
Fa0/17     notconnect 10          auto      auto    10/100BaseTX
Fa0/18     notconnect 10          auto      auto    10/100BaseTX
Fa0/19     notconnect 10          auto      auto    10/100BaseTX
Fa0/20     notconnect 10          auto      auto    10/100BaseTX
Fa0/21     notconnect 10          auto      auto    10/100BaseTX
Fa0/22     notconnect 10          auto      auto    10/100BaseTX
Fa0/23     notconnect 10          auto      auto    10/100BaseTX
Fa0/24     notconnect 10          auto      auto    10/100BaseTX
Gig0/1     notconnect trunk       auto      auto    10/100/1000BaseTX
Gig0/2     notconnect 1           auto      auto    10/100/1000BaseTX

SW-DC-SRV-01#
SW-DC-SRV-01#

```

```

SW-DC-SRV-01#
SW-DC-SRV-01#show spanning-tree
VLAN0020
  Spanning tree enabled protocol rstp
  Root ID    Priority    4116
            Address    000A.F354.D2B3
            Cost        23
            Port        1(FastEthernet0/1)
            Hello Time  2 sec Max Age 20 sec Forward Delay 15 sec

  Bridge ID  Priority    32788 (priority 32768 sys-id-ext 20)
            Address    0004.9A2A.A129
            Hello Time  2 sec Max Age 20 sec Forward Delay 15 sec
            Aging Time  20

Interface Role Sts Cost Prio.Nbr Type
-----
Fa0/1      Root FWD 19    128.1 P2p

VLAN0021
  Spanning tree enabled protocol rstp
  Root ID    Priority    4117
            Address    000A.F354.D2B3
            Cost        23
            Port        1(FastEthernet0/1)
            Hello Time  2 sec Max Age 20 sec Forward Delay 15 sec

  Bridge ID  Priority    32789 (priority 32768 sys-id-ext 21)
            Address    0004.9A2A.A129
            Hello Time  2 sec Max Age 20 sec Forward Delay 15 sec
            Aging Time  20

Interface Role Sts Cost Prio.Nbr Type
-----
Fa0/1      Root FWD 19    128.1 P2p
Fa0/10     Desg FWD 19    128.10 P2p

VLAN0022
  Spanning tree enabled protocol rstp
  Root ID    Priority    4118
            Address    000A.F354.D2B3
            Cost        23
            Port        1(FastEthernet0/1)
            Hello Time  2 sec Max Age 20 sec Forward Delay 15 sec

  Bridge ID  Priority    32790 (priority 32768 sys-id-ext 22)
            Address    0004.9A2A.A129
            Hello Time  2 sec Max Age 20 sec Forward Delay 15 sec
            Aging Time  20

Interface Role Sts Cost Prio.Nbr Type
-----
Fa0/1      Root FWD 19    128.1 P2p
Fa0/8      Desg FWD 19    128.8 P2p

VLAN0023
  Spanning tree enabled protocol rstp
  Root ID    Priority    4119
            Address    000A.F354.D2B3
            Cost        23
            Port        1(FastEthernet0/1)
            Hello Time  2 sec Max Age 20 sec Forward Delay 15 sec

  Bridge ID  Priority    32791 (priority 32768 sys-id-ext 23)
            Address    0004.9A2A.A129
            Hello Time  2 sec Max Age 20 sec Forward Delay 15 sec
            Aging Time  20

```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/3	Desg	FWD	19	128.3	P2p
Fa0/1	Root	FWD	19	128.1	P2p

VLAN0024

```

Spanning tree enabled protocol rstp
Root ID      Priority    4120
              Address    000A.F354.D2B3
              Cost       23
              Port       1(FastEthernet0/1)
              Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID    Priority    32792 (priority 32768 sys-id-ext 24)
              Address    0004.9A2A.A129
              Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time 20
  
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Root	FWD	19	128.1	P2p

VLAN0025

```

Spanning tree enabled protocol rstp
Root ID      Priority    4121
              Address    000A.F354.D2B3
              Cost       23
              Port       1(FastEthernet0/1)
              Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID    Priority    32793 (priority 32768 sys-id-ext 25)
              Address    0004.9A2A.A129
              Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time 20
  
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Root	FWD	19	128.1	P2p
Fa0/6	Desg	FWD	19	128.6	P2p

VLAN0099

```

Spanning tree enabled protocol rstp
Root ID      Priority    4195
              Address    000A.F354.D2B3
              Cost       23
              Port       1(FastEthernet0/1)
              Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID    Priority    32867 (priority 32768 sys-id-ext 99)
              Address    0004.9A2A.A129
              Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time 20
  
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Root	FWD	19	128.1	P2p

VLAN0110

```

Spanning tree enabled protocol rstp
Root ID      Priority    4206
              Address    000A.F354.D2B3
              Cost       23
              Port       1(FastEthernet0/1)
              Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID    Priority    32878 (priority 32768 sys-id-ext 110)
              Address    0004.9A2A.A129
              Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time 20
  
```

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Root	FWD	19	128.1	P2p

VLAN0111

Spanning tree enabled protocol rstp
Root ID Priority 4207
Address 000A.F354.D2B3
Cost 23
Port 1(FastEthernet0/1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Bridge ID Priority 32879 (priority 32768 sys-id-ext 111)
Address 0004.9A2A.A129
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Root	FWD	19	128.1	P2p

VLAN0112

Spanning tree enabled protocol rstp
Root ID Priority 4208
Address 000A.F354.D2B3
Cost 23
Port 1(FastEthernet0/1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Bridge ID Priority 32880 (priority 32768 sys-id-ext 112)
Address 0004.9A2A.A129
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Root	FWD	19	128.1	P2p

VLAN0113

Spanning tree enabled protocol rstp
Root ID Priority 4209
Address 000A.F354.D2B3
Cost 23
Port 1(FastEthernet0/1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Bridge ID Priority 32881 (priority 32768 sys-id-ext 113)
Address 0004.9A2A.A129
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Root	FWD	19	128.1	P2p

VLAN0999

Spanning tree enabled protocol rstp
Root ID Priority 5095
Address 000A.F354.D2B3
Cost 23
Port 1(FastEthernet0/1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Bridge ID Priority 33767 (priority 32768 sys-id-ext 999)

Bridge ID Priority 33767 (priority 32768 sys-id-ext 999)
Address 0004.9A2A.A129
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Root	FWD	19	128.1	P2p

SW-DC-SRV-01#

```
SW-DC-SRV-01#show mac address-table
Mac Address Table
```

Vlan	Mac Address	Type	Ports
20	0007.ec58.2402	DYNAMIC	Fa0/1
20	000a.f354.d203	DYNAMIC	Fa0/1
20	00d0.bc52.beca	DYNAMIC	Fa0/1
21	0007.ec58.2403	DYNAMIC	Fa0/1
21	000a.f354.d204	DYNAMIC	Fa0/1
21	00d0.bc52.beca	DYNAMIC	Fa0/1
23	0007.ec58.2405	DYNAMIC	Fa0/1
23	000a.f354.d206	DYNAMIC	Fa0/1
23	00d0.bc52.beca	DYNAMIC	Fa0/1
24	0007.ec58.2406	DYNAMIC	Fa0/1
24	000a.f354.d207	DYNAMIC	Fa0/1
24	00d0.bc52.beca	DYNAMIC	Fa0/1
25	0007.ec58.2407	DYNAMIC	Fa0/1
25	000a.f354.d208	DYNAMIC	Fa0/1
25	00d0.bc52.beca	DYNAMIC	Fa0/1
99	0007.ec58.240e	DYNAMIC	Fa0/1
99	000a.f354.d20d	DYNAMIC	Fa0/1
99	00d0.bc52.beca	DYNAMIC	Fa0/1
110	0007.ec58.240f	DYNAMIC	Fa0/1
110	000a.f354.d20e	DYNAMIC	Fa0/1
110	00d0.bc52.beca	DYNAMIC	Fa0/1
111	0007.ec58.2410	DYNAMIC	Fa0/1
111	000a.f354.d20f	DYNAMIC	Fa0/1
111	00d0.bc52.beca	DYNAMIC	Fa0/1
112	00d0.bc52.beca	DYNAMIC	Fa0/1
113	00d0.bc52.beca	DYNAMIC	Fa0/1

```
SW-DC-SRV-01#
```

SW-ACC-IT-01

```
SW-ACC-IT-01#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Gig0/1, Gig0/2
10	TRANSPORT-CORE	active	Fa0/2, Fa0/3, Fa0/4, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24
11	VLAN0011	active	
20	SERV-BACKOFFICE	active	
21	SERV-DNS	active	
22	SERV-DHCP	active	
23	SERV-MAIL	active	
24	SERV-NTP	active	
25	SERV-WEB	active	
30	SRV-DESARROLLO	active	
31	SRV-QA	active	
32	SRV-PREPROD	active	
33	SRV-PRODUCCION	active	
40	WIFI-CORP	active	
50	WIFI-VISIT	active	
99	MGMT-OOB	active	
110	Dept-Showroom	active	
111	CCTV-Showroom	active	
112	WIFI-C-SHOWROOM	active	
113	WIFI-V-SHOWROOM	active	
120	DEPT-VENTAS	active	
121	CCTV-VENTAS	active	
122	WIFI-C-VENTAS	active	
123	WIFI-V-VENTAS	active	
130	DEPT-AUDITORIO	active	
131	CCTV-AUDITORIO	active	
132	WIFI-C-AUDITORIO	active	
133	WIFI-V-AUDITORIO	active	
140	DEPT-IT	active	Fa0/5, Fa0/6, Fa0/7
141	CCTV-IT	active	
142	WIFI-C-IT	active	Fa0/8, Fa0/9, Fa0/10
143	WIFI-V-IT	active	
150	DEPT-RRHH	active	
151	CCTV-RRHH	active	
152	WIFI-C-RRHH	active	
153	WIFI-V-RRHH	active	
160	DEPT-MARKETING	active	
161	CCTV-MARKETING	active	
162	WIFI-C-MKT	active	
163	WIFI-V-MKT	active	
170	DEPT-AUDITORIA	active	
171	CCTV-AUDITORIA	active	
172	WIFI-C-AUDITORIA	active	
173	WIFI-V-AUDITORIA	active	
180	DEPT-FINANZAS	active	
181	CCTV-FINANZAS	active	
182	WIFI-C-FINANZAS	active	
183	WIFI-V-FINANZAS	active	
999	NATIVE-UNUSED	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
SW-ACC-IT-01#
```

```

SW-ACC-IT-01#show interfaces status
Port      Name      Status      Vlan      Duplex  Speed Type
Fa0/1     Enlace_Trunk_Hacia connected    trunk     a-full   a-100  10/100BaseTX
Fa0/2                                           auto     auto   10/100BaseTX
Fa0/3                                           notconnect 10     auto   10/100BaseTX
Fa0/4                                           notconnect 10     auto   10/100BaseTX
Fa0/5                                           connected  140    a-full a-100  10/100BaseTX
Fa0/6                                           notconnect 140    auto   10/100BaseTX
Fa0/7                                           notconnect 140    auto   10/100BaseTX
Fa0/8                                           notconnect 142    auto   10/100BaseTX
Fa0/9                                           connected  142    a-full a-100  10/100BaseTX
Fa0/10                                          notconnect 142    auto   10/100BaseTX
Fa0/11                                          notconnect 10     auto   10/100BaseTX
Fa0/12                                          notconnect 10     auto   10/100BaseTX
Fa0/13                                          notconnect 10     auto   10/100BaseTX
Fa0/14                                          notconnect 10     auto   10/100BaseTX
Fa0/15                                          notconnect 10     auto   10/100BaseTX
Fa0/16                                          notconnect 10     auto   10/100BaseTX
Fa0/17                                          notconnect 10     auto   10/100BaseTX
Fa0/18                                          notconnect 10     auto   10/100BaseTX
Fa0/19                                          notconnect 10     auto   10/100BaseTX
Fa0/20                                          notconnect 10     auto   10/100BaseTX
Fa0/21                                          notconnect 10     auto   10/100BaseTX
Fa0/22                                          notconnect 10     auto   10/100BaseTX
Fa0/23                                          notconnect 10     auto   10/100BaseTX
Fa0/24                                          notconnect 10     auto   10/100BaseTX
Gig0/1                                          notconnect 1      auto   10/100/1000BaseTX
Gig0/2                                          notconnect 1      auto   10/100/1000BaseTX

```

```

SW-ACC-IT-01#show spanning-tree
VLAN0010
  Spanning tree enabled protocol rstp
    Root ID    Priority      4106
              Address      000A.F354.D2B3
              Cost         23
              Port         1(FastEthernet0/1)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

    Bridge ID  Priority      32778 (priority 32768 sys-id-ext 10)
              Address      0060.2FCE.6EB3
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time   20

Interface                Role Sts Cost      Prio.Nbr Type
-----
Fa0/1                    Root FWD 19        128.1   P2p

VLAN0020
  Spanning tree enabled protocol rstp
    Root ID    Priority      4116
              Address      000A.F354.D2B3
              Cost         23
              Port         1(FastEthernet0/1)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

    Bridge ID  Priority      32788 (priority 32768 sys-id-ext 20)
              Address      0060.2FCE.6EB3
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time   20

Interface                Role Sts Cost      Prio.Nbr Type
-----
Fa0/1                    Root FWD 19        128.1   P2p

VLAN0030
  Spanning tree enabled protocol rstp
    Root ID    Priority      4126
              Address      000A.F354.D2B3
              Cost         23
              Port         1(FastEthernet0/1)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

    Bridge ID  Priority      32798 (priority 32768 sys-id-ext 30)
              Address      0060.2FCE.6EB3
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time   20

Interface                Role Sts Cost      Prio.Nbr Type
-----
Fa0/1                    Root FWD 19        128.1   P2p

VLAN0040
  Spanning tree enabled protocol rstp
    Root ID    Priority      32808
              Address      0003.E443.3E3B
              Cost         19
              Port         1(FastEthernet0/1)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

    Bridge ID  Priority      32808 (priority 32768 sys-id-ext 40)
              Address      0060.2FCE.6EB3
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time   20

Interface                Role Sts Cost      Prio.Nbr Type
--More--

```

```

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Root FWD 19        128.1   P2p

VLAN0050
Spanning tree enabled protocol rstp
Root ID        Priority 32818
Address        0003.E443.3E3B
Cost           19
Port           1(FastEthernet0/1)
Hello Time     2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID      Priority 32818 (priority 32768 sys-id-ext 50)
Address        0060.2FCE.6EB3
Hello Time     2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time     20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Root FWD 19        128.1   P2p

VLAN0099
Spanning tree enabled protocol rstp
Root ID        Priority 4195
Address        000A.F354.D2B3
Cost           23
Port           1(FastEthernet0/1)
Hello Time     2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID      Priority 32867 (priority 32768 sys-id-ext 99)
Address        0060.2FCE.6EB3
Hello Time     2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time     20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Root FWD 19        128.1   P2p

VLAN0140
Spanning tree enabled protocol rstp
Root ID        Priority 4236
Address        000A.F354.D2B3
Cost           23
Port           1(FastEthernet0/1)
Hello Time     2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID      Priority 32908 (priority 32768 sys-id-ext 140)
Address        0060.2FCE.6EB3
Hello Time     2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time     20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Root FWD 19        128.1   P2p
Fa0/5          Desg FWD 19        128.5   P2p

VLAN0141
Spanning tree enabled protocol rstp
Root ID        Priority 4237
Address        000A.F354.D2B3
Cost           23
Port           1(FastEthernet0/1)
Hello Time     2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID      Priority 32909 (priority 32768 sys-id-ext 141)
Address        0060.2FCE.6EB3

```

--More--

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```
Address      0060.2FCE.6EB3
Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
Aging Time   20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Root	FWD	19	128.1	P2p

```
VLAN0142
Spanning tree enabled protocol rstp
Root ID      Priority    4238
              Address     000A.F354.D2B3
              Cost        23
              Port        1(FastEthernet0/1)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID     Priority    32910 (priority 32768 sys-id-ext 142)
              Address     0060.2FCE.6EB3
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time   20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Root	FWD	19	128.1	P2p
Fa0/9	Desg	FWD	19	128.9	P2p

```
VLAN0143
Spanning tree enabled protocol rstp
Root ID      Priority    4239
              Address     000A.F354.D2B3
              Cost        23
              Port        1(FastEthernet0/1)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID     Priority    32911 (priority 32768 sys-id-ext 143)
              Address     0060.2FCE.6EB3
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time   20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Root	FWD	19	128.1	P2p

```
VLAN0999
Spanning tree enabled protocol rstp
Root ID      Priority    5095
              Address     000A.F354.D2B3
              Cost        23
              Port        1(FastEthernet0/1)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID     Priority    33767 (priority 32768 sys-id-ext 999)
              Address     0060.2FCE.6EB3
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time   20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Root	FWD	19	128.1	P2p

```
SW-ACC-IT-01#
SW-ACC-IT-01#
SW-ACC-IT-01#
SW-ACC-IT-01#
SW-ACC-IT-01#
SW-ACC-IT-01#
SW-ACC-IT-01#
```

SW-ACC-IT-01#show mac address-table
Mac Address Table

Vlan	Mac Address	Type	Ports
10	0001.97b0.b601	DYNAMIC	Fa0/1
10	0003.e443.3e01	DYNAMIC	Fa0/1
10	0007.ec58.2401	DYNAMIC	Fa0/1
10	000a.f337.c401	DYNAMIC	Fa0/1
10	000a.f354.d201	DYNAMIC	Fa0/1
10	000c.cf72.dc01	DYNAMIC	Fa0/1
10	0030.a3c6.a211	DYNAMIC	Fa0/1
10	0040.0b58.e301	DYNAMIC	Fa0/1
10	00d0.978d.ce01	DYNAMIC	Fa0/1
10	00e0.8fc3.2401	DYNAMIC	Fa0/1
20	0007.ec58.2402	DYNAMIC	Fa0/1
20	000a.f354.d203	DYNAMIC	Fa0/1
20	0030.a3c6.a211	DYNAMIC	Fa0/1
30	0007.ec58.2408	DYNAMIC	Fa0/1
30	000a.f354.d209	DYNAMIC	Fa0/1
30	0030.a3c6.a211	DYNAMIC	Fa0/1
40	0030.a3c6.a211	DYNAMIC	Fa0/1
50	0030.a3c6.a211	DYNAMIC	Fa0/1
99	0007.ec58.240e	DYNAMIC	Fa0/1
99	000a.f354.d20d	DYNAMIC	Fa0/1
99	0030.a3c6.a211	DYNAMIC	Fa0/1
140	0007.ec58.241b	DYNAMIC	Fa0/1
140	000a.f354.d21a	DYNAMIC	Fa0/1
140	0030.a3c6.a211	DYNAMIC	Fa0/1
141	0007.ec58.241c	DYNAMIC	Fa0/1
141	000a.f354.d21b	DYNAMIC	Fa0/1
141	0030.a3c6.a211	DYNAMIC	Fa0/1
142	0007.ec58.241d	DYNAMIC	Fa0/1
142	000a.f354.d21c	DYNAMIC	Fa0/1
142	0030.a3c6.a211	DYNAMIC	Fa0/1
143	0007.ec58.241e	DYNAMIC	Fa0/1
143	000a.f354.d21d	DYNAMIC	Fa0/1
143	0030.a3c6.a211	DYNAMIC	Fa0/1
999	0030.a3c6.a211	DYNAMIC	Fa0/1

SW-ACC-IT-01#

SW-ACC-IT-01#

SW-ACC-IT-01#

SW-ACC-RRH-01

SW-ACC-RRH-01#show vlan brief

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Gig0/1, Gig0/2
10	TRANSPORT-CORE	active	Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24
11	VLAN0011	active	
20	SERV-BACKOFFICE	active	
21	SERV-DNS	active	
22	SERV-DHCP	active	
23	SERV-MAIL	active	
24	SERV-NTP	active	
25	SERV-WEB	active	
30	SRV-DESARROLLO	active	
31	SRV-QA	active	
32	SRV-PREPROD	active	
33	SRV-PRODUCCION	active	
40	WIFI-CORP	active	
50	WIFI-VISIT	active	
99	MGMT-OOB	active	
110	Dept-Showroom	active	
111	CCTV-Showroom	active	
112	WIFI-C-SHOWROOM	active	
113	WIFI-V-SHOWROOM	active	
120	DEPT-VENTAS	active	
121	CCTV-VENTAS	active	
122	WIFI-C-VENTAS	active	
123	WIFI-V-VENTAS	active	
130	DEPT-AUDITORIO	active	
131	CCTV-AUDITORIO	active	
132	WIFI-C-AUDITORIO	active	
133	WIFI-V-AUDITORIO	active	
140	DEPT-IT	active	
141	CCTV-IT	active	
142	WIFI-C-IT	active	
143	WIFI-V-IT	active	
150	DEPT-RRHH	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6
151	CCTV-RRHH	active	
152	WIFI-C-RRHH	active	
153	WIFI-V-RRHH	active	
160	DEPT-MARKETING	active	
161	CCTV-MARKETING	active	
162	WIFI-C-MKT	active	
163	WIFI-V-MKT	active	
170	DEPT-AUDITORIA	active	
171	CCTV-AUDITORIA	active	
172	WIFI-C-AUDITORIA	active	
173	WIFI-V-AUDITORIA	active	
180	DEPT-FINANZAS	active	
181	CCTV-FINANZAS	active	
182	WIFI-C-FINANZAS	active	
183	WIFI-V-FINANZAS	active	
999	NATIVE-UNUSED	active	
1002	fdi-default	active	
1003	token-ring-default	active	
1004	fdinet-default	active	
1005	trnet-default	active	

SW-ACC-RRH-01#show interfaces status

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1		notconnect	trunk	auto	auto	10/100BaseTX
Fa0/2	Enlace_Trunk_Hacia	connected	trunk	a-full	a-100	10/100BaseTX
Fa0/3		notconnect	150	auto	auto	10/100BaseTX
Fa0/4		notconnect	150	auto	auto	10/100BaseTX
Fa0/5		connected	150	a-full	a-100	10/100BaseTX
Fa0/6		notconnect	150	auto	auto	10/100BaseTX
Fa0/7		notconnect	10	auto	auto	10/100BaseTX
Fa0/8		notconnect	10	auto	auto	10/100BaseTX
Fa0/9		notconnect	10	auto	auto	10/100BaseTX
Fa0/10		notconnect	10	auto	auto	10/100BaseTX
Fa0/11		notconnect	10	auto	auto	10/100BaseTX
Fa0/12		notconnect	10	auto	auto	10/100BaseTX
Fa0/13		notconnect	10	auto	auto	10/100BaseTX
Fa0/14		notconnect	10	auto	auto	10/100BaseTX
Fa0/15		notconnect	10	auto	auto	10/100BaseTX
Fa0/16		notconnect	10	auto	auto	10/100BaseTX
Fa0/17		notconnect	10	auto	auto	10/100BaseTX
Fa0/18		notconnect	10	auto	auto	10/100BaseTX
Fa0/19		notconnect	10	auto	auto	10/100BaseTX
Fa0/20		notconnect	10	auto	auto	10/100BaseTX
Fa0/21		notconnect	10	auto	auto	10/100BaseTX
Fa0/22		notconnect	10	auto	auto	10/100BaseTX
Fa0/23		notconnect	10	auto	auto	10/100BaseTX
Fa0/24		notconnect	10	auto	auto	10/100BaseTX
Gig0/1		notconnect	1	auto	auto	10/100/1000BaseTX
Gig0/2		notconnect	trunk	auto	auto	10/100/1000BaseTX

SW-ACC-RRH-01#

```

SW-ACC-RRH-01#show spanning-tree
VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    4106
             Address     000A.F354.D2B3
             Cost        23
             Port        2(FastEthernet0/2)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32778 (priority 32768 sys-id-ext 10)
             Address     0001.4251.1A27
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface    Role Sts Cost      Prio.Nbr Type
-----
Fa0/2        Root FWD 19        128.2   P2p

VLAN0020
  Spanning tree enabled protocol rstp
  Root ID    Priority    4116
             Address     000A.F354.D2B3
             Cost        23
             Port        2(FastEthernet0/2)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32788 (priority 32768 sys-id-ext 20)
             Address     0001.4251.1A27
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface    Role Sts Cost      Prio.Nbr Type
-----
Fa0/2        Root FWD 19        128.2   P2p

VLAN0030
  Spanning tree enabled protocol rstp
  Root ID    Priority    4126
             Address     000A.F354.D2B3
             Cost        23
             Port        2(FastEthernet0/2)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32798 (priority 32768 sys-id-ext 30)
             Address     0001.4251.1A27
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface    Role Sts Cost      Prio.Nbr Type
-----
Fa0/2        Root FWD 19        128.2   P2p

VLAN0040
  Spanning tree enabled protocol rstp
  Root ID    Priority    32808
             Address     0001.4251.1A27
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32808 (priority 32768 sys-id-ext 40)
             Address     0001.4251.1A27
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface    Role Sts Cost      Prio.Nbr Type
-----
--More--

```

```

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/2          Desg FWD 19       128.2   P2p

VLAN0050
Spanning tree enabled protocol rstp
Root ID        Priority    32818
                Address    0001.4251.1A27
                This bridge is the root
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID      Priority    32818 (priority 32768 sys-id-ext 50)
                Address    0001.4251.1A27
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
                Aging Time 20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/2          Desg FWD 19       128.2   P2p

VLAN0099
Spanning tree enabled protocol rstp
Root ID        Priority    4195
                Address    000A.F354.D2B3
                Cost        23
                Port        2(FastEthernet0/2)
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID      Priority    32867 (priority 32768 sys-id-ext 99)
                Address    0001.4251.1A27
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
                Aging Time 20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/2          Root FWD 19       128.2   P2p

VLAN0150
Spanning tree enabled protocol rstp
Root ID        Priority    4246
                Address    000A.F354.D2B3
                Cost        23
                Port        2(FastEthernet0/2)
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID      Priority    32918 (priority 32768 sys-id-ext 150)
                Address    0001.4251.1A27
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
                Aging Time 20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/5          Desg FWD 19       128.5   P2p
Fa0/2          Root FWD 19       128.2   P2p

VLAN0151
Spanning tree enabled protocol rstp
Root ID        Priority    4247
                Address    000A.F354.D2B3
                Cost        23
                Port        2(FastEthernet0/2)
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID      Priority    32919 (priority 32768 sys-id-ext 151)
                Address    0001.4251.1A27
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

--More--

```

```
Port 2(FastEthernet0/2)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32919 (priority 32768 sys-id-ext 151)
Address 0001.4251.1A27
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/2	Root	FWD	19	128.2	P2p

```
VLAN0152
Spanning tree enabled protocol rstp
Root ID Priority 4248
Address 000A.F354.D2B3
Cost 23
Port 2(FastEthernet0/2)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32920 (priority 32768 sys-id-ext 152)
Address 0001.4251.1A27
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/2	Root	FWD	19	128.2	P2p

```
VLAN0153
Spanning tree enabled protocol rstp
Root ID Priority 4249
Address 000A.F354.D2B3
Cost 23
Port 2(FastEthernet0/2)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32921 (priority 32768 sys-id-ext 153)
Address 0001.4251.1A27
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/2	Root	FWD	19	128.2	P2p

```
VLAN0999
Spanning tree enabled protocol rstp
Root ID Priority 5095
Address 000A.F354.D2B3
Cost 23
Port 2(FastEthernet0/2)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 33767 (priority 32768 sys-id-ext 999)
Address 0001.4251.1A27
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/2	Root	FWD	19	128.2	P2p

```
SW-ACC-RRH-01#
SW-ACC-RRH-01#
SW-ACC-RRH-01#
SW-ACC-RRH-01#
```

SW-ACC-RRH-01#show mac address-table
Mac Address Table

Vlan	Mac Address	Type	Ports
10	0001.97b0.b601	DYNAMIC	Fa0/2
10	0003.e443.3e01	DYNAMIC	Fa0/2
10	0007.ec58.2401	DYNAMIC	Fa0/2
10	000a.f337.c401	DYNAMIC	Fa0/2
10	000a.f354.d201	DYNAMIC	Fa0/2
10	000c.cf72.dc01	DYNAMIC	Fa0/2
10	0040.0b58.e301	DYNAMIC	Fa0/2
10	00d0.978d.ce01	DYNAMIC	Fa0/2
10	00e0.8f64.2221	DYNAMIC	Fa0/2
10	00e0.8fc3.2401	DYNAMIC	Fa0/2
20	0007.ec58.2402	DYNAMIC	Fa0/2
20	000a.f354.d203	DYNAMIC	Fa0/2
20	00e0.8f64.2221	DYNAMIC	Fa0/2
30	0007.ec58.2408	DYNAMIC	Fa0/2
30	000a.f354.d209	DYNAMIC	Fa0/2
30	00e0.8f64.2221	DYNAMIC	Fa0/2
99	0007.ec58.240e	DYNAMIC	Fa0/2
99	000a.f354.d20d	DYNAMIC	Fa0/2
99	00e0.8f64.2221	DYNAMIC	Fa0/2
150	0007.ec58.241f	DYNAMIC	Fa0/2
150	00e0.8f64.2221	DYNAMIC	Fa0/2
151	0007.ec58.2420	DYNAMIC	Fa0/2
151	000a.f354.d21f	DYNAMIC	Fa0/2
151	00e0.8f64.2221	DYNAMIC	Fa0/2
152	0007.ec58.2421	DYNAMIC	Fa0/2
152	000a.f354.d220	DYNAMIC	Fa0/2
152	00e0.8f64.2221	DYNAMIC	Fa0/2
153	0007.ec58.2422	DYNAMIC	Fa0/2
153	000a.f354.d221	DYNAMIC	Fa0/2
153	00e0.8f64.2221	DYNAMIC	Fa0/2
999	00e0.8f64.2221	DYNAMIC	Fa0/2

SW-ACC-RRH-01#

SW-ACC-RRH-01#

SW-ACC-RRH-01#

SW-ACC-MKT-01

SW-ACC-MKT-01#show vlan brief

VLAN	Name	Status	Ports
1	default	active	Gig0/1, Gig0/2
10	TRANSPORT-CORE	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24
11	VLAN0011	active	
20	SERV-BACKOFFICE	active	
21	SERV-DNS	active	
22	SERV-DHCP	active	
23	SERV-MAIL	active	
24	SERV-NTP	active	
25	SERV-WEB	active	
30	SRV-DESARROLLO	active	
31	SRV-QA	active	
32	SRV-FREPROD	active	
33	SRV-PRODUCCION	active	
40	WIFI-CORP	active	
50	WIFI-VISIT	active	
99	MGMT-OOB	active	
110	Dept-Showroom	active	
111	CCTV-Showroom	active	
112	WIFI-C-SHOWROOM	active	
113	WIFI-V-SHOWROOM	active	
120	DEPT-VENTAS	active	
121	CCTV-VENTAS	active	
122	WIFI-C-VENTAS	active	
123	WIFI-V-VENTAS	active	
130	DEPT-AUDITORIO	active	
131	CCTV-AUDITORIO	active	
132	WIFI-C-AUDITORIO	active	
133	WIFI-V-AUDITORIO	active	
140	DEPT-IT	active	
141	CCTV-IT	active	
142	WIFI-C-IT	active	
143	WIFI-V-IT	active	
150	DEPT-RRHH	active	
151	CCTV-RRHH	active	
152	WIFI-C-RRHH	active	
153	WIFI-V-RRHH	active	
160	DEPT-MARKETING	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5
161	CCTV-MARKETING	active	
162	WIFI-C-MKT	active	
163	WIFI-V-MKT	active	
170	DEPT-AUDITORIA	active	
171	CCTV-AUDITORIA	active	
172	WIFI-C-AUDITORIA	active	
173	WIFI-V-AUDITORIA	active	
180	DEPT-FINANZAS	active	
181	CCTV-FINANZAS	active	
182	WIFI-C-FINANZAS	active	
183	WIFI-V-FINANZAS	active	
999	NATIVE-UNUSED	active	
1002	fdi-default	active	
1003	token-ring-default	active	
1004	ddinet-default	active	
1005	trnet-default	active	
SW-ACC-MKT-01#			
SW-ACC-MKT-01#			

SW-ACC-MKT-01#show interfaces status

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1	Enlace_Trunk_Hacia	connected	trunk	a-full	a-100	10/100BaseTX
Fa0/2		notconnect	160	auto	auto	10/100BaseTX
Fa0/3		connected	160	a-full	a-100	10/100BaseTX
Fa0/4		notconnect	160	auto	auto	10/100BaseTX
Fa0/5		notconnect	160	auto	auto	10/100BaseTX
Fa0/6		notconnect	10	auto	auto	10/100BaseTX
Fa0/7		notconnect	10	auto	auto	10/100BaseTX
Fa0/8		notconnect	10	auto	auto	10/100BaseTX
Fa0/9		notconnect	10	auto	auto	10/100BaseTX
Fa0/10		notconnect	10	auto	auto	10/100BaseTX
Fa0/11		notconnect	10	auto	auto	10/100BaseTX
Fa0/12		notconnect	10	auto	auto	10/100BaseTX
Fa0/13		notconnect	10	auto	auto	10/100BaseTX
Fa0/14		notconnect	10	auto	auto	10/100BaseTX
Fa0/15		notconnect	10	auto	auto	10/100BaseTX
Fa0/16		notconnect	10	auto	auto	10/100BaseTX
Fa0/17		notconnect	10	auto	auto	10/100BaseTX
Fa0/18		notconnect	10	auto	auto	10/100BaseTX
Fa0/19		notconnect	10	auto	auto	10/100BaseTX
Fa0/20		notconnect	10	auto	auto	10/100BaseTX
Fa0/21		notconnect	10	auto	auto	10/100BaseTX
Fa0/22		notconnect	10	auto	auto	10/100BaseTX
Fa0/23		notconnect	10	auto	auto	10/100BaseTX
Fa0/24		notconnect	10	auto	auto	10/100BaseTX
Gig0/1		notconnect	1	auto	auto	10/100/1000BaseTX
Gig0/2		notconnect	1	auto	auto	10/100/1000BaseTX

```

SW-ACC-MKT-01#show spanning-tree
VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    4106
             Address     000A.F354.D2B3
             Cost        23
             Port        1(FastEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32778 (priority 32768 sys-id-ext 10)
             Address     0001.C78D.2DD9
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface                Role Sts Cost      Prio.Nbr Type
-----
Fa0/1                    Root FWD 19        128.1   P2p

VLAN0020
  Spanning tree enabled protocol rstp
  Root ID    Priority    4116
             Address     000A.F354.D2B3
             Cost        23
             Port        1(FastEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32788 (priority 32768 sys-id-ext 20)
             Address     0001.C78D.2DD9
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface                Role Sts Cost      Prio.Nbr Type
-----
Fa0/1                    Root FWD 19        128.1   P2p

VLAN0030
  Spanning tree enabled protocol rstp
  Root ID    Priority    4126
             Address     000A.F354.D2B3
             Cost        23
             Port        1(FastEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32798 (priority 32768 sys-id-ext 30)
             Address     0001.C78D.2DD9
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface                Role Sts Cost      Prio.Nbr Type
-----
Fa0/1                    Root FWD 19        128.1   P2p

VLAN0040
  Spanning tree enabled protocol rstp
  Root ID    Priority    32808
             Address     0001.4251.1A27
             Cost        38
             Port        1(FastEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32808 (priority 32768 sys-id-ext 40)
             Address     0001.C78D.2DD9
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface                Role Sts Cost      Prio.Nbr Type
--More--

```

```

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Root FWD 19        128.1   P2p

VLAN0050
Spanning tree enabled protocol rstp
Root ID        Priority 32818
                Address 0001.4251.1A27
                Cost    38
                Port    1(FastEthernet0/1)
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID      Priority 32818 (priority 32768 sys-id-ext 50)
                Address 0001.C78D.2DD9
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
                Aging Time 20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Root FWD 19        128.1   P2p

VLAN0099
Spanning tree enabled protocol rstp
Root ID        Priority 4195
                Address 000A.F354.D2B3
                Cost    23
                Port    1(FastEthernet0/1)
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID      Priority 32867 (priority 32768 sys-id-ext 99)
                Address 0001.C78D.2DD9
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
                Aging Time 20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Root FWD 19        128.1   P2p

VLAN0160
Spanning tree enabled protocol rstp
Root ID        Priority 4256
                Address 000A.F354.D2B3
                Cost    23
                Port    1(FastEthernet0/1)
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID      Priority 32928 (priority 32768 sys-id-ext 160)
                Address 0001.C78D.2DD9
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
                Aging Time 20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/3          Desg FWD 19        128.3   P2p
Fa0/1          Root FWD 19        128.1   P2p

VLAN0161
Spanning tree enabled protocol rstp
Root ID        Priority 4257
                Address 000A.F354.D2B3
                Cost    23
                Port    1(FastEthernet0/1)
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID      Priority 32929 (priority 32768 sys-id-ext 161)
                Address 0001.C78D.2DD9

```

--More--

```

        Address 000A.F354.D2B3
        Cost    23
        Port    1(FastEthernet0/1)
        Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

    Bridge ID Priority 32929 (priority 32768 sys-id-ext 161)
        Address 0001.C78D.2DD9
        Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
        Aging Time 20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Root FWD 19        128.1    P2p

VLAN0162
Spanning tree enabled protocol rstp
Root ID      Priority 4258
        Address 000A.F354.D2B3
        Cost    23
        Port    1(FastEthernet0/1)
        Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

    Bridge ID Priority 32930 (priority 32768 sys-id-ext 162)
        Address 0001.C78D.2DD9
        Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
        Aging Time 20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Root FWD 19        128.1    P2p

VLAN0163
Spanning tree enabled protocol rstp
Root ID      Priority 4259
        Address 000A.F354.D2B3
        Cost    23
        Port    1(FastEthernet0/1)
        Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

    Bridge ID Priority 32931 (priority 32768 sys-id-ext 163)
        Address 0001.C78D.2DD9
        Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
        Aging Time 20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Root FWD 19        128.1    P2p

VLAN0999
Spanning tree enabled protocol rstp
Root ID      Priority 5095
        Address 000A.F354.D2B3
        Cost    23
        Port    1(FastEthernet0/1)
        Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

    Bridge ID Priority 33767 (priority 32768 sys-id-ext 999)
        Address 0001.C78D.2DD9
        Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
        Aging Time 20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Root FWD 19        128.1    P2p

SW-ACC-MKT-01#
SW-ACC-MKT-01#

```

SW-ACC-MKT-01#show mac address-table
Mac Address Table

Vlan	Mac Address	Type	Ports
10	0001.97b0.b601	DYNAMIC	Fa0/1
10	0003.e443.3e01	DYNAMIC	Fa0/1
10	0007.ec58.2401	DYNAMIC	Fa0/1
10	000a.f337.c401	DYNAMIC	Fa0/1
10	000a.f354.d201	DYNAMIC	Fa0/1
10	000c.cf72.dc01	DYNAMIC	Fa0/1
10	0040.0b58.e301	DYNAMIC	Fa0/1
10	00d0.978d.ce01	DYNAMIC	Fa0/1
10	00d0.ff7d.807c	DYNAMIC	Fa0/1
10	00e0.8fc3.2401	DYNAMIC	Fa0/1
20	0007.ec58.2402	DYNAMIC	Fa0/1
20	000a.f354.d203	DYNAMIC	Fa0/1
20	00d0.ff7d.807c	DYNAMIC	Fa0/1
30	0007.ec58.2408	DYNAMIC	Fa0/1
30	000a.f354.d209	DYNAMIC	Fa0/1
30	00d0.ff7d.807c	DYNAMIC	Fa0/1
40	00d0.ff7d.807c	DYNAMIC	Fa0/1
50	00d0.ff7d.807c	DYNAMIC	Fa0/1
99	0007.ec58.240e	DYNAMIC	Fa0/1
99	000a.f354.d20d	DYNAMIC	Fa0/1
99	00d0.ff7d.807c	DYNAMIC	Fa0/1
160	0007.ec58.2423	DYNAMIC	Fa0/1
160	000a.f354.d222	DYNAMIC	Fa0/1
160	00d0.ff7d.807c	DYNAMIC	Fa0/1
161	0007.ec58.2424	DYNAMIC	Fa0/1
161	000a.f354.d223	DYNAMIC	Fa0/1
161	00d0.ff7d.807c	DYNAMIC	Fa0/1
162	0007.ec58.2425	DYNAMIC	Fa0/1
162	000a.f354.d224	DYNAMIC	Fa0/1
162	00d0.ff7d.807c	DYNAMIC	Fa0/1
163	0007.ec58.2426	DYNAMIC	Fa0/1
163	000a.f354.d225	DYNAMIC	Fa0/1
163	00d0.ff7d.807c	DYNAMIC	Fa0/1
999	00d0.ff7d.807c	DYNAMIC	Fa0/1

SW-ACC-MKT-01#

SW-ACC-AUD-01

SW-ACC-AUD-01#show vlan brief

VLAN	Name	Status	Ports
1	default	active	Gig0/1, Gig0/2
10	TRANSPORT-CORE	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24
11	VLAN0011	active	
20	SERV-BACKOFFICE	active	
21	SERV-DNS	active	
22	SERV-DHCP	active	
23	SERV-MAIL	active	
24	SERV-NTP	active	
25	SERV-WEB	active	
30	SRV-DESARROLLO	active	
31	SRV-QA	active	
32	SRV-PREPROD	active	
33	SRV-PRODUCCION	active	
40	WIFI-CORP	active	
50	WIFI-VISIT	active	
99	MGMT-OOB	active	
110	Dept-Showroom	active	
111	CCTV-Showroom	active	
112	WIFI-C-SHOWROOM	active	
113	WIFI-V-SHOWROOM	active	
120	DEPT-VENTAS	active	
121	CCTV-VENTAS	active	
122	WIFI-C-VENTAS	active	
123	WIFI-V-VENTAS	active	
130	DEPT-AUDITORIO	active	
131	CCTV-AUDITORIO	active	
132	WIFI-C-AUDITORIO	active	
133	WIFI-V-AUDITORIO	active	
140	DEPT-IT	active	
141	CCTV-IT	active	
142	WIFI-C-IT	active	
143	WIFI-V-IT	active	
150	DEPT-RRHH	active	
151	CCTV-RRHH	active	
152	WIFI-C-RRHH	active	
153	WIFI-V-RRHH	active	
160	DEPT-MARKETING	active	
161	CCTV-MARKETING	active	
162	WIFI-C-MKT	active	
163	WIFI-V-MKT	active	
170	DEPT-AUDITORIA	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5
171	CCTV-AUDITORIA	active	
172	WIFI-C-AUDITORIA	active	
173	WIFI-V-AUDITORIA	active	
180	DEPT-FINANZAS	active	
181	CCTV-FINANZAS	active	
182	WIFI-C-FINANZAS	active	
183	WIFI-V-FINANZAS	active	
999	NATIVE-UNUSED	active	
1002	fdi-default	active	
1003	token-ring-default	active	
1004	fdinet-default	active	
1005	trnet-default	active	

SW-ACC-AUD-01#show interfaces status

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1	Enlace_Trunk_Hacia	connected	trunk	a-full	a-100	10/100BaseTX
Fa0/2		notconnect	170	auto	auto	10/100BaseTX
Fa0/3		connected	170	a-full	a-100	10/100BaseTX
Fa0/4		notconnect	170	auto	auto	10/100BaseTX
Fa0/5		notconnect	170	auto	auto	10/100BaseTX
Fa0/6		notconnect	10	auto	auto	10/100BaseTX
Fa0/7		notconnect	10	auto	auto	10/100BaseTX
Fa0/8		notconnect	10	auto	auto	10/100BaseTX
Fa0/9		notconnect	10	auto	auto	10/100BaseTX
Fa0/10		notconnect	10	auto	auto	10/100BaseTX
Fa0/11		notconnect	10	auto	auto	10/100BaseTX
Fa0/12		notconnect	10	auto	auto	10/100BaseTX
Fa0/13		notconnect	10	auto	auto	10/100BaseTX
Fa0/14		notconnect	10	auto	auto	10/100BaseTX
Fa0/15		notconnect	10	auto	auto	10/100BaseTX
Fa0/16		notconnect	10	auto	auto	10/100BaseTX
Fa0/17		notconnect	10	auto	auto	10/100BaseTX
Fa0/18		notconnect	10	auto	auto	10/100BaseTX
Fa0/19		notconnect	10	auto	auto	10/100BaseTX
Fa0/20		notconnect	10	auto	auto	10/100BaseTX
Fa0/21		notconnect	10	auto	auto	10/100BaseTX
Fa0/22		notconnect	10	auto	auto	10/100BaseTX
Fa0/23		notconnect	10	auto	auto	10/100BaseTX
Fa0/24		notconnect	10	auto	auto	10/100BaseTX
Gig0/1		notconnect	1	auto	auto	10/100/1000BaseTX
Gig0/2		notconnect	1	auto	auto	10/100/1000BaseTX

SW-ACC-AUD-01#

```

SW-ACC-AUD-01#show spanning-tree
VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    4106
             Address     000A.F354.D2B3
             Cost        23
             Port        1(FastEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32778 (priority 32768 sys-id-ext 10)
             Address     0001.C9E0.A0BE
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface                Role Sts Cost      Prio.Nbr Type
-----
Fa0/1                    Root FWD 19       128.1    P2p

VLAN0020
  Spanning tree enabled protocol rstp
  Root ID    Priority    4116
             Address     000A.F354.D2B3
             Cost        23
             Port        1(FastEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32788 (priority 32768 sys-id-ext 20)
             Address     0001.C9E0.A0BE
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface                Role Sts Cost      Prio.Nbr Type
-----
Fa0/1                    Root FWD 19       128.1    P2p

VLAN0030
  Spanning tree enabled protocol rstp
  Root ID    Priority    4126
             Address     000A.F354.D2B3
             Cost        23
             Port        1(FastEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32798 (priority 32768 sys-id-ext 30)
             Address     0001.C9E0.A0BE
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface                Role Sts Cost      Prio.Nbr Type
-----
Fa0/1                    Root FWD 19       128.1    P2p

VLAN0040
  Spanning tree enabled protocol rstp
  Root ID    Priority    32808
             Address     0001.97B0.B634
             Cost        19
             Port        1(FastEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32808 (priority 32768 sys-id-ext 40)
             Address     0001.C9E0.A0BE
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface                Role Sts Cost      Prio.Nbr Type
-----
Fa0/1                    Root FWD 19       128.1    P2p

```

--More--

```

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Root FWD 19        128.1   P2p

VLAN0050
Spanning tree enabled protocol rstp
Root ID        Priority    32818
                Address    0001.97B0.B634
                Cost      19
                Port      1(FastEthernet0/1)
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID      Priority    32818 (priority 32768 sys-id-ext 50)
                Address    0001.C9E0.A0BE
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
                Aging Time 20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Root FWD 19        128.1   P2p

VLAN0099
Spanning tree enabled protocol rstp
Root ID        Priority    4195
                Address    000A.F354.D2B3
                Cost      23
                Port      1(FastEthernet0/1)
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID      Priority    32867 (priority 32768 sys-id-ext 99)
                Address    0001.C9E0.A0BE
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
                Aging Time 20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Root FWD 19        128.1   P2p

VLAN0170
Spanning tree enabled protocol rstp
Root ID        Priority    4266
                Address    000A.F354.D2B3
                Cost      23
                Port      1(FastEthernet0/1)
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID      Priority    32938 (priority 32768 sys-id-ext 170)
                Address    0001.C9E0.A0BE
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
                Aging Time 20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/3          Desg FWD 19        128.3   P2p
Fa0/1          Root FWD 19        128.1   P2p

VLAN0171
Spanning tree enabled protocol rstp
Root ID        Priority    4267
                Address    000A.F354.D2B3
                Cost      23
                Port      1(FastEthernet0/1)
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID      Priority    32939 (priority 32768 sys-id-ext 171)
                Address    0001.C9E0.A0BE
--More-- |

```

```

Address      000A.F354.D2B3
Cost         23
Port         1(FastEthernet0/1)
Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID    Priority 32939 (priority 32768 sys-id-ext 171)
Address      0001.C9E0.A0BE
Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
Aging Time   20

Interface    Role Sts Cost      Prio.Nbr Type
-----
Fa0/1        Root FWD 19        128.1    P2p

VLAN0172
Spanning tree enabled protocol rstp
Root ID      Priority 4268
Address      000A.F354.D2B3
Cost         23
Port         1(FastEthernet0/1)
Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID    Priority 32940 (priority 32768 sys-id-ext 172)
Address      0001.C9E0.A0BE
Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
Aging Time   20

Interface    Role Sts Cost      Prio.Nbr Type
-----
Fa0/1        Root FWD 19        128.1    P2p

VLAN0173
Spanning tree enabled protocol rstp
Root ID      Priority 4269
Address      000A.F354.D2B3
Cost         23
Port         1(FastEthernet0/1)
Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID    Priority 32941 (priority 32768 sys-id-ext 173)
Address      0001.C9E0.A0BE
Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
Aging Time   20

Interface    Role Sts Cost      Prio.Nbr Type
-----
Fa0/1        Root FWD 19        128.1    P2p

VLAN0999
Spanning tree enabled protocol rstp
Root ID      Priority 5095
Address      000A.F354.D2B3
Cost         23
Port         1(FastEthernet0/1)
Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID    Priority 33767 (priority 32768 sys-id-ext 999)
Address      0001.C9E0.A0BE
Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
Aging Time   20

Interface    Role Sts Cost      Prio.Nbr Type
-----
Fa0/1        Root FWD 19        128.1    P2p

SW-ACC-AUD-01#
SW-ACC-AUD-01#

```

SW-ACC-AUD-01#show mac address-table
Mac Address Table

Vlan	Mac Address	Type	Ports
10	0001.97b0.b601	DYNAMIC	Fa0/1
10	0003.e443.3e01	DYNAMIC	Fa0/1
10	0006.2aa4.28ad	DYNAMIC	Fa0/1
10	0007.ec58.2401	DYNAMIC	Fa0/1
10	000a.f337.c401	DYNAMIC	Fa0/1
10	000a.f354.d201	DYNAMIC	Fa0/1
10	000c.cf72.dc01	DYNAMIC	Fa0/1
10	0040.0b58.e301	DYNAMIC	Fa0/1
10	00d0.978d.ce01	DYNAMIC	Fa0/1
10	00e0.8fc3.2401	DYNAMIC	Fa0/1
20	0006.2aa4.28ad	DYNAMIC	Fa0/1
20	0007.ec58.2402	DYNAMIC	Fa0/1
20	000a.f354.d203	DYNAMIC	Fa0/1
30	0006.2aa4.28ad	DYNAMIC	Fa0/1
30	0007.ec58.2408	DYNAMIC	Fa0/1
30	000a.f354.d209	DYNAMIC	Fa0/1
40	0006.2aa4.28ad	DYNAMIC	Fa0/1
50	0006.2aa4.28ad	DYNAMIC	Fa0/1
99	0006.2aa4.28ad	DYNAMIC	Fa0/1
99	0007.ec58.240e	DYNAMIC	Fa0/1
99	000a.f354.d20d	DYNAMIC	Fa0/1
170	0006.2aa4.28ad	DYNAMIC	Fa0/1
170	0007.ec58.2427	DYNAMIC	Fa0/1
170	000a.f354.d226	DYNAMIC	Fa0/1
171	0006.2aa4.28ad	DYNAMIC	Fa0/1
171	0007.ec58.2428	DYNAMIC	Fa0/1
171	000a.f354.d227	DYNAMIC	Fa0/1
172	0006.2aa4.28ad	DYNAMIC	Fa0/1
172	0007.ec58.2429	DYNAMIC	Fa0/1
172	000a.f354.d228	DYNAMIC	Fa0/1
173	0006.2aa4.28ad	DYNAMIC	Fa0/1
173	0007.ec58.242a	DYNAMIC	Fa0/1
173	000a.f354.d229	DYNAMIC	Fa0/1
999	0006.2aa4.28ad	DYNAMIC	Fa0/1

SW-ACC-AUD-01#
SW-ACC-100-01#

SW-ACC-FIN-01

SW-ACC-FIN-01#show vlan brief

VLAN	Name	Status	Ports
1	default	active	Gig0/1, Gig0/2
10	TRANSPORT-CORE	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24
11	VLAN0011	active	
20	SERV-BACKOFFICE	active	
21	SERV-DNS	active	
22	SERV-DHCP	active	
23	SERV-MAIL	active	
24	SERV-NTP	active	
25	SERV-WEB	active	
30	SRV-DESARROLLO	active	
31	SRV-QA	active	
32	SRV-PREPROD	active	
33	SRV-PRODUCCION	active	
40	WIFI-CORP	active	
50	WIFI-VISIT	active	
99	MGMT-OOB	active	
110	Dept-Showroom	active	
111	CCTV-Showroom	active	
112	WIFI-C-SHOWROOM	active	
113	WIFI-V-SHOWROOM	active	
120	DEPT-VENTAS	active	
121	CCTV-VENTAS	active	
122	WIFI-C-VENTAS	active	
123	WIFI-V-VENTAS	active	
130	DEPT-AUDITORIO	active	
131	CCTV-AUDITORIO	active	
132	WIFI-C-AUDITORIO	active	
133	WIFI-V-AUDITORIO	active	
140	DEPT-IT	active	
141	CCTV-IT	active	
142	WIFI-C-IT	active	
143	WIFI-V-IT	active	
150	DEPT-RRHH	active	
151	CCTV-RRHH	active	
152	WIFI-C-RRHH	active	
153	WIFI-V-RRHH	active	
160	DEPT-MARKETING	active	
161	CCTV-MARKETING	active	
162	WIFI-C-MKT	active	
163	WIFI-V-MKT	active	
170	DEPT-AUDITORIA	active	
171	CCTV-AUDITORIA	active	
172	WIFI-C-AUDITORIA	active	
173	WIFI-V-AUDITORIA	active	
180	DEPT-FINANZAS	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5
181	CCTV-FINANZAS	active	
182	WIFI-C-FINANZAS	active	
183	WIFI-V-FINANZAS	active	
999	NATIVE-UNUSED	active	
1002	fdi-default	active	
1003	token-ring-default	active	
1004	fdinet-default	active	
1005	trnet-default	active	

SW-ACC-FIN-01#

SW-ACC-FIN-01#show interfaces status

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1	Enlace_Trunk_Hacia	connected	trunk	a-full	a-100	10/100BaseTX
Fa0/2		notconnect	180	auto	auto	10/100BaseTX
Fa0/3		notconnect	180	auto	auto	10/100BaseTX
Fa0/4		connected	180	a-full	a-100	10/100BaseTX
Fa0/5		notconnect	180	auto	auto	10/100BaseTX
Fa0/6		notconnect	10	auto	auto	10/100BaseTX
Fa0/7		notconnect	10	auto	auto	10/100BaseTX
Fa0/8		notconnect	10	auto	auto	10/100BaseTX
Fa0/9		notconnect	10	auto	auto	10/100BaseTX
Fa0/10		notconnect	10	auto	auto	10/100BaseTX
Fa0/11		notconnect	10	auto	auto	10/100BaseTX
Fa0/12		notconnect	10	auto	auto	10/100BaseTX
Fa0/13		notconnect	10	auto	auto	10/100BaseTX
Fa0/14		notconnect	10	auto	auto	10/100BaseTX
Fa0/15		notconnect	10	auto	auto	10/100BaseTX
Fa0/16		notconnect	10	auto	auto	10/100BaseTX
Fa0/17		notconnect	10	auto	auto	10/100BaseTX
Fa0/18		notconnect	10	auto	auto	10/100BaseTX
Fa0/19		notconnect	10	auto	auto	10/100BaseTX
Fa0/20		notconnect	10	auto	auto	10/100BaseTX
Fa0/21		notconnect	10	auto	auto	10/100BaseTX
Fa0/22		notconnect	10	auto	auto	10/100BaseTX
Fa0/23		notconnect	10	auto	auto	10/100BaseTX
Fa0/24		notconnect	10	auto	auto	10/100BaseTX
Gig0/1		notconnect	1	auto	auto	10/100/1000BaseTX
Gig0/2		notconnect	1	auto	auto	10/100/1000BaseTX

SW-ACC-FIN-01#

SW-ACC-FIN-01#

```

SW-ACC-FIN-01#show spanning-tree
VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    4106
             Address    000A.F354.D2B3
             Cost       23
             Port       1(FastEthernet0/1)
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32778 (priority 32768 sys-id-ext 10)
             Address    0002.4A29.9468
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 20

Interface    Role Sts Cost      Prio.Nbr Type
-----
Fa0/1        Root FWD 19        128.1    P2p

VLAN0011
  Spanning tree enabled protocol rstp
  Root ID    Priority    32779
             Address    0002.4A29.9468
             This bridge is the root
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32779 (priority 32768 sys-id-ext 11)
             Address    0002.4A29.9468
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 20

Interface    Role Sts Cost      Prio.Nbr Type
-----
Fa0/1        Desg FWD 19        128.1    P2p

VLAN0020
  Spanning tree enabled protocol rstp
  Root ID    Priority    4116
             Address    000A.F354.D2B3
             Cost       23
             Port       1(FastEthernet0/1)
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32788 (priority 32768 sys-id-ext 20)
             Address    0002.4A29.9468
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 20

Interface    Role Sts Cost      Prio.Nbr Type
-----
Fa0/1        Root FWD 19        128.1    P2p

VLAN0021
  Spanning tree enabled protocol rstp
  Root ID    Priority    4117
             Address    000A.F354.D2B3
             Cost       23
             Port       1(FastEthernet0/1)
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32789 (priority 32768 sys-id-ext 21)
             Address    0002.4A29.9468
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 20

Interface    Role Sts Cost      Prio.Nbr Type
-----
--More--

```

```

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Root FWD 19        128.1   P2p

VLAN0022
Spanning tree enabled protocol rstp
Root ID        Priority    4118
                Address    000A.F354.D2B3
                Cost      23
                Port      1(FastEthernet0/1)
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID      Priority    32790 (priority 32768 sys-id-ext 22)
                Address    0002.4A29.9468
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
                Aging Time 20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Root FWD 19        128.1   P2p

VLAN0023
Spanning tree enabled protocol rstp
Root ID        Priority    4119
                Address    000A.F354.D2B3
                Cost      23
                Port      1(FastEthernet0/1)
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID      Priority    32791 (priority 32768 sys-id-ext 23)
                Address    0002.4A29.9468
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
                Aging Time 20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Root FWD 19        128.1   P2p

VLAN0024
Spanning tree enabled protocol rstp
Root ID        Priority    4120
                Address    000A.F354.D2B3
                Cost      23
                Port      1(FastEthernet0/1)
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID      Priority    32792 (priority 32768 sys-id-ext 24)
                Address    0002.4A29.9468
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
                Aging Time 20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Root FWD 19        128.1   P2p

VLAN0025
Spanning tree enabled protocol rstp
Root ID        Priority    4121
                Address    000A.F354.D2B3
                Cost      23
                Port      1(FastEthernet0/1)
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID      Priority    32793 (priority 32768 sys-id-ext 25)
                Address    0002.4A29.9468
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

--More--

```

```

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Root FWD 19        128.1   P2p

VLAN0030
Spanning tree enabled protocol rstp
Root ID        Priority 4126
                Address 000A.F354.D2B3
                Cost    23
                Port    1(FastEthernet0/1)
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID      Priority 32798 (priority 32768 sys-id-ext 30)
                Address 0002.4A29.9468
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
                Aging Time 20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Root FWD 19        128.1   P2p

VLAN0031
Spanning tree enabled protocol rstp
Root ID        Priority 4127
                Address 000A.F354.D2B3
                Cost    23
                Port    1(FastEthernet0/1)
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID      Priority 32799 (priority 32768 sys-id-ext 31)
                Address 0002.4A29.9468
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
                Aging Time 20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Root FWD 19        128.1   P2p

VLAN0032
Spanning tree enabled protocol rstp
Root ID        Priority 4128
                Address 000A.F354.D2B3
                Cost    23
                Port    1(FastEthernet0/1)
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID      Priority 32800 (priority 32768 sys-id-ext 32)
                Address 0002.4A29.9468
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
                Aging Time 20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Root FWD 19        128.1   P2p

VLAN0033
Spanning tree enabled protocol rstp
Root ID        Priority 4129
                Address 000A.F354.D2B3
                Cost    23
                Port    1(FastEthernet0/1)
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

```

--More-- |

```

Spanning tree enabled protocol rstp
Root ID    Priority    32808
Address    0002.4A29.9468
This bridge is the root
Hello Time 2 sec    Max Age 20 sec    Forward Delay 15 sec

Bridge ID   Priority    32808 (priority 32768 sys-id-ext 40)
Address    0002.4A29.9468
Hello Time 2 sec    Max Age 20 sec    Forward Delay 15 sec
Aging Time 20

Interface   Role Sts Cost      Prio.Nbr Type
-----
Fa0/1       Desg FWD 19      128.1    P2p

VLAN0050
Spanning tree enabled protocol rstp
Root ID    Priority    32818
Address    0002.4A29.9468
This bridge is the root
Hello Time 2 sec    Max Age 20 sec    Forward Delay 15 sec

Bridge ID   Priority    32818 (priority 32768 sys-id-ext 50)
Address    0002.4A29.9468
Hello Time 2 sec    Max Age 20 sec    Forward Delay 15 sec
Aging Time 20

Interface   Role Sts Cost      Prio.Nbr Type
-----
Fa0/1       Desg FWD 19      128.1    P2p

VLAN0099
Spanning tree enabled protocol rstp
Root ID    Priority    4195
Address    000A.F354.D2B3
Cost       23
Port       1(FastEthernet0/1)
Hello Time 2 sec    Max Age 20 sec    Forward Delay 15 sec

Bridge ID   Priority    32867 (priority 32768 sys-id-ext 99)
Address    0002.4A29.9468
Hello Time 2 sec    Max Age 20 sec    Forward Delay 15 sec
Aging Time 20

Interface   Role Sts Cost      Prio.Nbr Type
-----
Fa0/1       Root FWD 19      128.1    P2p

VLAN0180
Spanning tree enabled protocol rstp
Root ID    Priority    4276
Address    000A.F354.D2B3
Cost       23
Port       1(FastEthernet0/1)
Hello Time 2 sec    Max Age 20 sec    Forward Delay 15 sec

Bridge ID   Priority    32948 (priority 32768 sys-id-ext 180)
Address    0002.4A29.9468
Hello Time 2 sec    Max Age 20 sec    Forward Delay 15 sec
Aging Time 20

Interface   Role Sts Cost      Prio.Nbr Type
-----
Fa0/1       Root FWD 19      128.1    P2p
Fa0/4       Desg FWD 19      128.4    P2p

VLAN0181
--More-- |

```

```

Spanning tree enabled protocol rstp
Root ID      Priority    4277
Address      000A.F354.D2B3
Cost         23
Port         1(FastEthernet0/1)
Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID     Priority    32949  (priority 32768 sys-id-ext 181)
Address       0002.4A29.9468
Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec
Aging Time    20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Root FWD 19        128.1   P2p

VLAN0182
Spanning tree enabled protocol rstp
Root ID      Priority    4278
Address      000A.F354.D2B3
Cost         23
Port         1(FastEthernet0/1)
Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID     Priority    32950  (priority 32768 sys-id-ext 182)
Address       0002.4A29.9468
Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec
Aging Time    20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Root FWD 19        128.1   P2p

VLAN0183
Spanning tree enabled protocol rstp
Root ID      Priority    4279
Address      000A.F354.D2B3
Cost         23
Port         1(FastEthernet0/1)
Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID     Priority    32951  (priority 32768 sys-id-ext 183)
Address       0002.4A29.9468
Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec
Aging Time    20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Root FWD 19        128.1   P2p

VLAN0999
Spanning tree enabled protocol rstp
Root ID      Priority    5095
Address      000A.F354.D2B3
Cost         23
Port         1(FastEthernet0/1)
Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID     Priority    33767  (priority 32768 sys-id-ext 999)
Address       0002.4A29.9468
Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec
Aging Time    20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Root FWD 19        128.1   P2p

```

SW-ACC-FIN-01#

SW-ACC-FIN-01#show mac address-table
Mac Address Table

Vlan	Mac Address	Type	Ports
10	0001.97b0.b601	DYNAMIC	Fa0/1
10	0002.4a75.babc	DYNAMIC	Fa0/1
10	0003.e443.3e01	DYNAMIC	Fa0/1
10	0007.ec58.2401	DYNAMIC	Fa0/1
10	000a.f354.d201	DYNAMIC	Fa0/1
10	000a.f354.d201	DYNAMIC	Fa0/1
10	000c.cf72.dc01	DYNAMIC	Fa0/1
10	0040.0b58.e301	DYNAMIC	Fa0/1
10	00d0.978d.ce01	DYNAMIC	Fa0/1
10	00e0.8fc3.2401	DYNAMIC	Fa0/1
20	0002.4a75.babc	DYNAMIC	Fa0/1
20	0007.ec58.2402	DYNAMIC	Fa0/1
20	000a.f354.d203	DYNAMIC	Fa0/1
21	0002.4a75.babc	DYNAMIC	Fa0/1
21	0007.ec58.2403	DYNAMIC	Fa0/1
21	000a.f354.d204	DYNAMIC	Fa0/1
22	0002.4a75.babc	DYNAMIC	Fa0/1
22	0007.ec58.2404	DYNAMIC	Fa0/1
22	000a.f354.d205	DYNAMIC	Fa0/1
23	0002.4a75.babc	DYNAMIC	Fa0/1
23	0007.ec58.2405	DYNAMIC	Fa0/1
23	000a.f354.d206	DYNAMIC	Fa0/1
24	0002.4a75.babc	DYNAMIC	Fa0/1
24	0007.ec58.2406	DYNAMIC	Fa0/1
24	000a.f354.d207	DYNAMIC	Fa0/1
25	0002.4a75.babc	DYNAMIC	Fa0/1
25	0007.ec58.2407	DYNAMIC	Fa0/1
25	000a.f354.d208	DYNAMIC	Fa0/1
30	0002.4a75.babc	DYNAMIC	Fa0/1
30	0007.ec58.2408	DYNAMIC	Fa0/1
30	000a.f354.d209	DYNAMIC	Fa0/1
31	0002.4a75.babc	DYNAMIC	Fa0/1
31	0007.ec58.2409	DYNAMIC	Fa0/1
31	000a.f354.d20a	DYNAMIC	Fa0/1
32	0002.4a75.babc	DYNAMIC	Fa0/1
32	0007.ec58.240a	DYNAMIC	Fa0/1
32	000a.f354.d20b	DYNAMIC	Fa0/1
33	0002.4a75.babc	DYNAMIC	Fa0/1
33	0007.ec58.240b	DYNAMIC	Fa0/1
33	000a.f354.d20c	DYNAMIC	Fa0/1
99	0002.4a75.babc	DYNAMIC	Fa0/1
99	0007.ec58.240e	DYNAMIC	Fa0/1
99	000a.f354.d20d	DYNAMIC	Fa0/1
180	0002.4a75.babc	DYNAMIC	Fa0/1
180	0007.ec58.242b	DYNAMIC	Fa0/1
180	000a.f354.d22a	DYNAMIC	Fa0/1
181	0002.4a75.babc	DYNAMIC	Fa0/1
181	0007.ec58.242c	DYNAMIC	Fa0/1
181	000a.f354.d22b	DYNAMIC	Fa0/1
182	0002.4a75.babc	DYNAMIC	Fa0/1
182	0007.ec58.242d	DYNAMIC	Fa0/1
182	000a.f354.d22c	DYNAMIC	Fa0/1
183	0002.4a75.babc	DYNAMIC	Fa0/1
183	0007.ec58.242e	DYNAMIC	Fa0/1
183	000a.f354.d22d	DYNAMIC	Fa0/1
999	0002.4a75.babc	DYNAMIC	Fa0/1

SW-ACC-FIN-01#

SW-SEDE-01

```
SW-SEDE-01#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/2
210	SEDE1-VENTAS	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4
211	SEDE1-SHOWROOM	active	
212	SEDE1-CCTV	active	
213	SEDE1-WIFI-CORP	active	
214	SEDE1-WIFI-VISIT	active	
215	SEDE1-MGMT	active	
999	NATIVE-UNUSED	active	
1002	fdi-default	active	
1003	token-ring-default	active	
1004	fdinet-default	active	
1005	trinet-default	active	

```
SW-SEDE-01#
SW-SEDE-01#
```

```
SW-SEDE-01#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Gig0/1	on	802.1q	trunking	999

Port Vlan allowed on trunk
Gig0/1 1-1005

Port Vlan allowed and active in management domain
Gig0/1 1,210,211,212,213,214,215,999

Port Vlan in spanning tree forwarding state and not pruned
Gig0/1 1,210,211,212,213,214,215,999

```
SW-SEDE-01#
```

```
SW-SEDE-01#show interfaces status
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1	PC-PRUEBA-SEDE1	connected	210	a-full	a-100	10/100BaseTX
Fa0/2		notconnect	210	auto	auto	10/100BaseTX
Fa0/3		notconnect	210	auto	auto	10/100BaseTX
Fa0/4		notconnect	210	auto	auto	10/100BaseTX
Fa0/5		notconnect	1	auto	auto	10/100BaseTX
Fa0/6		notconnect	1	auto	auto	10/100BaseTX
Fa0/7		notconnect	1	auto	auto	10/100BaseTX
Fa0/8		notconnect	1	auto	auto	10/100BaseTX
Fa0/9		notconnect	1	auto	auto	10/100BaseTX
Fa0/10		notconnect	1	auto	auto	10/100BaseTX
Fa0/11		notconnect	1	auto	auto	10/100BaseTX
Fa0/12		notconnect	1	auto	auto	10/100BaseTX
Fa0/13		notconnect	1	auto	auto	10/100BaseTX
Fa0/14		notconnect	1	auto	auto	10/100BaseTX
Fa0/15		notconnect	1	auto	auto	10/100BaseTX
Fa0/16		notconnect	1	auto	auto	10/100BaseTX
Fa0/17		notconnect	1	auto	auto	10/100BaseTX
Fa0/18		notconnect	1	auto	auto	10/100BaseTX
Fa0/19		notconnect	1	auto	auto	10/100BaseTX
Fa0/20		notconnect	1	auto	auto	10/100BaseTX
Fa0/21		notconnect	1	auto	auto	10/100BaseTX
Fa0/22		notconnect	1	auto	auto	10/100BaseTX
Fa0/23		notconnect	1	auto	auto	10/100BaseTX
Fa0/24		notconnect	1	auto	auto	10/100BaseTX
Gig0/1		connected	trunk	a-full	a-100	10/100/1000BaseTX
Gig0/2		notconnect	1	auto	auto	10/100/1000BaseTX

SW-SEDE-02

```
SW-SEDE-02>show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/2
220 SEDE2-VENTAS	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4
221 SEDE2-SHOWROOM	active	Fa0/5, Fa0/6, Fa0/7, Fa0/8
222 SEDE2-CCTV	active	
223 SEDE2-WIFI-CORP	active	
224 SEDE2-WIFI-VISIT	active	
225 SEDE2-MGMT	active	
999 NATIVE-UNUSED	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
SW-SEDE-02>
SW-SEDE-02>
```

```
SW-SEDE-02#show interfaces status
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1		connected	220	a-full	a-100	10/100BaseTX
Fa0/2		notconnect	220	auto	auto	10/100BaseTX
Fa0/3		notconnect	220	auto	auto	10/100BaseTX
Fa0/4		notconnect	220	auto	auto	10/100BaseTX
Fa0/5		connected	221	a-full	a-100	10/100BaseTX
Fa0/6		notconnect	221	auto	auto	10/100BaseTX
Fa0/7		notconnect	221	auto	auto	10/100BaseTX
Fa0/8		notconnect	221	auto	auto	10/100BaseTX
Fa0/9		notconnect	1	auto	auto	10/100BaseTX
Fa0/10		notconnect	1	auto	auto	10/100BaseTX
Fa0/11		notconnect	1	auto	auto	10/100BaseTX
Fa0/12		notconnect	1	auto	auto	10/100BaseTX
Fa0/13		notconnect	1	auto	auto	10/100BaseTX
Fa0/14		notconnect	1	auto	auto	10/100BaseTX
Fa0/15		notconnect	1	auto	auto	10/100BaseTX
Fa0/16		notconnect	1	auto	auto	10/100BaseTX
Fa0/17		notconnect	1	auto	auto	10/100BaseTX
Fa0/18		notconnect	1	auto	auto	10/100BaseTX
Fa0/19		notconnect	1	auto	auto	10/100BaseTX
Fa0/20		notconnect	1	auto	auto	10/100BaseTX
Fa0/21		notconnect	1	auto	auto	10/100BaseTX
Fa0/22		notconnect	1	auto	auto	10/100BaseTX
Fa0/23		notconnect	1	auto	auto	10/100BaseTX
Fa0/24		notconnect	1	auto	auto	10/100BaseTX
Gig0/1		connected	trunk	a-full	a-100	10/100/1000BaseTX
Gig0/2		notconnect	1	auto	auto	10/100/1000BaseTX

```
SW-SEDE-02#
```

```
SW-SEDE-02#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Gig0/1	on	802.1q	trunking	999

Port Gig0/1 Vlans allowed on trunk 1-1005

Port Gig0/1 Vlans allowed and active in management domain 1,220,221,222,223,224,225,999

Port Gig0/1 Vlans in spanning tree forwarding state and not pruned 1,220,221,222,223,224,225,999

```
SW-SEDE-02#
```

SW-SEDE-03

```
SW-SEDE-03#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/2
230 SEDE3-VENTAS	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5
231 SEDE3-SHOWROOM	active	
232 SEDE3-CCTV	active	
233 SEDE3-WIFI-CORP	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10
234 SEDE3-WIFI-VISIT	active	
235 SEDE3-MGMT	active	
999 NATIVE-UNUSED	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	

```
SW-SEDE-03#show inter
```

```
SW-SEDE-03#show interfaces trun
```

```
SW-SEDE-03#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Gig0/1	on	802.1q	trunking	999
Port	Vlans allowed on trunk			
Gig0/1	1-1005			
Port	Vlans allowed and active in management domain			
Gig0/1	1,230,231,232,233,234,235,999			
Port	Vlans in spanning tree forwarding state and not pruned			
Gig0/1	1,230,231,232,233,234,235,999			

```
SW-SEDE-03#show interfaces status
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1		connected	230	a-full	a-100	10/100BaseTX
Fa0/2		notconnect	230	auto	auto	10/100BaseTX
Fa0/3		notconnect	230	auto	auto	10/100BaseTX
Fa0/4		notconnect	230	auto	auto	10/100BaseTX
Fa0/5		notconnect	230	auto	auto	10/100BaseTX
Fa0/6		connected	233	a-full	a-100	10/100BaseTX
Fa0/7		notconnect	233	auto	auto	10/100BaseTX
Fa0/8		notconnect	233	auto	auto	10/100BaseTX
Fa0/9		notconnect	233	auto	auto	10/100BaseTX
Fa0/10		notconnect	233	auto	auto	10/100BaseTX
Fa0/11		notconnect	1	auto	auto	10/100BaseTX
Fa0/12		notconnect	1	auto	auto	10/100BaseTX
Fa0/13		notconnect	1	auto	auto	10/100BaseTX
Fa0/14		notconnect	1	auto	auto	10/100BaseTX
Fa0/15		notconnect	1	auto	auto	10/100BaseTX
Fa0/16		notconnect	1	auto	auto	10/100BaseTX
Fa0/17		notconnect	1	auto	auto	10/100BaseTX
Fa0/18		notconnect	1	auto	auto	10/100BaseTX
Fa0/19		notconnect	1	auto	auto	10/100BaseTX
Fa0/20		notconnect	1	auto	auto	10/100BaseTX
Fa0/21		notconnect	1	auto	auto	10/100BaseTX
Fa0/22		notconnect	1	auto	auto	10/100BaseTX
Fa0/23		notconnect	1	auto	auto	10/100BaseTX
Fa0/24		notconnect	1	auto	auto	10/100BaseTX
Gig0/1		connected	trunk	a-full	a-100	10/100/1000BaseTX
Gig0/2		notconnect	1	auto	auto	10/100/1000BaseTX

```
SW-SEDE-03#
```

RT-BORDER-01

```
RT-BORDER-01#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

Gateway of last resort is 200.0.0.2 to network 0.0.0.0

```
10.0.0.0/8 is variably subnetted, 44 subnets, 4 masks
S    10.0.0.0/8 is directly connected, Null0
C    10.0.0.0/30 is directly connected, GigabitEthernet0/0/0
L    10.0.0.1/32 is directly connected, GigabitEthernet0/0/0
C    10.0.0.4/30 is directly connected, GigabitEthernet0/0/1
L    10.0.0.5/32 is directly connected, GigabitEthernet0/0/1
O    10.10.10.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    10.10.11.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    10.10.20.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    10.10.21.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    10.10.22.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    10.10.23.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    10.10.30.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    10.10.31.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    10.10.32.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    10.10.33.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    10.10.40.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    10.10.41.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    10.10.42.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    10.10.43.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    10.10.50.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    10.10.51.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    10.10.52.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    10.10.53.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    10.10.60.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    10.10.61.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    10.10.62.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    10.10.63.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    10.10.70.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    10.10.71.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    10.10.72.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    10.10.73.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    10.10.80.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    10.10.81.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    10.10.82.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    10.10.83.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
C    10.255.0.0/30 is directly connected, Serial0/1/0
L    10.255.0.1/32 is directly connected, Serial0/1/0
C    10.255.0.2/32 is directly connected, Serial0/1/0
C    10.255.0.4/30 is directly connected, Serial0/1/1
L    10.255.0.5/32 is directly connected, Serial0/1/1
C    10.255.0.6/32 is directly connected, Serial0/1/1
C    10.255.0.8/30 is directly connected, Serial0/2/1
L    10.255.0.9/32 is directly connected, Serial0/2/1
C    10.255.0.10/32 is directly connected, Serial0/2/1
172.16.0.0/24 is subnetted, 18 subnets
B    172.16.10.0/24 [20/0] via 10.255.0.2, 00:00:00
B    172.16.11.0/24 [20/0] via 10.255.0.2, 00:00:00
B    172.16.12.0/24 [20/0] via 10.255.0.2, 00:00:00
B    172.16.13.0/24 [20/0] via 10.255.0.2, 00:00:00
B    172.16.14.0/24 [20/0] via 10.255.0.2, 00:00:00
B    172.16.15.0/24 [20/0] via 10.255.0.2, 00:00:00
B    172.16.20.0/24 [20/0] via 10.255.0.6, 00:00:00
B    172.16.21.0/24 [20/0] via 10.255.0.6, 00:00:00
B    172.16.22.0/24 [20/0] via 10.255.0.6, 00:00:00
--More--
```

```
B    172.16.24.0/24 [20/0] via 10.255.0.6, 00:00:00
B    172.16.23.0/24 [20/0] via 10.255.0.6, 00:00:00
B    172.16.24.0/24 [20/0] via 10.255.0.6, 00:00:00
B    172.16.25.0/24 [20/0] via 10.255.0.6, 00:00:00
B    172.16.30.0/24 [20/0] via 10.255.0.10, 00:00:00
B    172.16.31.0/24 [20/0] via 10.255.0.10, 00:00:00
B    172.16.32.0/24 [20/0] via 10.255.0.10, 00:00:00
B    172.16.33.0/24 [20/0] via 10.255.0.10, 00:00:00
B    172.16.34.0/24 [20/0] via 10.255.0.10, 00:00:00
B    172.16.35.0/24 [20/0] via 10.255.0.10, 00:00:00
S    192.168.0.0/16 is directly connected, Null0
O    192.168.10.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    192.168.20.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    192.168.21.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    192.168.22.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    192.168.23.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    192.168.24.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    192.168.25.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    192.168.30.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    192.168.31.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    192.168.32.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    192.168.33.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
O    192.168.99.0/24 [110/2] via 10.0.0.2, 00:11:40, GigabitEthernet0/0/0
200.0.0.0/24 is variably subnetted, 3 subnets, 2 masks
C    200.0.0.0/30 is directly connected, Serial0/2/0
L    200.0.0.1/32 is directly connected, Serial0/2/0
C    200.0.0.2/32 is directly connected, Serial0/2/0
S*   0.0.0.0/0 [1/0] via 200.0.0.2
```

RT-BORDER-01#

```
RT-BORDER-01#show ip ospf nei
RT-BORDER-01#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.99.2	1	FULL/BDR	00:00:32	10.0.0.2	GigabitEthernet0/0/0
192.168.99.3	1	FULL/BDR	00:00:32	10.0.0.6	GigabitEthernet0/0/1

```
RT-BORDER-01#show ip bgp summary
BGP router identifier 1.1.1.1, local AS number 65000
BGP table version is 21, main routing table version 6
20 network entries using 2640 bytes of memory
20 path entries using 1040 bytes of memory
18/18 BGP path/bestpath attribute entries using 3312 bytes of memory
4 BGP AS-PATH entries using 96 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 7120 total bytes of memory
BGP activity 20/0 prefixes, 20/0 paths, scan interval 60 secs
```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
10.255.0.2	4	65010	19	13	21	0	0	00:11:05	4
10.255.0.6	4	65020	21	15	21	0	0	00:13:03	4
10.255.0.10	4	65030	21	15	21	0	0	00:13:03	4

```
RT-BORDER-01#
```

```
RT-BORDER-01#show ip bgp
BGP table version is 21, local router ID is 1.1.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete
```

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.0.0.0/8	0.0.0.0	0	0	32768	i
*> 172.16.10.0/24	10.255.0.2	0	0	0	65010 i
*> 172.16.11.0/24	10.255.0.2	0	0	0	65010 i
*> 172.16.12.0/24	10.255.0.2	0	0	0	65010 i
*> 172.16.13.0/24	10.255.0.2	0	0	0	65010 i
*> 172.16.14.0/24	10.255.0.2	0	0	0	65010 i
*> 172.16.15.0/24	10.255.0.2	0	0	0	65010 i
*> 172.16.20.0/24	10.255.0.6	0	0	0	65020 i
*> 172.16.21.0/24	10.255.0.6	0	0	0	65020 i
*> 172.16.22.0/24	10.255.0.6	0	0	0	65020 i
*> 172.16.23.0/24	10.255.0.6	0	0	0	65020 i
*> 172.16.24.0/24	10.255.0.6	0	0	0	65020 i
*> 172.16.25.0/24	10.255.0.6	0	0	0	65020 i
*> 172.16.30.0/24	10.255.0.10	0	0	0	65030 i
*> 172.16.31.0/24	10.255.0.10	0	0	0	65030 i
*> 172.16.32.0/24	10.255.0.10	0	0	0	65030 i
*> 172.16.33.0/24	10.255.0.10	0	0	0	65030 i
*> 172.16.34.0/24	10.255.0.10	0	0	0	65030 i
*> 172.16.35.0/24	10.255.0.10	0	0	0	65030 i
*> 192.168.0.0/16	0.0.0.0	0	0	0	65000 i

```
RT-BORDER-01#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0/0  10.0.0.1        YES manual up          up
GigabitEthernet0/0/1  10.0.0.5        YES manual up          up
GigabitEthernet0/0/2  unassigned      YES unset  administratively down down
Serial0/1/0         10.255.0.1      YES manual up          up
Serial0/1/1         10.255.0.5      YES manual up          up
Serial0/2/0         200.0.0.1       YES manual up          up
Serial0/2/1         10.255.0.9      YES manual up          up
Vlan1             unassigned      YES unset  administratively down down
RT-BORDER-01#
```

RT-BORDER-02

```
RT-BORDER-02#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 200.0.0.6 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.4/30 is directly connected, GigabitEthernet0/0/0
L       10.0.0.5/32 is directly connected, GigabitEthernet0/0/0
    200.0.0.0/24 is variably subnetted, 3 subnets, 2 masks
C       200.0.0.4/30 is directly connected, Serial0/1/0
L       200.0.0.5/32 is directly connected, Serial0/1/0
C       200.0.0.6/32 is directly connected, Serial0/1/0
    200.0.1.0/30 is subnetted, 1 subnets
B       200.0.1.0/30 [20/0] via 200.0.0.6, 00:00:00
S*    0.0.0.0/0 [1/0] via 200.0.0.6
```

```
RT-BORDER-02#show ip bgp summary
BGP router identifier 2.2.2.2, local AS number 65001
BGP table version is 2, main routing table version 6
1 network entries using 132 bytes of memory
1 path entries using 52 bytes of memory
1/1 BGP path/bestpath attribute entries using 184 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 424 total bytes of memory
BGP activity 1/0 prefixes, 1/0 paths, scan interval 60 secs
```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
200.0.0.6	4	65002	17	16	2	0	0	00:14:10	4

RT-BORDER-02#

```
RT-BORDER-02#
RT-BORDER-02#show ip bgp
BGP table version is 2, local router ID is 2.2.2.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete
```

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 200.0.1.0/30	200.0.0.6	0	0	0	65002 i

```
RT-BORDER-02#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0/0  10.0.0.5        YES manual up          up
GigabitEthernet0/0/1  unassigned      YES unset  administratively down down
GigabitEthernet0/0/2  unassigned      YES unset  administratively down down
Serial0/1/0        200.0.0.5        YES manual up          up
Serial0/1/1        unassigned      YES unset  administratively down down
Serial0/2/0        unassigned      YES unset  administratively down down
Serial0/2/1        unassigned      YES unset  administratively down down
Vlan1            unassigned      YES unset  administratively down down
```

RT-BORDER-02#

RT-SEDES-01

```
RT-SEDE-01#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
```

Gateway of last resort is not set

```
10.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
B 10.0.0.0/8 [20/0] via 10.255.0.1, 00:00:00
C 10.255.0.0/30 is directly connected, Serial0/2/0
C 10.255.0.1/32 is directly connected, Serial0/2/0
L 10.255.0.2/32 is directly connected, Serial0/2/0
172.16.0.0/16 is variably subnetted, 24 subnets, 2 masks
C 172.16.10.0/24 is directly connected, GigabitEthernet0/1.210
L 172.16.10.1/32 is directly connected, GigabitEthernet0/1.210
C 172.16.11.0/24 is directly connected, GigabitEthernet0/1.211
L 172.16.11.1/32 is directly connected, GigabitEthernet0/1.211
C 172.16.12.0/24 is directly connected, GigabitEthernet0/1.212
L 172.16.12.1/32 is directly connected, GigabitEthernet0/1.212
C 172.16.13.0/24 is directly connected, GigabitEthernet0/1.213
L 172.16.13.1/32 is directly connected, GigabitEthernet0/1.213
C 172.16.14.0/24 is directly connected, GigabitEthernet0/1.214
L 172.16.14.1/32 is directly connected, GigabitEthernet0/1.214
C 172.16.15.0/24 is directly connected, GigabitEthernet0/1.215
L 172.16.15.1/32 is directly connected, GigabitEthernet0/1.215
B 172.16.20.0/24 [20/0] via 10.255.0.1, 00:00:00
B 172.16.21.0/24 [20/0] via 10.255.0.1, 00:00:00
B 172.16.22.0/24 [20/0] via 10.255.0.1, 00:00:00
B 172.16.23.0/24 [20/0] via 10.255.0.1, 00:00:00
B 172.16.24.0/24 [20/0] via 10.255.0.1, 00:00:00
B 172.16.25.0/24 [20/0] via 10.255.0.1, 00:00:00
B 172.16.30.0/24 [20/0] via 10.255.0.1, 00:00:00
B 172.16.31.0/24 [20/0] via 10.255.0.1, 00:00:00
B 172.16.32.0/24 [20/0] via 10.255.0.1, 00:00:00
B 172.16.33.0/24 [20/0] via 10.255.0.1, 00:00:00
B 172.16.34.0/24 [20/0] via 10.255.0.1, 00:00:00
B 172.16.35.0/24 [20/0] via 10.255.0.1, 00:00:00
B 192.168.0.0/16 [20/0] via 10.255.0.1, 00:00:00
```

RT-SEDE-01#

```
RT-SEDE-01#show ip ospf neighbor
```

```
RT-SEDE-01#
RT-SEDE-01#show ip inter
RT-SEDE-01#show ip interface bri
RT-SEDE-01#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	unset	administratively down	down
GigabitEthernet0/1	unassigned	YES	unset	up	up
GigabitEthernet0/1.210	172.16.10.1	YES	manual	up	up
GigabitEthernet0/1.211	172.16.11.1	YES	manual	up	up
GigabitEthernet0/1.212	172.16.12.1	YES	manual	up	up
GigabitEthernet0/1.213	172.16.13.1	YES	manual	up	up
GigabitEthernet0/1.214	172.16.14.1	YES	manual	up	up
GigabitEthernet0/1.215	172.16.15.1	YES	manual	up	up
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
GigabitEthernet0/0/0	unassigned	YES	unset	administratively down	down
GigabitEthernet0/1/0	unassigned	YES	unset	administratively down	down
Serial0/2/0	10.255.0.2	YES	manual	up	up
Serial0/2/1	unassigned	YES	unset	administratively down	down
Serial0/3/0	unassigned	YES	unset	administratively down	down
Serial0/3/1	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

RT-SEDE-01#

```

RT-SEDE-01#show interfaces
GigabitEthernet0/0 is administratively down, line protocol is down (disabled)
Hardware is CN Gigabit Ethernet, address is 0001.9759.e701 (bia 0001.9759.e701)
MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
Full-duplex, 100Mb/s, media type is RJ45
output flow-control is unsupported, input flow-control is unsupported
ARP type: ARPA, ARP Timeout 04:00:00,
Last input 00:00:08, output 00:00:08, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: fifo
Output queue : 0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 watchdog, 1017 multicast, 0 pause input
    0 input packets with dribble condition detected
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 2 interface resets
    0 unknown protocol drops
    0 babbles, 0 late collision, 0 deferred
    0 lost carrier, 0 no carrier
    0 output buffer failures, 0 output buffers swapped out
GigabitEthernet0/1 is up, line protocol is up (connected)
Hardware is CN Gigabit Ethernet, address is 0001.9759.e702 (bia 0001.9759.e702)
MTU 1500 bytes, BW 1000000 Kbit, DLY 10 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
Full-duplex, 100Mb/s, media type is RJ45
output flow-control is unsupported, input flow-control is unsupported
ARP type: ARPA, ARP Timeout 04:00:00,
Last input 00:00:08, output 00:00:08, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: fifo
Output queue : 0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 watchdog, 1017 multicast, 0 pause input
    0 input packets with dribble condition detected
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 1 interface resets
    0 unknown protocol drops
    0 babbles, 0 late collision, 0 deferred
    0 lost carrier, 0 no carrier
    0 output buffer failures, 0 output buffers swapped out
GigabitEthernet0/1.210 is up, line protocol is up (connected)
Hardware is PQICC_FEC, address is 0001.9759.e702 (bia 0001.9759.e702)
Internet address is 172.16.10.1/24
MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation 802.1Q Virtual LAN, Vlan ID 210
ARP type: ARPA, ARP Timeout 04:00:00,
Last clearing of "show interface" counters never
GigabitEthernet0/1.211 is up, line protocol is up (connected)
Hardware is PQICC_FEC, address is 0001.9759.e702 (bia 0001.9759.e702)
Internet address is 172.16.11.1/24
--More--

```

```

MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation 802.1Q Virtual LAN, Vlan ID 211
ARP type: ARPA, ARP Timeout 04:00:00,
Last clearing of "show interface" counters never
GigabitEthernet0/1.212 is up, line protocol is up (connected)
Hardware is PQUICC_FEC, address is 0001.9759.e702 (bia 0001.9759.e702)
Internet address is 172.16.12.1/24
MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation 802.1Q Virtual LAN, Vlan ID 212
ARP type: ARPA, ARP Timeout 04:00:00,
Last clearing of "show interface" counters never
GigabitEthernet0/1.213 is up, line protocol is up (connected)
Hardware is PQUICC_FEC, address is 0001.9759.e702 (bia 0001.9759.e702)
Internet address is 172.16.13.1/24
MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation 802.1Q Virtual LAN, Vlan ID 213
ARP type: ARPA, ARP Timeout 04:00:00,
Last clearing of "show interface" counters never
GigabitEthernet0/1.214 is up, line protocol is up (connected)
Hardware is PQUICC_FEC, address is 0001.9759.e702 (bia 0001.9759.e702)
Internet address is 172.16.14.1/24
MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation 802.1Q Virtual LAN, Vlan ID 214
ARP type: ARPA, ARP Timeout 04:00:00,
Last clearing of "show interface" counters never
GigabitEthernet0/1.215 is up, line protocol is up (connected)
Hardware is PQUICC_FEC, address is 0001.9759.e702 (bia 0001.9759.e702)
Internet address is 172.16.15.1/24
MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation 802.1Q Virtual LAN, Vlan ID 215
ARP type: ARPA, ARP Timeout 04:00:00,
Last clearing of "show interface" counters never
GigabitEthernet0/2 is administratively down, line protocol is down (disabled)
Hardware is CN Gigabit Ethernet, address is 0001.9759.e703 (bia 0001.9759.e703)
MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
Full-duplex, 100Mb/s, media type is RJ45
output flow-control is unsupported, input flow-control is unsupported
ARP type: ARPA, ARP Timeout 04:00:00,
Last input 00:00:08, output 00:00:05, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: fifo
Output queue :0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
  0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
  0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
  0 watchdog, 1017 multicast, 0 pause input
  0 input packets with dribble condition detected
  0 packets output, 0 bytes, 0 underruns
  0 output errors, 0 collisions, 2 interface resets
  0 unknown protocol drops
  0 babbles, 0 late collision, 0 deferred
  0 lost carrier, 0 no carrier
  0 output buffer failures, 0 output buffers swapped out
GigabitEthernet0/0/0 is administratively down, line protocol is down (disabled)
Hardware is CN Gigabit Ethernet, address is 00d0.9786.018d (bia 00d0.9786.018d)
--More--

```

```
MTU 1500 bytes, BW 1000000 Kbit, DLY 10 usec,  
    reliability 255/255, txload 1/255, rxload 1/255  
Encapsulation ARPA, loopback not set  
Keepalive set (10 sec)  
Full-duplex, 100Mb/s, media type is RJ45  
output flow-control is unsupported, input flow-control is unsupported  
ARP type: ARPA, ARP Timeout 04:00:00,  
Last input 00:00:08, output 00:00:05, output hang never  
Last clearing of "show interface" counters never  
Input queue: 0/75/0 (size/max/drops); Total output drops: 0  
Queueing strategy: fifo  
Output queue :0/40 (size/max)  
5 minute input rate 0 bits/sec, 0 packets/sec  
5 minute output rate 0 bits/sec, 0 packets/sec  
    0 packets input, 0 bytes, 0 no buffer  
    Received 0 broadcasts, 0 runs, 0 giants, 0 throttles  
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort  
    0 watchdog, 1017 multicast, 0 pause input  
    0 input packets with dribble condition detected  
    0 packets output, 0 bytes, 0 underruns  
    0 output errors, 0 collisions, 2 interface resets  
    0 unknown protocol drops  
    0 babbles, 0 late collision, 0 deferred  
    0 lost carrier, 0 no carrier  
    0 output buffer failures, 0 output buffers swapped out  
GigabitEthernet0/1/0 is administratively down, line protocol is down (disabled)  
Hardware is CN Gigabit Ethernet, address is 0002.4ace.e153 (bia 0002.4ace.e153)  
MTU 1500 bytes, BW 1000000 Kbit, DLY 10 usec,  
    reliability 255/255, txload 1/255, rxload 1/255  
Encapsulation ARPA, loopback not set  
Keepalive set (10 sec)  
Full-duplex, 100Mb/s, media type is RJ45  
output flow-control is unsupported, input flow-control is unsupported  
ARP type: ARPA, ARP Timeout 04:00:00,  
Last input 00:00:08, output 00:00:05, output hang never  
Last clearing of "show interface" counters never  
Input queue: 0/75/0 (size/max/drops); Total output drops: 0  
Queueing strategy: fifo  
Output queue :0/40 (size/max)  
5 minute input rate 0 bits/sec, 0 packets/sec  
5 minute output rate 0 bits/sec, 0 packets/sec  
    0 packets input, 0 bytes, 0 no buffer  
    Received 0 broadcasts, 0 runs, 0 giants, 0 throttles  
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort  
    0 watchdog, 1017 multicast, 0 pause input  
    0 input packets with dribble condition detected  
    0 packets output, 0 bytes, 0 underruns  
    0 output errors, 0 collisions, 2 interface resets  
    0 unknown protocol drops  
    0 babbles, 0 late collision, 0 deferred  
    0 lost carrier, 0 no carrier  
    0 output buffer failures, 0 output buffers swapped out  
Serial0/2/0 is up, line protocol is up (connected)  
Hardware is HD64570  
Description: ENLACE A DATACENTER  
Internet address is 10.255.0.2/30  
MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,  
    reliability 255/255, txload 1/255, rxload 1/255  
Encapsulation PPP, loopback not set, keepalive set (10 sec)  
LCP Open  
Open: IPCP, CDPCP  
Last input never, output never, output hang never  
Last clearing of "show interface" counters never  
Input queue: 0/75/0 (size/max/drops); Total output drops: 0  
Queueing strategy: weighted fair  
Output queue: 0/1000/64/0 (size/max total/threshold/drops)  
--More--
```

```
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: weighted fair
Output queue: 0/1000/64/0 (size/max total/threshold/drops)
  Conversations 0/1/256 (active/max active/max total)
  Reserved Conversations 0/0 (allocated/max allocated)
  Available Bandwidth 1158 kilobits/sec
5 minute input rate 5 bits/sec, 0 packets/sec
5 minute output rate 57 bits/sec, 0 packets/sec
  54 packets input, 2206 bytes, 0 no buffer
  Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
  0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
  114 packets output, 6619 bytes, 0 underruns
  0 output errors, 0 collisions, 0 interface resets
  0 output buffer failures, 0 output buffers swapped out
  0 carrier transitions
DCD=up DSR=up DTR=up RTS=up CTS=up
Serial0/2/1 is administratively down, line protocol is down (disabled)
Hardware is HD64570
MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
  reliability 255/255, txload 1/255, rxload 1/255
Encapsulation HDLC, loopback not set, keepalive set (10 sec)
Last input never, output never, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: weighted fair
Output queue: 0/1000/64/0 (size/max total/threshold/drops)
  Conversations 0/0/256 (active/max active/max total)
  Reserved Conversations 0/0 (allocated/max allocated)
  Available Bandwidth 1158 kilobits/sec
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
  0 packets input, 0 bytes, 0 no buffer
  Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
  0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
  0 packets output, 0 bytes, 0 underruns
  0 output errors, 0 collisions, 2 interface resets
  0 output buffer failures, 0 output buffers swapped out
  0 carrier transitions
DCD=down DSR=down DTR=down RTS=down CTS=down
Serial0/3/0 is administratively down, line protocol is down (disabled)
Hardware is HD64570
MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
  reliability 255/255, txload 1/255, rxload 1/255
Encapsulation HDLC, loopback not set, keepalive set (10 sec)
Last input never, output never, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: weighted fair
Output queue: 0/1000/64/0 (size/max total/threshold/drops)
  Conversations 0/0/256 (active/max active/max total)
  Reserved Conversations 0/0 (allocated/max allocated)
  Available Bandwidth 1158 kilobits/sec
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
  0 packets input, 0 bytes, 0 no buffer
  Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
  0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
  0 packets output, 0 bytes, 0 underruns
  0 output errors, 0 collisions, 2 interface resets
  0 output buffer failures, 0 output buffers swapped out
  0 carrier transitions
DCD=down DSR=down DTR=down RTS=down CTS=down
Serial0/3/1 is administratively down, line protocol is down (disabled)
Hardware is HD64570
MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
  reliability 255/255, txload 1/255, rxload 1/255
--More--
```

```
Serial0/3/1 is administratively down, line protocol is down (disabled)
Hardware is HD64570
MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
  reliability 255/255, txload 1/255, rxload 1/255
Encapsulation HDLC, loopback not set, keepalive set (10 sec)
Last input never, output never, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: weighted fair
Output queue: 0/1000/64/0 (size/max total/threshold/drops)
  Conversations 0/0/256 (active/max active/max total)
  Reserved Conversations 0/0 (allocated/max allocated)
  Available Bandwidth 1158 kilobits/sec
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
  0 packets input, 0 bytes, 0 no buffer
  Received 0 broadcasts, 0 runs, 0 giants, 0 throttles
  0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
  0 packets output, 0 bytes, 0 underruns
  0 output errors, 0 collisions, 2 interface resets
  0 output buffer failures, 0 output buffers swapped out
  0 carrier transitions
DCD=down DSR=down DTR=down RTS=down CTS=down
Vlan1 is administratively down, line protocol is down
Hardware is CPU Interface, address is 0050.0f15.74b9 (bia 0050.0f15.74b9)
MTU 1500 bytes, BW 1000000 Kbit, DLY 1000000 usec,
  reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
ARP type: ARPA, ARP Timeout 04:00:00
Last input 21:40:21, output never, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue: 0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
  1682 packets input, 530955 bytes, 0 no buffer
  Received 0 broadcasts (0 IP multicast)
  0 runs, 0 giants, 0 throttles
  0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
  563859 packets output, 0 bytes, 0 underruns
  0 output errors, 23 interface resets
  0 output buffer failures, 0 output buffers swapped out
```

RT-SEDE-01#

RT-SEDES-02

```
RT-SEDE-02#show ip rou
RT-SEDE-02#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

Gateway of last resort is 10.255.0.5 to network 0.0.0.0

```
10.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
B 10.0.0.0/8 [20/0] via 10.255.0.5, 00:00:00
C 10.255.0.4/30 is directly connected, Serial0/3/0
C 10.255.0.5/32 is directly connected, Serial0/3/0
L 10.255.0.6/32 is directly connected, Serial0/3/0
172.16.0.0/16 is variably subnetted, 24 subnets, 2 masks
B 172.16.10.0/24 [20/0] via 10.255.0.5, 00:00:00
B 172.16.11.0/24 [20/0] via 10.255.0.5, 00:00:00
B 172.16.12.0/24 [20/0] via 10.255.0.5, 00:00:00
B 172.16.13.0/24 [20/0] via 10.255.0.5, 00:00:00
B 172.16.14.0/24 [20/0] via 10.255.0.5, 00:00:00
B 172.16.15.0/24 [20/0] via 10.255.0.5, 00:00:00
C 172.16.20.0/24 is directly connected, GigabitEthernet0/1.220
L 172.16.20.1/32 is directly connected, GigabitEthernet0/1.220
C 172.16.21.0/24 is directly connected, GigabitEthernet0/1.221
L 172.16.21.1/32 is directly connected, GigabitEthernet0/1.221
C 172.16.22.0/24 is directly connected, GigabitEthernet0/1.222
L 172.16.22.1/32 is directly connected, GigabitEthernet0/1.222
C 172.16.23.0/24 is directly connected, GigabitEthernet0/1.223
L 172.16.23.1/32 is directly connected, GigabitEthernet0/1.223
C 172.16.24.0/24 is directly connected, GigabitEthernet0/1.224
L 172.16.24.1/32 is directly connected, GigabitEthernet0/1.224
C 172.16.25.0/24 is directly connected, GigabitEthernet0/1.225
L 172.16.25.1/32 is directly connected, GigabitEthernet0/1.225
B 172.16.30.0/24 [20/0] via 10.255.0.5, 00:00:00
B 172.16.31.0/24 [20/0] via 10.255.0.5, 00:00:00
B 172.16.32.0/24 [20/0] via 10.255.0.5, 00:00:00
B 172.16.33.0/24 [20/0] via 10.255.0.5, 00:00:00
B 172.16.34.0/24 [20/0] via 10.255.0.5, 00:00:00
B 172.16.35.0/24 [20/0] via 10.255.0.5, 00:00:00
B 192.168.0.0/16 [20/0] via 10.255.0.5, 00:00:00
S* 0.0.0.0/0 [1/0] via 10.255.0.5
```

RT-SEDE-02#

```
RT-SEDE-02#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  unassigned      YES unset   administratively down down
GigabitEthernet0/1  unassigned      YES unset   up          up
GigabitEthernet0/1.220  172.16.20.1     YES manual  up          up
GigabitEthernet0/1.221  172.16.21.1     YES manual  up          up
GigabitEthernet0/1.222  172.16.22.1     YES manual  up          up
GigabitEthernet0/1.223  172.16.23.1     YES manual  up          up
GigabitEthernet0/1.224  172.16.24.1     YES manual  up          up
GigabitEthernet0/1.225  172.16.25.1     YES manual  up          up
GigabitEthernet0/2    unassigned      YES unset   administratively down down
GigabitEthernet0/0/0  unassigned      YES unset   administratively down down
GigabitEthernet0/1/0  unassigned      YES unset   administratively down down
Serial0/2/0          unassigned      YES unset   administratively down down
Serial0/2/1          unassigned      YES unset   administratively down down
Serial0/3/0          10.255.0.6      YES manual  up          up
Serial0/3/1          unassigned      YES unset   administratively down down
Vlan1                unassigned      YES unset   administratively down down
RT-SEDE-02#
RT-SEDE-02#
```

```

RT-SEDE-02#show interfaces
GigabitEthernet0/0 is administratively down, line protocol is down (disabled)
Hardware is CN Gigabit Ethernet, address is 0003.e412.eace (bia 0003.e412.eace)
MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
Full-duplex, 100Mb/s, media type is RJ45
output flow-control is unsupported, input flow-control is unsupported
ARP type: ARPA, ARP Timeout 04:00:00,
Last input 00:00:08, output 00:00:05, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: fifo
Output queue :0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 watchdog, 1017 multicast, 0 pause input
    0 input packets with dribble condition detected
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 2 interface resets
    0 unknown protocol drops
    0 babbles, 0 late collision, 0 deferred
    0 lost carrier, 0 no carrier
    0 output buffer failures, 0 output buffers swapped out
GigabitEthernet0/1 is up, line protocol is up (connected)
Hardware is CN Gigabit Ethernet, address is 0002.4ac5.2980 (bia 0002.4ac5.2980)
MTU 1500 bytes, BW 1000000 Kbit, DLY 10 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
Full-duplex, 100Mb/s, media type is RJ45
output flow-control is unsupported, input flow-control is unsupported
ARP type: ARPA, ARP Timeout 04:00:00,
Last input 00:00:08, output 00:00:05, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: fifo
Output queue :0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 watchdog, 1017 multicast, 0 pause input
    0 input packets with dribble condition detected
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 1 interface resets
    0 unknown protocol drops
    0 babbles, 0 late collision, 0 deferred
    0 lost carrier, 0 no carrier
    0 output buffer failures, 0 output buffers swapped out
GigabitEthernet0/1.220 is up, line protocol is up (connected)
Hardware is PQUICC_FEC, address is 0002.4ac5.2980 (bia 0002.4ac5.2980)
Internet address is 172.16.20.1/24
MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation 802.1Q Virtual LAN, Vlan ID 220
ARP type: ARPA, ARP Timeout 04:00:00,
Last clearing of "show interface" counters never
GigabitEthernet0/1.221 is up, line protocol is up (connected)
Hardware is PQUICC_FEC, address is 0002.4ac5.2980 (bia 0002.4ac5.2980)
Internet address is 172.16.21.1/24
--More--

```

```

MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation 802.1Q Virtual LAN, Vlan ID 221
ARP type: ARPA, ARP Timeout 04:00:00,
    Last clearing of "show interface" counters never
GigabitEthernet0/1.222 is up, line protocol is up (connected)
Hardware is PQUICC_FEC, address is 0002.4ac5.2980 (bia 0002.4ac5.2980)
Internet address is 172.16.22.1/24
MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation 802.1Q Virtual LAN, Vlan ID 222
ARP type: ARPA, ARP Timeout 04:00:00,
    Last clearing of "show interface" counters never
GigabitEthernet0/1.223 is up, line protocol is up (connected)
Hardware is PQUICC_FEC, address is 0002.4ac5.2980 (bia 0002.4ac5.2980)
Internet address is 172.16.23.1/24
MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation 802.1Q Virtual LAN, Vlan ID 223
ARP type: ARPA, ARP Timeout 04:00:00,
    Last clearing of "show interface" counters never
GigabitEthernet0/1.224 is up, line protocol is up (connected)
Hardware is PQUICC_FEC, address is 0002.4ac5.2980 (bia 0002.4ac5.2980)
Internet address is 172.16.24.1/24
MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation 802.1Q Virtual LAN, Vlan ID 224
ARP type: ARPA, ARP Timeout 04:00:00,
    Last clearing of "show interface" counters never
GigabitEthernet0/1.225 is up, line protocol is up (connected)
Hardware is PQUICC_FEC, address is 0002.4ac5.2980 (bia 0002.4ac5.2980)
Internet address is 172.16.25.1/24
MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation 802.1Q Virtual LAN, Vlan ID 225
ARP type: ARPA, ARP Timeout 04:00:00,
    Last clearing of "show interface" counters never
GigabitEthernet0/2 is administratively down, line protocol is down (disabled)
Hardware is CN Gigabit Ethernet, address is 0002.4a80.7279 (bia 0002.4a80.7279)
MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
Full-duplex, 100Mb/s, media type is RJ45
output flow-control is unsupported, input flow-control is unsupported
ARP type: ARPA, ARP Timeout 04:00:00,
    Last input 00:00:08, output 00:00:05, output hang never
    Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: fifo
Output queue :0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 watchdog, 1017 multicast, 0 pause input
    0 input packets with dribble condition detected
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 2 interface resets
    0 unknown protocol drops
    0 babbles, 0 late collision, 0 deferred
    0 lost carrier, 0 no carrier
    0 output buffer failures, 0 output buffers swapped out
GigabitEthernet0/0/0 is administratively down, line protocol is down (disabled)
Hardware is CN Gigabit Ethernet, address is 0001.97b6.98d4 (bia 0001.97b6.98d4)
--More--

```

```
MTU 1500 bytes, BW 1000000 Kbit, DLY 10 usec,  
    reliability 255/255, txload 1/255, rxload 1/255  
Encapsulation ARPA, loopback not set  
Keepalive set (10 sec)  
Full-duplex, 100Mb/s, media type is RJ45  
output flow-control is unsupported, input flow-control is unsupported  
ARP type: ARPA, ARP Timeout 04:00:00,  
Last input 00:00:08, output 00:00:05, output hang never  
Last clearing of "show interface" counters never  
Input queue: 0/75/0 (size/max/drops); Total output drops: 0  
Queueing strategy: fifo  
Output queue :0/40 (size/max)  
5 minute input rate 0 bits/sec, 0 packets/sec  
5 minute output rate 0 bits/sec, 0 packets/sec  
    0 packets input, 0 bytes, 0 no buffer  
    Received 0 broadcasts, 0 runs, 0 giants, 0 throttles  
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort  
    0 watchdog, 1017 multicast, 0 pause input  
    0 input packets with dribble condition detected  
    0 packets output, 0 bytes, 0 underruns  
    0 output errors, 0 collisions, 2 interface resets  
    0 unknown protocol drops  
    0 babbles, 0 late collision, 0 deferred  
    0 lost carrier, 0 no carrier  
    0 output buffer failures, 0 output buffers swapped out  
GigabitEthernet0/1/0 is administratively down, line protocol is down (disabled)  
Hardware is CN Gigabit Ethernet, address is 0006.2aeb.6570 (bia 0006.2aeb.6570)  
MTU 1500 bytes, BW 1000000 Kbit, DLY 10 usec,  
    reliability 255/255, txload 1/255, rxload 1/255  
Encapsulation ARPA, loopback not set  
Keepalive set (10 sec)  
Full-duplex, 100Mb/s, media type is RJ45  
output flow-control is unsupported, input flow-control is unsupported  
ARP type: ARPA, ARP Timeout 04:00:00,  
Last input 00:00:08, output 00:00:05, output hang never  
Last clearing of "show interface" counters never  
Input queue: 0/75/0 (size/max/drops); Total output drops: 0  
Queueing strategy: fifo  
Output queue :0/40 (size/max)  
5 minute input rate 0 bits/sec, 0 packets/sec  
5 minute output rate 0 bits/sec, 0 packets/sec  
    0 packets input, 0 bytes, 0 no buffer  
    Received 0 broadcasts, 0 runs, 0 giants, 0 throttles  
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort  
    0 watchdog, 1017 multicast, 0 pause input  
    0 input packets with dribble condition detected  
    0 packets output, 0 bytes, 0 underruns  
    0 output errors, 0 collisions, 2 interface resets  
    0 unknown protocol drops  
    0 babbles, 0 late collision, 0 deferred  
    0 lost carrier, 0 no carrier  
    0 output buffer failures, 0 output buffers swapped out  
Serial0/2/0 is administratively down, line protocol is down (disabled)  
Hardware is HD64570  
MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,  
    reliability 255/255, txload 1/255, rxload 1/255  
Encapsulation HDLC, loopback not set, keepalive set (10 sec)  
Last input never, output never, output hang never  
Last clearing of "show interface" counters never  
Input queue: 0/75/0 (size/max/drops); Total output drops: 0  
Queueing strategy: weighted fair  
Output queue: 0/1000/64/0 (size/max total/threshold/drops)  
    Conversations 0/0/256 (active/max active/max total)  
    Reserved Conversations 0/0 (allocated/max allocated)  
    Available Bandwidth 1158 kilobits/sec  
5 minute input rate 0 bits/sec, 0 packets/sec  
--More--
```

```
5 minute input rate 0 bits/sec, 0 packets/sec
0 packets input, 0 bytes, 0 no buffer
Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
0 packets output, 0 bytes, 0 underruns
0 output errors, 0 collisions, 2 interface resets
0 output buffer failures, 0 output buffers swapped out
0 carrier transitions
DCD=down DSR=down DTR=down RTS=down CTS=down
Serial0/2/1 is administratively down, line protocol is down (disabled)
Hardware is HD64570
MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation HDLC, loopback not set, keepalive set (10 sec)
Last input never, output never, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: weighted fair
Output queue: 0/1000/64/0 (size/max total/threshold/drops)
Conversations 0/0/256 (active/max active/max total)
Reserved Conversations 0/0 (allocated/max allocated)
Available Bandwidth 1158 kilobits/sec
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
0 packets input, 0 bytes, 0 no buffer
Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
0 packets output, 0 bytes, 0 underruns
0 output errors, 0 collisions, 2 interface resets
0 output buffer failures, 0 output buffers swapped out
0 carrier transitions
DCD=down DSR=down DTR=down RTS=down CTS=down
Serial0/3/0 is up, line protocol is up (connected)
Hardware is HD64570
Description: ENLACE A DATACENTER
Internet address is 10.255.0.6/30
MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation PPP, loopback not set, keepalive set (10 sec)
LCP Open
Open: IPCP, CDPCP
Last input never, output never, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: weighted fair
Output queue: 0/1000/64/0 (size/max total/threshold/drops)
Conversations 0/0/256 (active/max active/max total)
Reserved Conversations 0/0 (allocated/max allocated)
Available Bandwidth 1158 kilobits/sec
5 minute input rate 6 bits/sec, 0 packets/sec
5 minute output rate 55 bits/sec, 0 packets/sec
93 packets input, 5528 bytes, 0 no buffer
Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
160 packets output, 9403 bytes, 0 underruns
0 output errors, 0 collisions, 0 interface resets
0 output buffer failures, 0 output buffers swapped out
0 carrier transitions
DCD=up DSR=up DTR=up RTS=up CTS=up
Serial0/3/1 is administratively down, line protocol is down (disabled)
Hardware is HD64570
MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation HDLC, loopback not set, keepalive set (10 sec)
Last input never, output never, output hang never
Last clearing of "show interface" counters never
--More--
```

```
Last input never, output never, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: weighted fair
Output queue: 0/1000/64/0 (size/max total/threshold/drops)
Conversations 0/0/255 (active/max active/max total)
Reserved Conversations 0/0 (allocated/max allocated)
Available Bandwidth 1158 kilobits/sec
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
0 packets input, 0 bytes, 0 no buffer
Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
0 packets output, 0 bytes, 0 underruns
0 output errors, 0 collisions, 2 interface resets
0 output buffer failures, 0 output buffers swapped out
0 carrier transitions
DCD=down DSR=down DTR=down RTS=down CTS=down
Vlan1 is administratively down, line protocol is down
Hardware is CPU Interface, address is 0004.9a76.c8ae (bia 0004.9a76.c8ae)
MTU 1500 bytes, BW 100000 Kbit, DLY 1000000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
ARP type: ARPA, ARP Timeout 04:00:00
Last input 21:40:21, output never, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue: 0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
1682 packets input, 530355 bytes, 0 no buffer
Received 0 broadcasts (0 IP multicast)
0 runts, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
563859 packets output, 0 bytes, 0 underruns
0 output errors, 23 interface resets
0 output buffer failures, 0 output buffers swapped out

RT-SEDE-02#
RT-SEDE-02#
RT-SEDE-02#
--
```

RT-SEDES-03

```
RT-SEDE-03#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
```

Gateway of last resort is not set

```
10.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
B 10.0.0.0/8 [20/0] via 10.255.0.9, 00:00:00
C 10.255.0.8/30 is directly connected, Serial0/2/0
C 10.255.0.9/32 is directly connected, Serial0/2/0
L 10.255.0.10/32 is directly connected, Serial0/2/0
172.16.0.0/16 is variably subnetted, 24 subnets, 2 masks
B 172.16.10.0/24 [20/0] via 10.255.0.9, 00:00:00
B 172.16.11.0/24 [20/0] via 10.255.0.9, 00:00:00
B 172.16.12.0/24 [20/0] via 10.255.0.9, 00:00:00
B 172.16.13.0/24 [20/0] via 10.255.0.9, 00:00:00
B 172.16.14.0/24 [20/0] via 10.255.0.9, 00:00:00
B 172.16.15.0/24 [20/0] via 10.255.0.9, 00:00:00
B 172.16.20.0/24 [20/0] via 10.255.0.9, 00:00:00
B 172.16.21.0/24 [20/0] via 10.255.0.9, 00:00:00
B 172.16.22.0/24 [20/0] via 10.255.0.9, 00:00:00
B 172.16.23.0/24 [20/0] via 10.255.0.9, 00:00:00
B 172.16.24.0/24 [20/0] via 10.255.0.9, 00:00:00
B 172.16.25.0/24 [20/0] via 10.255.0.9, 00:00:00
C 172.16.30.0/24 is directly connected, GigabitEthernet0/1.230
L 172.16.30.1/32 is directly connected, GigabitEthernet0/1.230
C 172.16.31.0/24 is directly connected, GigabitEthernet0/1.231
L 172.16.31.1/32 is directly connected, GigabitEthernet0/1.231
C 172.16.32.0/24 is directly connected, GigabitEthernet0/1.232
L 172.16.32.1/32 is directly connected, GigabitEthernet0/1.232
C 172.16.33.0/24 is directly connected, GigabitEthernet0/1.233
L 172.16.33.1/32 is directly connected, GigabitEthernet0/1.233
C 172.16.34.0/24 is directly connected, GigabitEthernet0/1.234
L 172.16.34.1/32 is directly connected, GigabitEthernet0/1.234
C 172.16.35.0/24 is directly connected, GigabitEthernet0/1.235
L 172.16.35.1/32 is directly connected, GigabitEthernet0/1.235
B 192.168.0.0/16 [20/0] via 10.255.0.9, 00:00:00
```

RT-SEDE-03#

RT-SEDE-03#

RT-SEDE-03#

```
RT-SEDE-03#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 unassigned YES unset administratively down down
GigabitEthernet0/1 unassigned YES unset up up
GigabitEthernet0/1.230 172.16.30.1 YES manual up up
GigabitEthernet0/1.231 172.16.31.1 YES manual up up
GigabitEthernet0/1.232 172.16.32.1 YES manual up up
GigabitEthernet0/1.233 172.16.33.1 YES manual up up
GigabitEthernet0/1.234 172.16.34.1 YES manual up up
GigabitEthernet0/1.235 172.16.35.1 YES manual up up
GigabitEthernet0/2 unassigned YES unset administratively down down
GigabitEthernet0/0/0 unassigned YES unset administratively down down
GigabitEthernet0/1/0 unassigned YES unset administratively down down
Serial0/2/0 10.255.0.10 YES manual up up
Serial0/2/1 unassigned YES manual administratively down down
Serial0/3/0 unassigned YES unset administratively down down
Serial0/3/1 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
RT-SEDE-03#
RT-SEDE-03#
RT-SEDE-03#
RT-SEDE-03#
```

```
RT-SEDE-03#show interfaces
GigabitEthernet0/0 is administratively down, line protocol is down (disabled)
Hardware is CN Gigabit Ethernet, address is 0001.420e.5384 (bia 0001.420e.5384)
MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
Full-duplex, 100Mb/s, media type is RJ45
output flow-control is unsupported, input flow-control is unsupported
ARP type: ARPA, ARP Timeout 04:00:00,
Last input 00:00:08, output 00:00:05, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: fifo
Output queue : 0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 watchdog, 1017 multicast, 0 pause input
    0 input packets with dribble condition detected
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 2 interface resets
    0 unknown protocol drops
    0 babbles, 0 late collision, 0 deferred
    0 lost carrier, 0 no carrier
    0 output buffer failures, 0 output buffers swapped out
GigabitEthernet0/1 is up, line protocol is up (connected)
Hardware is CN Gigabit Ethernet, address is 0090.0c44.e6a3 (bia 0090.0c44.e6a3)
MTU 1500 bytes, BW 1000000 Kbit, DLY 10 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
Full-duplex, 100Mb/s, media type is RJ45
output flow-control is unsupported, input flow-control is unsupported
ARP type: ARPA, ARP Timeout 04:00:00,
Last input 00:00:08, output 00:00:05, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: fifo
Output queue : 0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 watchdog, 1017 multicast, 0 pause input
    0 input packets with dribble condition detected
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 1 interface resets
    0 unknown protocol drops
    0 babbles, 0 late collision, 0 deferred
    0 lost carrier, 0 no carrier
    0 output buffer failures, 0 output buffers swapped out
GigabitEthernet0/1.230 is up, line protocol is up (connected)
Hardware is PQUICC_FEC, address is 0090.0c44.e6a3 (bia 0090.0c44.e6a3)
Internet address is 172.16.30.1/24
MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation 802.1Q Virtual LAN, Vlan ID 230
ARP type: ARPA, ARP Timeout 04:00:00,
Last clearing of "show interface" counters never
GigabitEthernet0/1.231 is up, line protocol is up (connected)
Hardware is PQUICC_FEC, address is 0090.0c44.e6a3 (bia 0090.0c44.e6a3)
Internet address is 172.16.31.1/24
--More--
```

```
MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation 802.1Q Virtual LAN, Vlan ID 231
ARP type: ARPA, ARP Timeout 04:00:00,
Last clearing of "show interface" counters never
GigabitEthernet0/1.232 is up, line protocol is up (connected)
Hardware is PQUICC_FEC, address is 0090.0c44.e6a3 (bia 0090.0c44.e6a3)
Internet address is 172.16.32.1/24
MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation 802.1Q Virtual LAN, Vlan ID 232
ARP type: ARPA, ARP Timeout 04:00:00,
Last clearing of "show interface" counters never
GigabitEthernet0/1.233 is up, line protocol is up (connected)
Hardware is PQUICC_FEC, address is 0090.0c44.e6a3 (bia 0090.0c44.e6a3)
Internet address is 172.16.33.1/24
MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation 802.1Q Virtual LAN, Vlan ID 233
ARP type: ARPA, ARP Timeout 04:00:00,
Last clearing of "show interface" counters never
GigabitEthernet0/1.234 is up, line protocol is up (connected)
Hardware is PQUICC_FEC, address is 0090.0c44.e6a3 (bia 0090.0c44.e6a3)
Internet address is 172.16.34.1/24
MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation 802.1Q Virtual LAN, Vlan ID 234
ARP type: ARPA, ARP Timeout 04:00:00,
Last clearing of "show interface" counters never
GigabitEthernet0/1.235 is up, line protocol is up (connected)
Hardware is PQUICC_FEC, address is 0090.0c44.e6a3 (bia 0090.0c44.e6a3)
Internet address is 172.16.35.1/24
MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation 802.1Q Virtual LAN, Vlan ID 235
ARP type: ARPA, ARP Timeout 04:00:00,
Last clearing of "show interface" counters never
GigabitEthernet0/2 is administratively down, line protocol is down (disabled)
Hardware is CN Gigabit Ethernet, address is 00d0.976d.23d6 (bia 00d0.976d.23d6)
MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
Full-duplex, 100Mb/s, media type is RJ45
output flow-control is unsupported, input flow-control is unsupported
ARP type: ARPA, ARP Timeout 04:00:00,
Last input 00:00:08, output 00:00:05, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: fifo
Output queue :0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
  0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
  0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
  0 watchdog, 1017 multicast, 0 pause input
  0 input packets with dribble condition detected
  0 packets output, 0 bytes, 0 underruns
  0 output errors, 0 collisions, 2 interface resets
  0 unknown protocol drops
  0 babbles, 0 late collision, 0 deferred
  0 lost carrier, 0 no carrier
  0 output buffer failures, 0 output buffers swapped out
GigabitEthernet0/0/0 is administratively down, line protocol is down (disabled)
Hardware is CN Gigabit Ethernet, address is 00e0.f910.bcc0 (bia 00e0.f910.bcc0)
--More--
```

```
MTU 1500 bytes, BW 1000000 Kbit, DLY 10 usec,  
    reliability 255/255, txload 1/255, rxload 1/255  
Encapsulation ARPA, loopback not set  
Keepalive set (10 sec)  
Full-duplex, 100Mb/s, media type is RJ45  
output flow-control is unsupported, input flow-control is unsupported  
ARP type: ARPA, ARP Timeout 04:00:00,  
Last input 00:00:08, output 00:00:05, output hang never  
Last clearing of "show interface" counters never  
Input queue: 0/75/0 (size/max/drops); Total output drops: 0  
Queueing strategy: fifo  
Output queue :0/40 (size/max)  
5 minute input rate 0 bits/sec, 0 packets/sec  
5 minute output rate 0 bits/sec, 0 packets/sec  
    0 packets input, 0 bytes, 0 no buffer  
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles  
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort  
    0 watchdog, 1017 multicast, 0 pause input  
    0 input packets with dribble condition detected  
    0 packets output, 0 bytes, 0 underruns  
    0 output errors, 0 collisions, 2 interface resets  
    0 unknown protocol drops  
    0 babbles, 0 late collision, 0 deferred  
    0 lost carrier, 0 no carrier  
    0 output buffer failures, 0 output buffers swapped out  
GigabitEthernet0/1/0 is administratively down, line protocol is down (disabled)  
Hardware is CN Gigabit Ethernet, address is 000c.cfa3.5e69 (bia 000c.cfa3.5e69)  
MTU 1500 bytes, BW 1000000 Kbit, DLY 10 usec,  
    reliability 255/255, txload 1/255, rxload 1/255  
Encapsulation ARPA, loopback not set  
Keepalive set (10 sec)  
Full-duplex, 100Mb/s, media type is RJ45  
output flow-control is unsupported, input flow-control is unsupported  
ARP type: ARPA, ARP Timeout 04:00:00,  
Last input 00:00:08, output 00:00:05, output hang never  
Last clearing of "show interface" counters never  
Input queue: 0/75/0 (size/max/drops); Total output drops: 0  
Queueing strategy: fifo  
Output queue :0/40 (size/max)  
5 minute input rate 0 bits/sec, 0 packets/sec  
5 minute output rate 0 bits/sec, 0 packets/sec  
    0 packets input, 0 bytes, 0 no buffer  
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles  
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort  
    0 watchdog, 1017 multicast, 0 pause input  
    0 input packets with dribble condition detected  
    0 packets output, 0 bytes, 0 underruns  
    0 output errors, 0 collisions, 2 interface resets  
    0 unknown protocol drops  
    0 babbles, 0 late collision, 0 deferred  
    0 lost carrier, 0 no carrier  
    0 output buffer failures, 0 output buffers swapped out  
Serial0/2/0 is up, line protocol is up (connected)  
Hardware is HD64570  
Description: ENLACE A DATACENTER  
Internet address is 10.255.0.10/30  
MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,  
    reliability 255/255, txload 1/255, rxload 1/255  
Encapsulation PPP, loopback not set, keepalive set (10 sec)  
LCP Open  
Open: IPCP, CDPCP  
Last input never, output never, output hang never  
Last clearing of "show interface" counters never  
Input queue: 0/75/0 (size/max/drops); Total output drops: 0  
Queueing strategy: weighted fair  
Output queue: 0/1000/64/0 (size/max total/threshold/drops)  
--More--
```

```
Conversations 0/0/256 (active/max active/max total)
Reserved Conversations 0/0 (allocated/max allocated)
Available Bandwidth 1158 kilobits/sec
5 minute input rate 6 bits/sec, 0 packets/sec
5 minute output rate 57 bits/sec, 0 packets/sec
  90 packets input, 5400 bytes, 0 no buffer
Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
  0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
 161 packets output, 9544 bytes, 0 underruns
   0 output errors, 0 collisions, 0 interface resets
   0 output buffer failures, 0 output buffers swapped out
   0 carrier transitions
DCD=up DSR=up DTR=up RTS=up CTS=up
Serial0/2/1 is administratively down, line protocol is down (disabled)
Hardware is HD64570
Description: CONEXION AL CORPORATIVO
MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation PPP, loopback not set, keepalive set (10 sec)
LCP Closed
Closed: LEXCP, BRIDGECP, IPCP, CCP, CDPCP, LLC2, BACP
Last input never, output never, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: weighted fair
Output queue: 0/1000/64/0 (size/max total/threshold/drops)
Conversations 0/0/256 (active/max active/max total)
Reserved Conversations 0/0 (allocated/max allocated)
Available Bandwidth 1158 kilobits/sec
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
  0 packets input, 0 bytes, 0 no buffer
Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
  0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
  0 packets output, 0 bytes, 0 underruns
   0 output errors, 0 collisions, 2 interface resets
   0 output buffer failures, 0 output buffers swapped out
   0 carrier transitions
DCD=down DSR=down DTR=down RTS=down CTS=down
Serial0/3/0 is administratively down, line protocol is down (disabled)
Hardware is HD64570
MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation HDLC, loopback not set, keepalive set (10 sec)
Last input never, output never, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: weighted fair
Output queue: 0/1000/64/0 (size/max total/threshold/drops)
Conversations 0/0/256 (active/max active/max total)
Reserved Conversations 0/0 (allocated/max allocated)
Available Bandwidth 1158 kilobits/sec
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
  0 packets input, 0 bytes, 0 no buffer
Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
  0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
  0 packets output, 0 bytes, 0 underruns
   0 output errors, 0 collisions, 2 interface resets
   0 output buffer failures, 0 output buffers swapped out
   0 carrier transitions
DCD=down DSR=down DTR=down RTS=down CTS=down
Serial0/3/1 is administratively down, line protocol is down (disabled)
Hardware is HD64570
MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
--More--
```

```
MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation HDLC, loopback not set, keepalive set (10 sec)
Last input never, output never, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: weighted fair
Output queue: 0/1000/64/0 (size/max total/threshold/drops)
    Conversations 0/0/256 (active/max active/max total)
    Reserved Conversations 0/0 (allocated/max allocated)
    Available Bandwidth 1158 kilobits/sec
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runs, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 2 interface resets
    0 output buffer failures, 0 output buffers swapped out
    0 carrier transitions
DCD=down DSR=down DTR=down RTS=down CTS=down
Vlan1 is administratively down, line protocol is down
Hardware is CPU Interface, address is 0009.7cb2.2bcd (bia 0009.7cb2.2bcd)
MTU 1500 bytes, BW 100000 Kbit, DLY 1000000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
ARP type: ARPA, ARP Timeout 04:00:00
Last input 21:40:21, output never, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue: 0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
    1682 packets input, 530955 bytes, 0 no buffer
    Received 0 broadcasts (0 IP multicast)
    0 runs, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
    563859 packets output, 0 bytes, 0 underruns
    0 output errors, 23 interface resets
    0 output buffer failures, 0 output buffers swapped out

RT-SEDE-03#
RT-SEDE-03#
RT-SEDE-03#
RT-SEDE-03#
```

RT-ISP

```
RT-ISP-01#show ip bgp summary
BGP router identifier 4.4.4.4, local AS number 65002
BGP table version is 1, main routing table version 6
0 network entries using 0 bytes of memory
0 path entries using 0 bytes of memory
0/0 BGP path/bestpath attribute entries using 0 bytes of memory
0 BGP AS-PATH entries using 0 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 32 total bytes of memory
BGP activity 0/0 prefixes, 0/0 paths, scan interval 60 secs

Neighbor      V    AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
200.0.0.1      4 65001      0       0        1    0    0 00:19:54      4

RT-ISP-01#
RT-ISP-01#
RT-ISP-01#
```

```
RT-ISP-01#show ip bgp summary
BGP router identifier 4.4.4.4, local AS number 65002
BGP table version is 1, main routing table version 6
0 network entries using 0 bytes of memory
0 path entries using 0 bytes of memory
0/0 BGP path/bestpath attribute entries using 0 bytes of memory
0 BGP AS-PATH entries using 0 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 32 total bytes of memory
BGP activity 0/0 prefixes, 0/0 paths, scan interval 60 secs

Neighbor      V    AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
200.0.0.1      4 65001      0       0        1    0    0 00:19:54      4

RT-ISP-01#
RT-ISP-01#
RT-ISP-01#
RT-ISP-01#show ip bgp
BGP table version is 1, local router ID is 4.4.4.4
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               x RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

Network        Next Hop        Metric LocPrf Weight Path

```

```
RT-ISP-01#
RT-ISP-01#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 200.0.0.1 to network 0.0.0.0

200.0.0.0/24 is variably subnetted, 3 subnets, 2 masks
C    200.0.0.0/30 is directly connected, Serial0/2/0
C    200.0.0.1/32 is directly connected, Serial0/2/0
L    200.0.0.2/32 is directly connected, Serial0/2/0
L    200.0.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    200.0.1.0/30 is directly connected, GigabitEthernet0/1
L    200.0.1.1/32 is directly connected, GigabitEthernet0/1
S*   0.0.0.0/0 [1/0] via 200.0.0.1

RT-ISP-01#
RT-ISP-01#
RT-ISP-01#
```

```
RT-ISP-01#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0 unassigned      YES unset    administratively down down
GigabitEthernet0/1 200.0.1.1       YES manual    up          up
GigabitEthernet0/2 unassigned      YES unset    administratively down down
GigabitEthernet0/0/0 unassigned      YES unset    administratively down down
GigabitEthernet0/1/0 unassigned      YES unset    administratively down down
Serial0/2/0       200.0.0.2       YES manual    up          up
Serial0/2/1       unassigned      YES unset    administratively down down
Serial0/3/0       unassigned      YES unset    administratively down down
Serial0/3/1       unassigned      YES unset    administratively down down
Vlan1            unassigned      YES unset    administratively down down

RT-ISP-01#
RT-ISP-01#
RT-ISP-01#
RT-ISP-01#
```

RT-CLOUD

```
RT-CLOUD-01#show ip bgp summary
BGP router identifier 3.3.3.3, local AS number 65002
BGP table version is 2, main routing table version 6
1 network entries using 132 bytes of memory
1 path entries using 52 bytes of memory
0/0 BGP path/bestpath attribute entries using 0 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 240 total bytes of memory
BGP activity 1/0 prefixes, 1/0 paths, scan interval 60 secs

Neighbor      V   AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down State/PfxRcd
200.0.0.5      4 65001    22     21       2    0  0 00:20:35      4

RT-CLOUD-01#
```

```
RT-CLOUD-01#show ip bgp
BGP table version is 2, local router ID is 3.3.3.3
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop        Metric LocPrf Weight Path
*> 200.0.1.0/30      0.0.0.0          0         0 32768 i

RT-CLOUD-01#
RT-CLOUD-01#
```

```
RT-CLOUD-01#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 200.0.0.5 to network 0.0.0.0

200.0.0.0/24 is variably subnetted, 3 subnets, 2 masks
C    200.0.0.4/30 is directly connected, Serial0/2/0
C    200.0.0.5/32 is directly connected, Serial0/2/0
L    200.0.0.6/32 is directly connected, Serial0/2/0
200.0.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    200.0.1.0/30 is directly connected, GigabitEthernet0/2
L    200.0.1.2/32 is directly connected, GigabitEthernet0/2
S*   0.0.0.0/0 [1/0] via 200.0.0.5

RT-CLOUD-01#
```

```
RT-CLOUD-01#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0 unassigned      YES unset    administratively down down
GigabitEthernet0/1 unassigned      YES unset    administratively down down
GigabitEthernet0/2 200.0.1.2       YES manual    up          up
GigabitEthernet0/0/0 unassigned      YES unset    administratively down down
GigabitEthernet0/1/0 unassigned      YES unset    administratively down down
Serial0/2/0      200.0.0.6       YES manual    up          up
Serial0/2/1      unassigned      YES unset    administratively down down
Serial0/3/0      unassigned      YES unset    administratively down down
Serial0/3/1      unassigned      YES unset    administratively down down
Vlan1           unassigned      YES unset    administratively down down

RT-CLOUD-01#
RT-CLOUD-01#
RT-CLOUD-01#
RT-CLOUD-01#
```